

The Macro Alibi: Subjective Risk Attribution in Analyst Scenarios

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Abstract

Sell-side analysts over-attribute a stock's downside risk to macroeconomic forces. We document this *Macro Bear Bias* in 63,191 Morgan Stanley scenario-based valuation reports. Bear-case narratives assign 6.5 percentage points more weight to macro forces than base-case narratives by the same analyst, about the same firm, on the same date, while bull-case narratives show no such gap. The bear–base gap is largely insensitive to market conditions. Yet, conditional on the same market state, realized bear outcomes are no more market-driven than realized base outcomes. We explain the bias with a memory-based mechanism combining anchoring and representativeness. Analysts overgeneralize a remembered historical regularity—that severe declines are more macro-driven on average—to firm-specific downside scenarios under current market conditions, where the regularity need not hold. Consistent with this mechanism, the gap widened more after the Global Financial Crisis in the sectors most exposed to the crisis. The bias is consequential: the bear–base gap predicts pessimistic errors in subsequent base-case forecasts, and portfolios formed on a nonlinear bias-adjusted signal earn monthly CAPM alphas of up to 1.9%. Narratives analysts use to rationalize downside risk can distort numerical forecasts and asset prices.

Keywords: macro bear bias, narrative finance, downside risk, analyst beliefs, large language model

JEL: G11, G12, G14, G23

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1 Introduction

Risk in financial markets is not only a quantity, but also a story. When sell-side equity analysts publish a price target, they do not simply report a number; they explain it, naming the forces they expect to move cash flows up or down, the channels through which uncertainty operates, and the contingencies that would change their view. These narrative attributions matter. They are read by portfolio managers, cited in investment committees, and embedded—through analysts’ own forecasts—into the prices that shape and aggregate market beliefs. Whether the stories that professional information intermediaries tell about risk correctly identify the *sources* of risk is therefore a first-order question for understanding belief formation in financial markets.

One particularly visible expression of this question arises in how equity analysts describe firms’ upside and downside states. Bull cases tend to emphasize firm-specific opportunities, such as product cycles, margin expansion, and competitive position. Bear cases, by contrast, routinely emphasize aggregate forces, such as recessions, financial contagion, and monetary tightening. A natural interpretation is that this asymmetry reflects the true structure of risk. Conditional asset pricing argues that systematic risk is elevated in bad states: downside correlations rise and conditional betas widen ([Ang and Chen, 2002](#); [Ang, Chen, and Xing, 2006](#); [Lettau, Maggiori, and Weber, 2014](#)). Under this view, an analyst who leans on macro forces in the bear cases may simply be tracking where systematic risk is most concentrated. A skeptic who doubts that analysts track risk so precisely might still expect their macro emphasis to be conditional on another time-varying signal. In canonical behavioral accounts, vivid crises should make macro risks more salient, while their influence should fade as those crises recede ([Shiller, 2019](#); [Bordalo, Gennaioli, and Shleifer, 2020](#); [Bordalo et al., 2023](#)). Although these rational and behavioral readings differ, they share a common premise: bear-case macro emphasis should vary with an identifiable state variable.

Three findings, taken together, call this premise into question. First, macro emphasis in base, bear, and bull scenarios is essentially flat across market states. It neither rises in weak markets nor recedes in strong ones, even though the true conditional importance of systematic risk does. This pattern is difficult to reconcile with a rational benchmark: analysts do not appear to track where systematic risk concentrates, because their emphasis is insensitive to the market state on which that concentration depends. Second, in the time series, bear-case macro attribution rises only mildly

after vivid crises, and its year-to-year variation is small relative to the average gap. The direction is consistent with the canonical memory account, but the magnitude is not: salience shifts the emphasis at the margin rather than serving as the first-order force. Once both conditional signals are accounted for, a third fact remains: bear narratives consistently assign a higher macro share than base or bull narratives. This gap is wide, stable, and present across all market states. Belief formation about the downside therefore appears to follow a default script rather than a state-specific calculation. The distortion is behavioral in origin, in the sense that it reflects a story analysts reach for whenever a downside case is required, but it is near-unconditional in form and far stickier than canonical memory models allow.

To study this distortion, we exploit the institutional setting of the Morgan Stanley Risk-Reward Framework reports from 2007 to 2025, comprising 63,191 scenario-based analyst reports. The Risk-Reward Framework is unique in requiring analysts to articulate three explicit scenarios—bear, base, and bull—for the same firm at the same point in time, each accompanied by a structured narrative explaining the conditions under which that scenario obtains. This within-report, across-scenario design holds the analyst, the firm, the calendar date, and the prevailing macro environment fixed, so the primary variation we exploit is the directional state of the world being described. The design also supplies an object that the representativeness literature usually has to assume or estimate: the comparison against which the bear case is judged. Here the base case plays that role, and the protocol reports it.

We use large language models to extract, from each scenario narrative, the share of analyst attention allocated to macroeconomic, industry-level, and firm-specific drivers. We then build ex post counterparts for the same reports: we classify which scenario is ultimately realized—bear if the stock subsequently falls below the analyst’s bear target, base if it lands between the bear and bull targets, and bull if it exceeds the bull target. We then compute realized CAPM R^2 within the corresponding market-state-by-realized-scenario subsample as the natural quantitative analog to the share of attribution devoted to macro factors, following Roll’s (1988) canonical approach of using R^2 to attribute return variation to market-wide forces. Under the rational benchmark, the ex ante macro attribution an analyst assigns to a given scenario should track the ex post realized macro content of that scenario within the same conditioning set. A persistent disconnect between these two objects is our measure of narrative misattribution.

The data reveal a sharp and systematic asymmetry. Bear-case narratives devote significantly more (6.5 percentage points) attribution to macroeconomic forces than base-case narratives—a within-report gap that absorbs all analyst-, firm-, and time-level heterogeneity by construction. The corresponding bull–base gap is essentially zero, an order of magnitude smaller. The bear-narrative tilt is also remarkably stable across market states: it does not collapse in calm years and is not amplified in crises. The ex post benchmark behaves very differently. Conditional on the same forward market state, the bear–base difference in realized CAPM R^2 is close to zero in weak market states and turns negative in good market states. Realized downside outcomes are therefore not disproportionately explained by market-wide variation; bear narratives describe them as if they were. Most severe declines are not market events: for about three-quarters of the reports whose stock ends below the bear target with a negative excess return, the market component accounts for less than half of the decline. We refer to this dissociation between ex ante narrative attribution and ex post realized variation as *Macro Bear Bias*: analysts exhibit a tendency to over-attribute downside outcomes to aggregate macroeconomic risks in bear-case narratives, even when the realized data do not support the attribution. In this sense, macro risk can serve as a “macro alibi” for the bear case.

The bias has economic consequences that extend well beyond the bear narrative itself. Scenario narratives serve as the cognitive scaffolding for the analyst’s valuation judgment, and a macro-heavy bear case spills over into the base-case forecast on which prices are anchored. We propose a discount-rate channel: a bear narrative that over-attributes downside risk to systematic factors may lead the analyst to perceive the firm as having greater aggregate exposure. This, in turn, raises the implicit discount rate applied to future cash flows and lowers the base-case price target. In this channel the distortion scales multiplicatively with the firm’s true systematic exposure, so high-beta firms should display larger forecast-error responses to a given bear–base macro-attribution gap than low-beta firms; our specification weights the gap by beta but does not separately identify that gradient. Empirically, a one-standard-deviation increase in the beta-weighted bear–base macro-attribution gap predicts a positive base-case forecast error of roughly 2–4%. This suggests that when analysts disproportionately attribute downside risk to macro forces in the bear case, they are systematically too pessimistic about the base-case price target.

The forecast-error pattern translates into return predictability, more strongly under a nonlinear

specification than a linear one.¹ Sorting on analysts' raw base-case forecasts generates no abnormal performance. Adjusting those forecasts for the predictable bias component using a flexible approximation to the nonlinear conditional bias produces a value-weighted long–short portfolio with an average monthly return of 2.02%, a CAPM alpha of 1.91%, and an annualized Sharpe ratio of 1.10.² The narrative asymmetry, in other words, is not a stylistic feature of analyst writing; it carries first-order information about subsequent forecast errors and cross-sectional returns.

We next study the mechanism behind Macro Bear Bias. What cognitive or institutional process produces a bear-case template applied so independently of the market state? A purely rational benchmark would predict bear-case macro emphasis that tracks the conditional importance of systematic risk; a canonical memory-based model would predict emphasis that rises and falls sharply with the salience of recent crises. The data are broadly consistent with the latter, but only at the margin. Bear-case macro attribution and the bear–base gap itself do tick up in the aftermath of vivid crash episodes such as the Global Financial Crisis (GFC) and the COVID-19 crash, which is the direction a salience or memory model predicts. In the aggregate, however, the movement is mild (from 5.7 to 7.2 percentage points between 2007 and 2008, and from 6.0 to 7.2 between 2019 and 2020; Figure 11). It is a second-order effect layered on top of a sticky baseline across time. Salience nudges a standing template rather than generating a sharp, transient pattern. The mechanism we identify therefore serves as a refinement of, rather than an alternative to, standard memory-based explanations.

We formalize the mechanism in a model of how analysts attribute a scenario to its causes. Its starting point is a kernel of truth. Pooled across history, severe declines really are more market-driven than ordinary outcomes: within firm, realized R^2 is 3.45 percentage points higher in bear outcomes than in base outcomes, and the difference comes entirely from timing, because bear outcomes cluster in years when the market falls. Holding the market state fixed, it is -0.68 points. In the model, analysts build the bear case from memory, where that regularity is stored and rehearsed, rather than from the distribution that applies at the report date. Two channels draw on the

¹This is consistent with the broader theoretical observation that belief-driven asset price dynamics are generically nonlinear and state-dependent. Dew-Becker, Giglio, and Molavi (2025) show that under Bayesian filtering—including under behavioral departures from full rationality—the response of subjective beliefs to signals is governed by the higher-order cumulants of agents' posterior distributions, so that the mapping between subjective belief objects and realized outcomes is nonlinear by construction.

²The equal-weighted counterparts are 1.80%, 1.74%, and 0.99.

same memory. Through anchoring, the analyst partially substitutes the remembered composition of severe declines for the case-specific assessment. Through a diagnostic tilt, once the bear case is cued, analysts overweight the macro feature most representative of severe declines. This is the stereotype mechanism of [Bordalo et al. \(2016\)](#), applied to the attribution of an outcome to its causes. Exaggeration alone cannot produce the pattern: a tilt evaluated under current conditions would vanish wherever the rational gap vanishes. It is the slowly moving memory that lets a true regularity be applied to dates that do not warrant it. The analyst data pin down the combined strength of the two channels, but not the split between them.

The model delivers four predictions:

1. Bear narratives carry more macro attribution than base narratives written by the same analyst, about the same firm, on the same date.
2. The bear–base gap moves little with the market state, even when the realized macro content of downside outcomes moves a great deal.
3. The bull case shows no matching gap: its macro attribution stays close to that of the base case.
4. The gap responds to which crises analysts remember: it widens after a macro crisis, and it is wider for analysts who lived through one.

A rational analyst also delivers the first prediction whenever macro forces are genuinely more diagnostic of the downside at the report date. A rational analyst delivers none of the other three. A rational gap moves with the market state; a rational analyst has no reason to treat the downside differently from the upside once both are measured against their benchmarks; and the set of crises an analyst happens to remember plays no role in a rational assessment.

The evidence lines up with the three predictions that distinguish our mechanism from the rational benchmark. Across report years, the bear–base gap varies a third as much as the realized bear–base difference in systematic variation and is uncorrelated with it. Bull-case attribution stays within one percentage point of base-case attribution in every market state, while the bear-side excess over the realized benchmark lies between 3 and 9 percentage points. After the Global Financial Crisis, the gap widened by 4.9 percentage points more in financials, energy, and materials—the sectors most exposed to the systematic downturn the crisis made salient—holding the firm and the year fixed, even though the aggregate gap moved little. News coverage of a macro topic raises

bear-case attention to it, and that attention lingers after coverage fades, though both effects are only marginally significant. Analysts who lived through the crisis at Morgan Stanley write wider gaps, an estimate significant only at the 10 percent level. A further prediction, that the gap scales with how macro-diagnostic a firm’s downside genuinely is, cannot be tested in our sample: that diagnosticity varies mainly over time rather than across firms, and the sample spans few years.

Related literature. Our paper first contributes to the literature on subjective risk perceptions. A central insight of this literature is that perceived risk need not coincide with statistical risk: investors’ beliefs about uncertainty are shaped by experience, memory, attention, and extrapolation, and these subjective perceptions matter for prices and portfolio choices (Malmendier and Nagel, 2011; Guiso, Sapienza, and Zingales, 2018; Greenwood and Shleifer, 2014; Giglio et al., 2021; Lochstoer and Muir, 2022; Jensen, 2022; Bohren et al., 2024). We document a new form of distorted subjective risk perception: analysts systematically misattribute the source of downside risk, over-weighting macroeconomic forces even when realized downside outcomes are not disproportionately explained by market-wide variation. In contemporaneous work, Chen, Li, and Yang (2025) use scenario-based analyst reports to study the higher moments of subjective return distributions. The two papers are complementary: they characterize the shape of the subjective return distribution analysts assign to each firm, while we study how analysts attribute the *source* of risk across scenarios.

A second strand studies downside risk and state-dependent systematic risk. Ang and Chen (2002) and Ang, Chen, and Xing (2006) document and price downside risk in U.S. equities; Lettau, Maggiori, and Weber (2014) show that a downside-risk CAPM, with betas that differ in bad market states, helps explain currency and equity returns; Kacperczyk, Van Nieuwerburgh, and Veldkamp (2014, 2016) show that fund managers reallocate attention between aggregate and firm-specific information over the business cycle, devoting more attention to macro conditions in recessions.³ We contribute by studying not realized downside comovement itself but analysts’ subjective attribution of downside risk. Analysts appear to overgeneralize a valid empirical regularity: while systematic risk can be more important in bad states, they apply macro-heavy bear

³Lewellen and Nagel (2006) and Lewellen, Nagel, and Shanken (2010) caution that time variation in betas is unlikely to fully resolve standard asset-pricing anomalies and that conventional cross-sectional tests have limited power to distinguish among conditional factor models. Our argument does not rest on conditional asset pricing fully accounting for the realized data; we show that analysts’ bear-case macro emphasis is largely unresponsive even to the state-dependent variation that conditional asset pricing does identify.

narratives near-unconditionally, even when conditional realized R^2 does not justify the emphasis. Macro Bear Bias is therefore not evidence against state-dependent downside risk; it is evidence that analysts fail to condition their risk narratives on the prevailing market state.

Methodologically, our focus on a nonlinear, state-dependent mapping from analyst narratives to forecast errors connects to two complementary literatures. On the theoretical side, [Dew-Becker, Giglio, and Molavi \(2025\)](#) show that under (Bayesian and behavioral) learning belief dynamics are generically nonlinear, so that linear specifications should be expected to understate the predictive content of subjective beliefs. On the empirical side, [Rahimi and Recht \(2007, 2008\)](#) introduce Random Fourier Features (RFF) regression as a tractable approximation to nonlinear kernel methods, and [Kelly, Malamud, and Zhou \(2022, 2024\)](#); [Didisheim et al. \(2024\)](#); [Kelly and Malamud \(2025\)](#) show that highly parameterized RFF specifications can outperform sparse linear models in return prediction.⁴ We bring these together by identifying, in a specific institutional setting, a subjective-belief distortion whose return content linear specifications understate and a flexible nonlinear approximation recovers more fully.

A third strand studies the qualitative narratives in analyst reports and the broader role of narratives in shaping beliefs and prices. [Joos, Piotroski, and Srinivasan \(2016\)](#) show that analyst risk scenarios contain information beyond quantitative forecasts; [Ke \(2024\)](#); [Bradshaw et al. \(2025\)](#); [Bastianello, Décaire, and Guenzel \(2024\)](#) apply textual analysis to study how analysts form subjective beliefs, frame risk discussions, and bring mental models to valuation.⁵ At a broader level, [Shiller \(2019\)](#) and [Hirshleifer \(2020\)](#) argue that contagious economic narratives and social processes shape beliefs, asset prices, and economic outcomes. We extend this literature by exploiting within-report, across-scenario variation in narrative content: the Morgan Stanley Risk-Reward Framework lets us compare how the same analyst explains bull, base, and bear outcomes for the same firm at the same point in time, identifying the incremental macro attribution analysts add when moving from the expected case to the downside case—the specific miscalibration that single-scenario textual analyses cannot detect.

⁴More broadly, a growing asset-pricing literature shows that nonlinear methods can uncover predictive relations and interaction effects that are difficult to detect with linear models; see, for example, [Gu, Kelly, and Xiu \(2020\)](#); [Freyberger, Neuhierl, and Weber \(2020\)](#); [Gu, Kelly, and Xiu \(2021\)](#); [Chen, Pelger, and Zhu \(2024\)](#); [Feng et al. \(2024\)](#); [Bryzgalova, Pelger, and Zhu \(2025\)](#). For a broad overview, see [Kelly and Xiu \(2023\)](#).

⁵A complementary literature studies subjective beliefs in analysts' numerical forecasts; see, for example, [La Porta \(1996\)](#), [Bordalo et al. \(2019\)](#), [Bordalo et al. \(2024\)](#), [Bouchaud et al. \(2019\)](#), and [De la O and Myers \(2021\)](#).

Finally, our mechanism connects to a long line of work in behavioral economics and finance on the cognitive foundations of belief formation (Hirshleifer, 2001; Barberis, 2018), particularly related to representativeness and availability heuristics (Barberis, Shleifer, and Vishny, 1998; Gennaioli and Shleifer, 2010; Bordalo, Gennaioli, and Shleifer, 2018; Bordalo et al., 2019). Tversky and Kahneman (1973) introduce the availability heuristic as a fundamental bias in subjective probability judgments, and Bordalo, Gennaioli, and Shleifer (2020) and Bordalo et al. (2023) formalize how cues from the current environment trigger retrieval of similar past experiences, generating beliefs that systematically over-weight vivid and recently activated memories. We bring representativeness to a different object—the attribution of an outcome to its causes rather than the distribution of future outcomes—in a professional setting where the comparison group is reported rather than assumed, and we show that the kernel of truth behind the macro tilt lies in the timing of severe declines.⁶

The rest of the paper proceeds as follows. Section 2 documents Macro Bear Bias. We introduce the Morgan Stanley Risk-Reward reports, construct LLM-based narrative attribution measures, and compare analysts’ ex ante macro attributions with an ex post realized CAPM R^2 benchmark across market states. Section 3 develops the model, and Section 4 takes its predictions to the data. Section 5 shows that the attribution gap predicts base-case forecast errors and that bias-adjusted forecasts predict returns. Section 6 examines alternative explanations, and Section 7 concludes.

2 Data and the Macro Bear Bias

This section poses the paper’s central puzzle, which has two arms. The first is that analysts assign markedly more macroeconomic attribution to bear narratives than to base narratives written by the same analyst, about the same firm, on the same date, and that they do so near-unconditionally: the gap barely moves with the market state. The second is that the realized data do not support the attribution. Conditional on the market state, the bear–base difference in realized systematic

⁶Our findings also relate to the literature on career concerns and herding (Holmström, 1999; Scharfstein and Stein, 1990; Hong, Kubik, and Solomon, 2000; Hong and Kubik, 2003; Chevalier and Ellison, 1999). A natural alternative to our cognitive interpretation is that macro narratives function as reputationally safer deflections from sharp firm-specific calls, because aggregate explanations are harder to falsify and less damaging to management relationships. The data provide little support for this institutional, career-concerns form of the view: junior analysts do not display stronger Macro Bear Bias, recent Morgan Stanley investment-banking relationships do not amplify it, and standard analyst-level career-incentive proxies explain little of its variation.

variation is close to zero, and negative in good states, so the macro emphasis analysts place on the downside is not tracking the macro content of downside outcomes.

We build to this in four steps. We begin by introducing the Morgan Stanley Risk-Reward reports and our text-based measures of narrative risk attribution. We then state the rational benchmark that ex ante attribution should satisfy, and construct its ex post analogue, the realized CAPM R^2 . Comparing the two delivers the two arms of the puzzle. Finally, we ask whether the macro emphasis in bear narratives has any basis at all, and show that it does—pooled across market states, severe declines really are more macro-driven than ordinary outcomes—which sharpens the puzzle rather than resolving it.

What the evidence in this section cannot settle is whether the attribution is wrong given the true source likelihoods an analyst faces at the report date. The realized R^2 benchmark is an ex post proxy for that object, not the object itself, and the divergence between ex ante emphasis and ex post macro content is evidence of miscalibration only under assumptions about what analysts could have known when they wrote. Section 3 takes up that question.

2.1 Data

Our paper uses data from the Risk-Reward Framework in Morgan Stanley analyst reports downloaded from the LSEG (formerly Refinitiv) Workspace. In total, we collected 63,191 Risk-Reward reports from January 2007 to May 2025. Some examples are shown in Appendix Section B.

Traditional sell-side research often relied on single-point price targets or coarse recommendations (e.g., “buy,” “hold,” or “sell”), which can convey a false sense of precision and provide limited information about the range of possible future outcomes. To address this limitation, Morgan Stanley introduced the Risk-Reward Framework in 2007, requiring analysts to present and justify 12-month stock-price expectations under three scenarios: bull, base, and bear. This framework was designed to provide a more informative and differentiated research product for institutional investors, who care not only about expected return but also about the distribution of potential outcomes and the sources of upside and downside risk. By explicitly presenting upside and downside scenarios, these reports help investors assess the skewness of expected payoffs, understand the assumptions underlying each case, and incorporate analyst views more directly into portfolio decisions. The

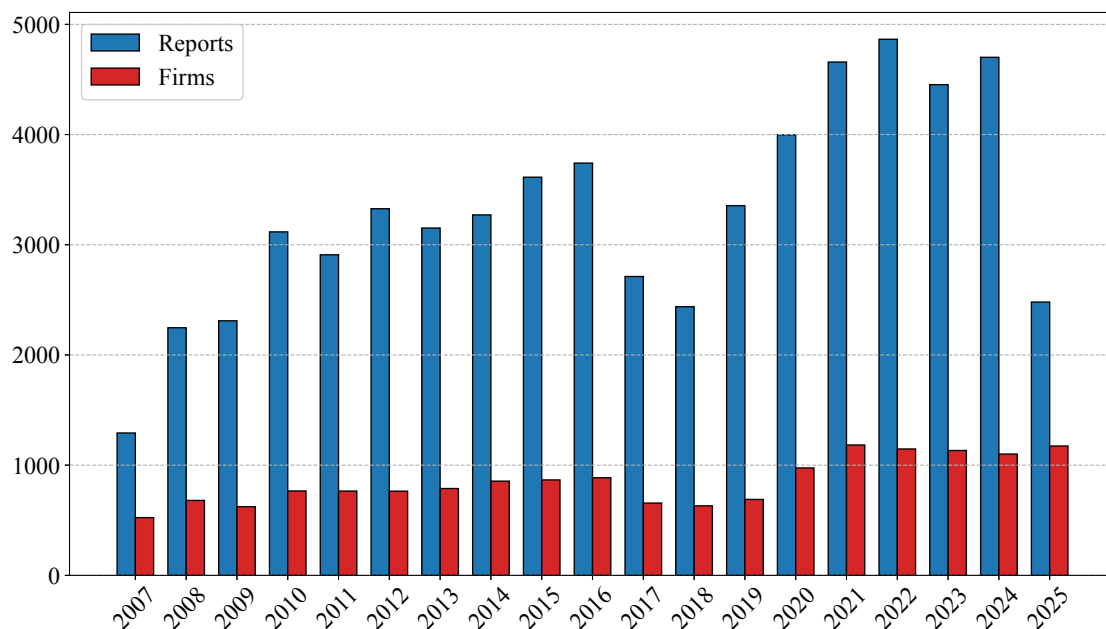


Figure 1 Histogram of Reports and Firms by Year.

Note: This figure shows the yearly count of Morgan Stanley Risk-Reward analyst reports and covered firms from January 2007 to May 2025.

reports are particularly valuable for our purposes because they provide both structured scenario-specific price targets and narrative discussion of the reasoning, key drivers, and investment debate behind those targets. ⁷

This dataset offers two main advantages. First, in addition to the base-case forecast, it contains analyst expectations under both bull and bear scenarios, which allows us to measure subjective upside and downside risk expectations separately. Second, the accompanying narratives explain the assumptions and reasoning behind each scenario. Using natural language processing (NLP) and large language models (LLMs), we can systematically extract and analyze these texts to identify the drivers of perceived upside and downside risks.

Figure 1 reports the number of available Risk-Reward reports and the number of covered firms by year. Because the Risk-Reward Framework is specific to Morgan Stanley and covers primarily large-capitalization, analyst-covered U.S. equities, the results should be interpreted with this sample scope in mind.

⁷For introductions to the framework, see [Weyns et al. \(2011\)](#) and [Srinivasan and Lane \(2011\)](#).

2.2 Textual Identification of Subjective Risks

We focus on the narrative text in each Morgan Stanley Risk-Reward report to understand how analysts justify their bull-, base-, and bear-case expectations. Specifically, we examine whether analysts exhibit a downside–upside asymmetry in the types of risks they emphasize across scenarios, distinguishing among macro-level, industry-level, and firm-level risks. To quantify the content of these narratives, we apply the GPT prompt described in Appendix Section C using the gpt-4o-mini model. Our main finding is that analysts place significantly greater emphasis on macro-level risk in bear scenarios than in base or bull scenarios. As a robustness check, we also examine the distinction between systematic and idiosyncratic risk using the prompt described in Appendix Section C.3.

Figure 2 presents the report-level distributions of risk attribution across scenarios. Panel A shows the distribution of attribution to macro-level risks in bull, base, and bear cases. Panel B reports the corresponding distribution for firm-level risks, and Panel C does the same for industry-level risks. The figure reveals a clear pattern: relative to bull and base scenarios, bear scenarios assign substantially greater weight to macro-level risks and significantly less weight to firm-level risks. By contrast, the distributions of macro-level attribution in bull and base scenarios are similar to each other, as are the distributions of firm-level attribution across these two scenarios. Industry-level attribution shows broadly similar distributions across all three scenarios.

Figure 3 complements this distribution evidence by showing the time-series evolution of average attribution to macro-, firm-, and industry-level risks from 2007 to 2025. Panel A plots macro-level attribution, Panel B plots firm-level attribution, and Panel C plots industry-level attribution. The time-series evidence reinforces the distributional results: throughout the sample period, analysts consistently devote more attribution to macro-level risks in bear scenarios than in bull or base scenarios, while placing less attribution on firm-level risks in bear scenarios. In contrast, attribution patterns in bull and base scenarios remain broadly similar over time for both macro- and firm-level risks. Industry-level attribution again appears relatively stable and similar across bull, base, and bear cases. For completeness, Figure A.1 reports the distribution at each point in time. Table 1 quantifies the asymmetry. With firm fixed effects, bear-case narratives carry 6.47 percentage points more macro attribution than base-case narratives ($t = 34.09$) and 6.07 points less firm attribution, while the bull–base difference in macro attribution is -0.40 points.

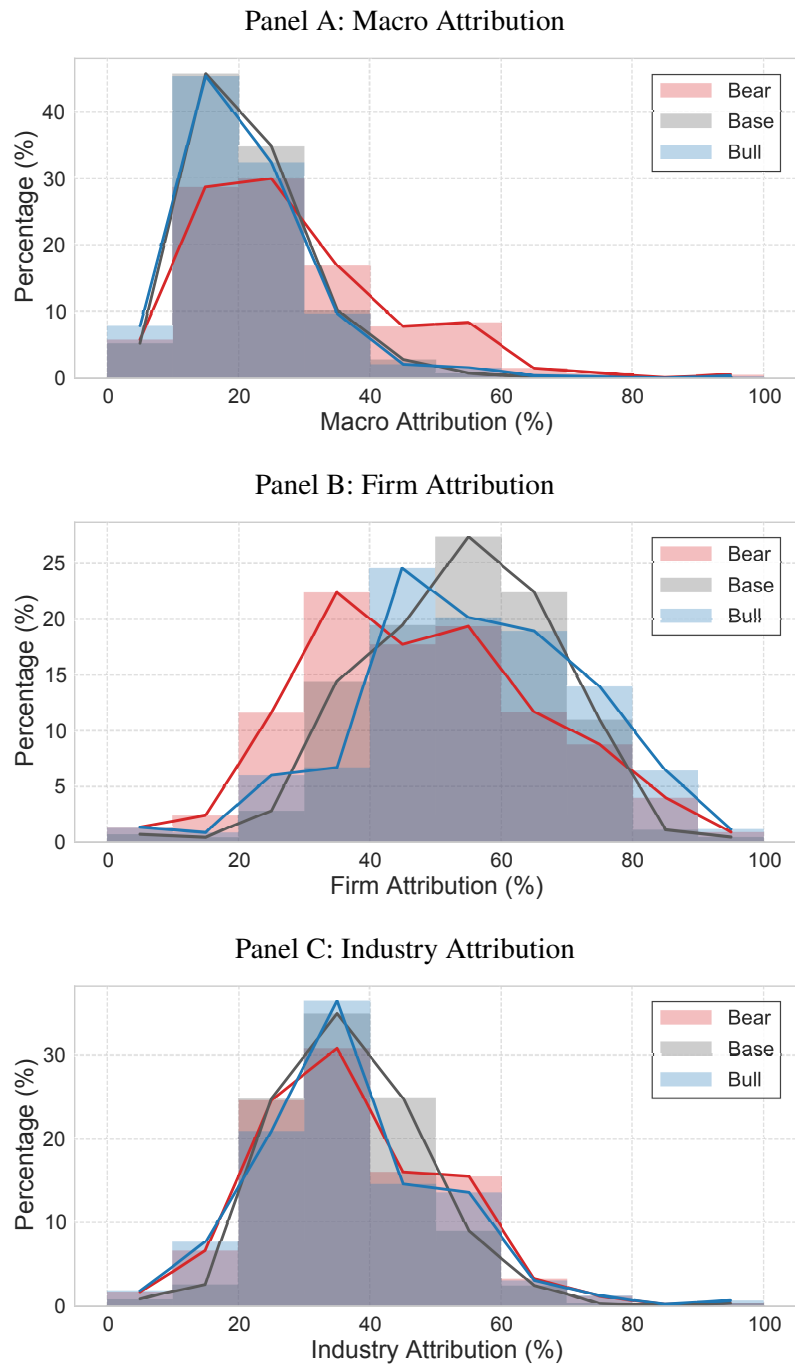


Figure 2 Attribution Distribution to Macro, Firm, and Industry News.

Note: Report-level distribution of attribution shares to firm-, industry-, and macro-level news across bull, base, and bear cases.

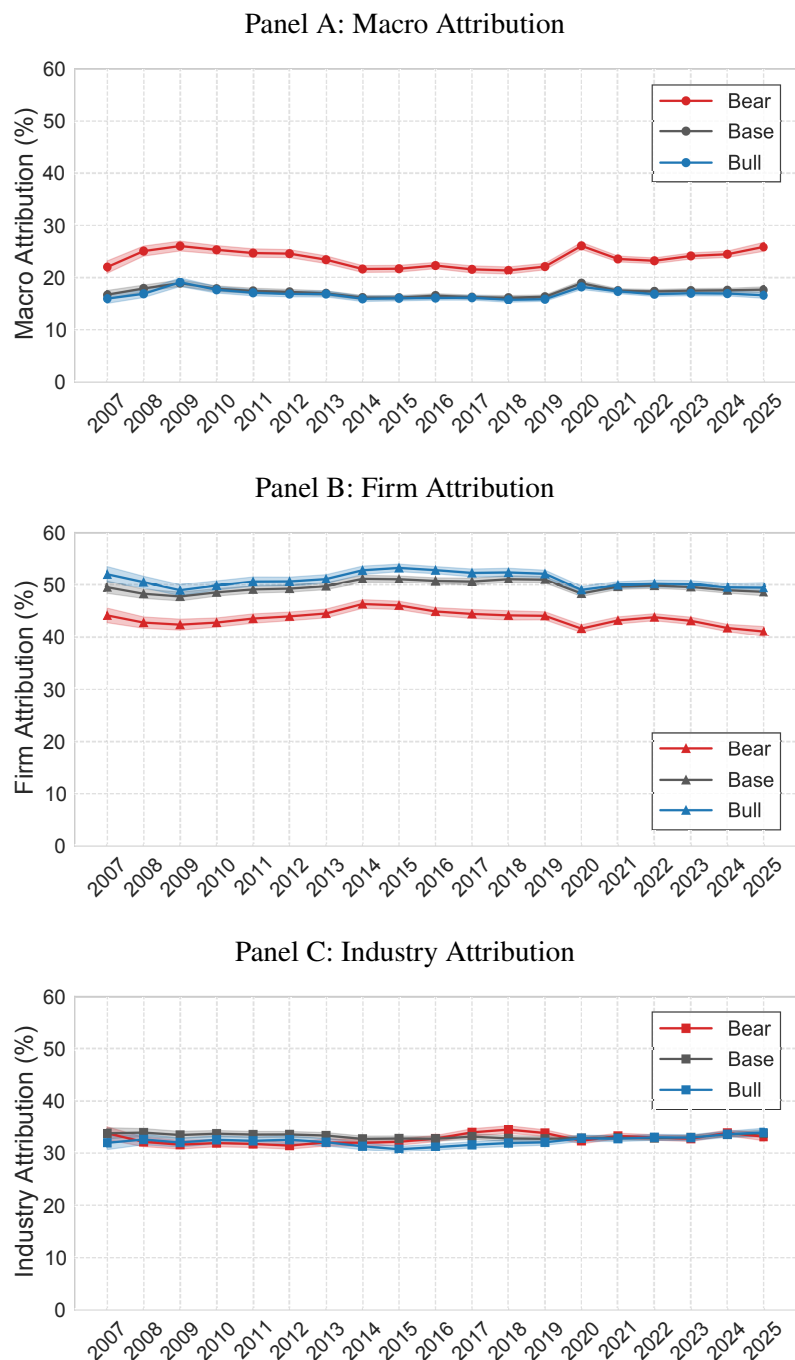


Figure 3 Time Series of Average Attribution to Macro, Firm, and Industry News.

Note: Time series of report-level annual average attribution shares to macro-, firm-, industry-level news. The shaded areas are 99% confidence interval of the average.

Table 1 Scenario Dummies and Attribution Shares

Note: This table reports panel regressions of attribution shares on scenario indicators. The dependent variables are macro attribution (cols 1–2), firm attribution (cols 3–4), and industry attribution (cols 5–6). Base scenario is the omitted category; the constant reports the mean attribution share for the Base scenario. Bear and Bull coefficients measure the difference relative to Base. All specifications include firm fixed effects; even columns add year-month fixed effects. Standard errors are clustered by firm and month. t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
	Macro	Macro	Firm	Firm	Industry	Industry
Bear	6.4687*** (34.09)	6.4687*** (34.07)	-6.0672*** (-31.54)	-6.0672*** (-31.53)	-0.4027** (-2.55)	-0.4027** (-2.55)
Bull	-0.3982*** (-5.54)	-0.3982*** (-5.54)	1.1609*** (10.20)	1.1609*** (10.19)	-0.7654*** (-7.21)	-0.7654*** (-7.21)
Constant	17.1843*** (443.27)	17.1843*** (443.27)	49.6454*** (858.57)	49.6454*** (858.57)	33.1654*** (722.96)	33.1654*** (722.96)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE		Yes		Yes		Yes
Observations	187,917	187,917	187,917	187,917	187,917	187,917
Within R^2	0.0781	0.0863	0.0431	0.0521	0.0006	0.0053

2.3 The Rational Benchmark

The benchmark asks how much of a firm's return variation macro forces would explain if a given scenario materialized. Write firm i 's excess return as

$$R_{i,t}^{\text{realized}} = \alpha_i + \beta_{i,t} M_t + \varepsilon_{i,t}, \quad (1)$$

where M_t denotes the market excess return, $\beta_{i,t}$ the firm's exposure to aggregate macro conditions, and $\varepsilon_{i,t}$ firm-specific shocks. The importance of macro risk is conditional: it should be judged against the market state prevailing when the analyst writes. Let z_t denote the information available at time t about that state, and define macro importance in scenario $k \in \{\text{bear, base, bull}\}$ as the share of return variance attributable to the market within that state and scenario,

$$\text{Macro Importance}_{i,t} \mid z_t, k \equiv \frac{\beta_{i,t}^2 \text{Var}(M_t \mid z_t, k)}{\text{Var}(R_{i,t} \mid z_t, k)}.$$

Here $\beta_{i,t}$ is defined as the population coefficient from projecting $R_{i,t}$ on M_t within the conditioning set defined by (z_t, k) . Equivalently, the residual $\varepsilon_{i,t}$ is orthogonal to M_t within that conditioning set, so the ratio has a natural interpretation as a variance share.⁸ A rational analyst's macro emphasis in scenario k tracks the expectation of this share; Section 3 formalizes it as the rational attribution share $A_{i,k,t}^*$, reserving $A_{i,k,t}$ for what the analyst actually reports. Two properties follow. Macro attribution should move with the market environment, and differences across scenarios should reflect differences in the conditional relevance of macro risk, not a template that treats bad outcomes as macro-driven. The realized CAPM R^2 of Section 2.4 is the sample analogue of macro importance within a market-state, realized-scenario cell.

2.4 Realized CAPM R^2

The evidence so far establishes that analysts assign substantially more macro attribution to bear scenarios than to base or bull scenarios. But is this differential attribution warranted? To assess whether analysts are correct to place greater weight on macro risks in bear scenarios, we construct an *ex post* realized benchmark based on the realized CAPM R^2 in each case, following Roll (1988)'s classic argument that R^2 captures the share of a firm's return variation attributable to market-wide forces. If macro risks truly matter more in downside states, a firm's realized return variation should be more strongly explained by market-wide movements when the report ultimately realizes a bear outcome than when it realizes a base or bull outcome. We test whether, conditional on the market state, the realized CAPM R^2 is significantly higher for realized-bear reports than for realized-base or realized-bull reports. A higher realized R^2 in bear outcomes would support analysts' macro emphasis in downside scenarios; otherwise, it would suggest overweighting.

Two features of this benchmark are worth mentioning. First, our attribution measure captures macroeconomic drivers while the benchmark is a *market* model; we treat market-model R^2 as the natural quantitative analog of macro attribution because both are variance-attribution shares, and because market returns aggregate the priced component of macroeconomic shocks. Second, a one-period R^2 measures contemporaneous comovement, whereas macroeconomic shocks may propagate over time, so returns at any horizon reflect the cumulated effect of past macro innovations. Our

⁸Without this conditional orthogonality, the variance decomposition would also include a $2\beta_{i,t}\text{Cov}(M, \varepsilon \mid z_t, k)$ term. The conditioning therefore makes the object a clean variance share rather than a covariance-laden index.

identification is robust to this distinction because the object of interest is *differential*: we compare realized R^2 across bear, base, and bull outcomes within the same firm and the same market state, so any common low-frequency propagation component differences out unless it varies systematically with the realized scenario.

Our empirical procedure has three steps. First, for each report, we compute the realized ex post CAPM R^2 by regressing the firm's daily excess returns on daily market excess returns over the 252 trading days following the report release. This R^2 measures the extent to which the firm's subsequent stock return variation is explained by systematic market movements.

Second, we classify each report as realized-bear, realized-base, or realized-bull according to the relationship between the analyst's target prices and the stock's realized price one year after the report release. A report is classified as realized-bear if the realized price is less than or equal to the bear-case target price. It is classified as realized-bull if the realized price is greater than or equal to the bull-case target price. If the realized price lies strictly between the bear and bull targets, the report is classified as realized-base.

Third, we aggregate these report-level realized R^2 measures to the firm level. For each firm, the realized-bear R^2 is the average realized CAPM R^2 across all of its realized-bear reports. Likewise, the realized-base and realized-bull R^2 are defined as the average realized CAPM R^2 across the firm's realized-base and realized-bull reports, respectively. Constructing the measures at the firm level allows us to compare whether the realized R^2 is higher in bear outcomes than in base or bull outcomes for the same firm, thereby reducing concerns that cross-firm differences drive the results.

To make these comparisons within a common market environment, we further construct firm-level realized R^2 measures separately by market condition. Specifically, under a given market state, we restrict the sample to firms that experience all three realized report types—bear, base, and bull—within that same market condition. For example, when analyzing periods in which the market excess return over the 252 trading days after the report release lies between -5% and 5% , we include only firms that have realized-bear, realized-base, and realized-bull reports within that market-return range. This restriction ensures that comparisons across realized cases are not mechanically driven by differences in overall market environments.

We obtain stock return data from the CRSP database. After matching Morgan Stanley reports to 252-trading-day forward returns following the report release date, the sample contains 45,765

reports in total, with report dates from January 2007 to December 2023. Figure A.2 in the Appendix plots the annual number of reports and covered firms in the matched sample.

To mitigate distortions from stock splits and reverse splits, we construct an alternative measure of the one-year-ahead realized price by compounding daily returns excluding dividends, denoted by $r_{t,x}$, over the 252 trading days after the report release:

$$\text{constructed realized price} = \text{current price} \times \prod_{t=1}^{252} (1 + r_{t,x}).$$

If the realized price in one year differs from the constructed realized price by more than 10% in either direction, we replace the observed realized price with the constructed realized price. This adjustment helps correct cases in which stock splits or reverse splits mechanically distort the reported price level.

Table 2 Summary Statistics

Note: This table reports summary statistics for the return-matched Morgan Stanley Risk-Reward report sample, with report dates from 2007 to 2023. Panel A reports report-level attribution shares (percentages) by scenario. Panel B reports firm characteristics at the report date and the one-year-ahead return and forecast variables. All attribution shares are extracted by GPT using the prompt in Appendix C.

Variable	N	Mean	Median	Std	Min	P25	P75	Max
Panel A: Narrative Attention								
Bull: Macro	45765	16.70	10.00	11.11	0.00	10.00	20.00	100.00
Bull: Industry	45765	32.29	30.00	14.33	0.00	20.00	40.00	100.00
Bull: Firm	45765	51.01	50.00	17.32	0.00	40.00	60.00	100.00
Base: Macro	45765	17.02	15.00	9.46	0.00	10.00	20.00	100.00
Base: Industry	45765	33.14	30.00	11.49	0.00	25.00	40.00	100.00
Base: Firm	45765	49.84	50.00	14.43	0.00	40.00	60.00	100.00
Bear: Macro	45765	23.33	20.00	15.10	0.00	10.00	30.00	100.00
Bear: Industry	45765	32.64	30.00	14.06	0.00	20.00	40.00	100.00
Bear: Firm	45765	44.03	40.00	18.32	0.00	30.00	60.00	100.00
Panel B: Returns and Forecast Variables								
Return	45765	0.12	0.07	0.74	-3.29	-0.17	0.31	74.14
Subjective Return	45765	0.20	0.10	1.79	-4.67	-0.01	0.22	115.33
Forecast Error	45765	-0.08	-0.02	1.91	-115.26	-0.30	0.26	74.03
$\hat{\beta}_{\text{CAPM}}$	43008	1.13	1.08	0.44	-1.04	0.83	1.37	5.29
R^2_{CAPM}	43008	0.32	0.30	0.18	0.00	0.18	0.44	0.86

Table 2 reports summary statistics of core narrative and firm-level data. Panel A highlights the narrative asymmetry across scenarios. Bear-case macro attribution averages 23.3%, compared with 17.0% and 16.7% in the base and bull cases, respectively—a bear–base gap of 6.3 percentage points in the return-matched sample.⁹ The mirror image appears in firm-level attribution: bear-case firm attribution averages 44.0%, roughly 6 percentage points below the base-case mean of 49.8%, while industry attribution is nearly identical across all three scenarios. Panel B shows that the coverage universe skews toward high-beta firms (mean $\hat{\beta}_{\text{CAPM}} = 1.13$).

Figure 4 shows how realized outcomes depend on the market. We group reports by the forward 252-day market excess return, using a centered 10% window around each point on the horizontal axis.

Panel A reports the share of bear, base, and bull outcomes within each market state. As the market return rises, the share of realized-bear outcomes falls and the share of realized-bull outcomes rises; the base share is roughly stable. In a weak market, more stocks end near the analyst’s downside scenario. In a strong market, more reach the upside. Panel B reports how each outcome type is spread across market states. Realized-bear outcomes are spread broadly, while realized-base and especially realized-bull outcomes concentrate in strong markets, because forward market returns are mostly positive. Which outcome a report realizes therefore depends on the market, so comparisons across realized outcomes must hold the market state fixed. We do so throughout.

2.5 Empirical Evidence on Macro Bear Bias

With the realized CAPM R^2 benchmark and market-state conditioning framework in hand, we now examine whether the elevated macro emphasis in bear-case narratives reflects the actual macro content of bear-case outcomes. We show that it does not. Instead, analysts systematically overweight macro risk in bear scenarios. We refer to this distortion as Macro Bear Bias: the tendency to over-attribute downside outcomes to aggregate macro risks in bear-case narratives.

To ensure an apples-to-apples comparison with realized R^2 , we construct firm-level ex ante macro attribution for each case (bear, base, and bull) as the average macro attribution across all

⁹In the full sample of Table 1, the bear–base gap is 6.47 percentage points. Every report contributes one bear, one base, and one bull narrative, so the Bear coefficient equals the difference in means; the two numbers differ only because the samples differ.

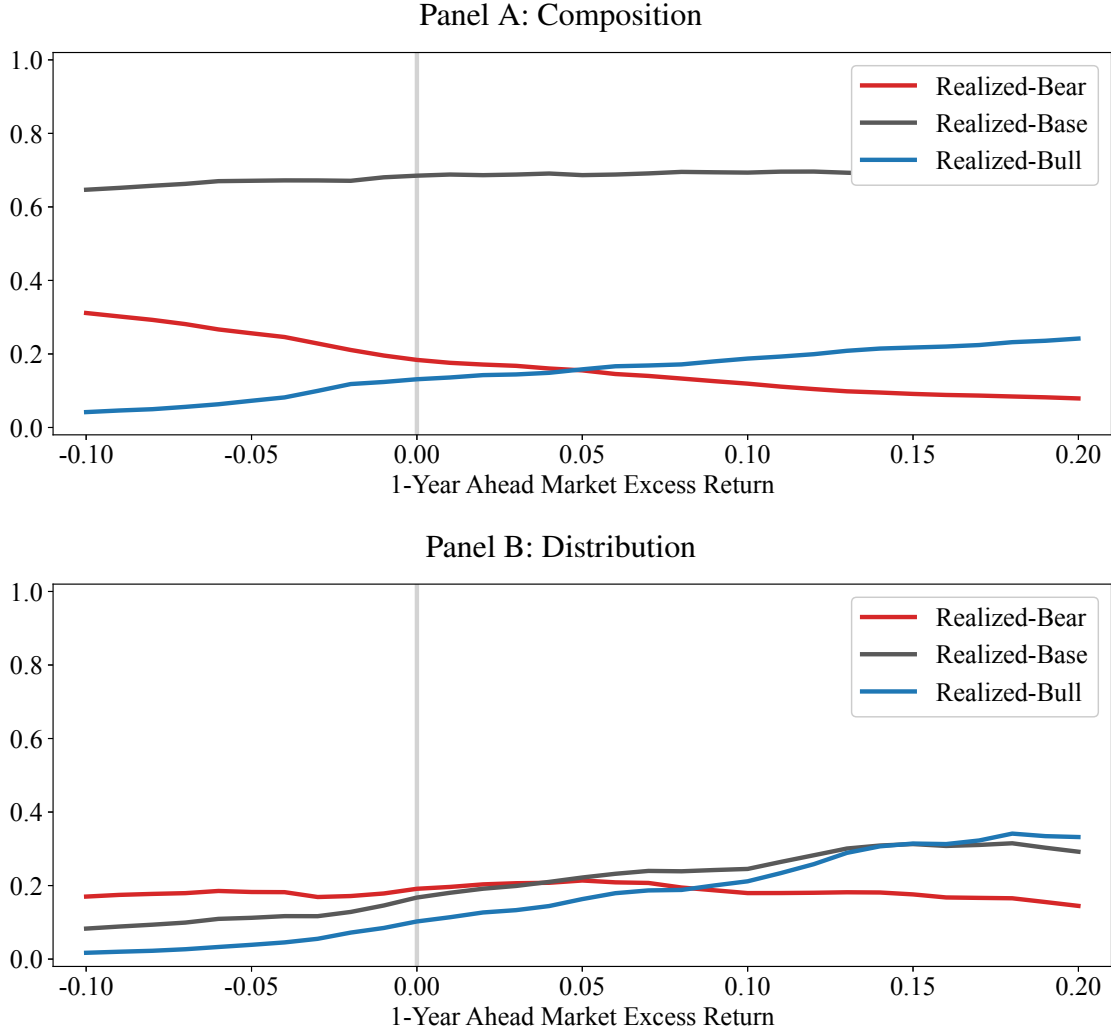


Figure 4 Composition and Distribution of Realized Bear, Base, and Bull Reports Across Market States.

Note: This figure plots the composition and distribution of realized bear, base, and bull reports across market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. Panel A shows the composition of bear, base, and bull reports within each return state:

$$\frac{\text{Number of realized bear/base/bull reports in a return state}}{\text{Total number of reports in that market excess return}}.$$

Panel B shows the distribution for realized bear, base, and bull reports:

$$\frac{\text{Number of realized bear/base/bull reports in a market excess return state}}{\text{Total number of realized bear/base/bull reports}}.$$

reports of the firm that fall within the same market state. Thus, the comparison with the ex post

realized R^2 is based on the exact same subset of reports. Figure 5 plots realized R^2 and macro attribution conditional on market state.¹⁰ For each point x on the horizontal axis, we consider reports whose forward 252-day market excess returns fall within a centered 10% window, $[x - 5\%, x + 5\%]$, around that point. Panel A shows the average CAPM R^2 over the 252 trading days after report release for realized bear, realized base, and realized bull cases. Panel B shows the average ex ante macro attribution that analysts assign to the bear, base, and bull cases.¹¹

Two findings stand out. First, macro attribution in the bear case is consistently and substantially higher than in the base or bull case, even though the realized R^2 is similar across the realized bear, base, and bull cases. This bear-base attribution gap is the empirical signature of Macro Bear Bias: even when the realized macro component of risk is similar across scenarios conditional on market state, analysts continue to describe downside outcomes as disproportionately macro-driven. Second, the three attribution series are relatively flat across market states, in sharp contrast to the realized R^2 benchmarks, which decline broadly as the market state improves. This suggests that analysts do not sufficiently adjust their attribution of macro risk to the future one-year realized market environment.

Figure 6 makes this pattern especially clear by plotting the bear–base gap in macro attribution together with the corresponding bear–base gap in realized R^2 across market states.¹² The realized R^2 gap is statistically indistinguishable from zero when the subsequent annual market excess return is below about 16%, and significantly negative above it. In contrast, the bear–base gap in macro attribution remains significantly positive throughout. In other words, analysts consistently portray the bear case as much more macro-driven than the base case, even though the realized difference in macro risk is economically small or even negative, depending on the market state. This divergence provides direct evidence of Macro Bear Bias.

The pattern is not a forecast of a market crash. If the scenarios simply encoded different expected market states—every bear case tied to a large market decline—then base-case attribution would track realized base-case R^2 across market states, and bull-case attribution would sit well

¹⁰As a robustness check, we find similar results when classifying risks into systematic and idiosyncratic components; see Figure A.3.

¹¹Figure A.4 shows average macro attribution conditional on the realization of bear, base, and bull outcomes. In each bin, macro attribution is measured within the corresponding realized scenario. For example, Macro Bear attribution includes a report’s Macro Bear attribution only when the report’s outcome is realized as a bear case. The pattern is similar to that in Figure 5.

¹²Figure A.5 shows that the results are similar when risks are classified into systematic and idiosyncratic.

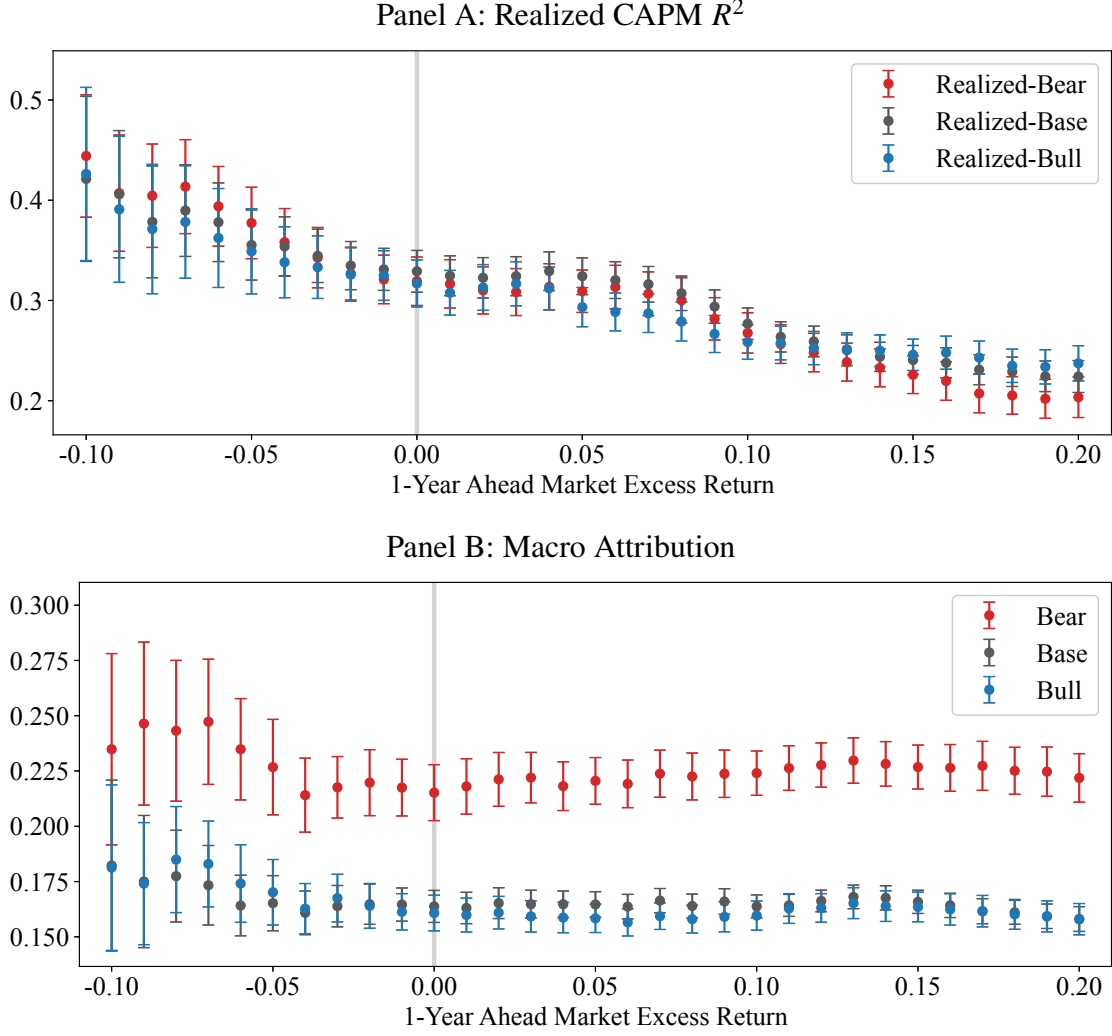


Figure 5 Realized CAPM R^2 and Macro Attribution Conditional on Market States.

Note: This figure shows the realized R^2 and macro attribution conditional on different market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. We apply the same-firms rule within each bin, requiring firms to have realized-bear, realized-base, and realized-bull reports. Macro attribution is measured as the firm-level average macro attribution within each ex-ante scenario (bear, base, and bull). Realized CAPM R^2 is computed as the firm-level average R^2 over the 252 trading days following the report release for reports ex post classified as realized bear/base/bull cases. Error bars show 95% confidence intervals from constant-only regressions.

below base-case attribution. Neither holds. Base-case attribution is flat (Panel B of Figure 5), and the bull–base gap is -0.40 percentage points against 6.47 for bear–base (Table 1). Section 3 develops the account the evidence points to, in which bear narratives draw on remembered macro crises rather than on current conditions.

A natural objection is that the pattern could be mechanical: bear cases describe worse outcomes,

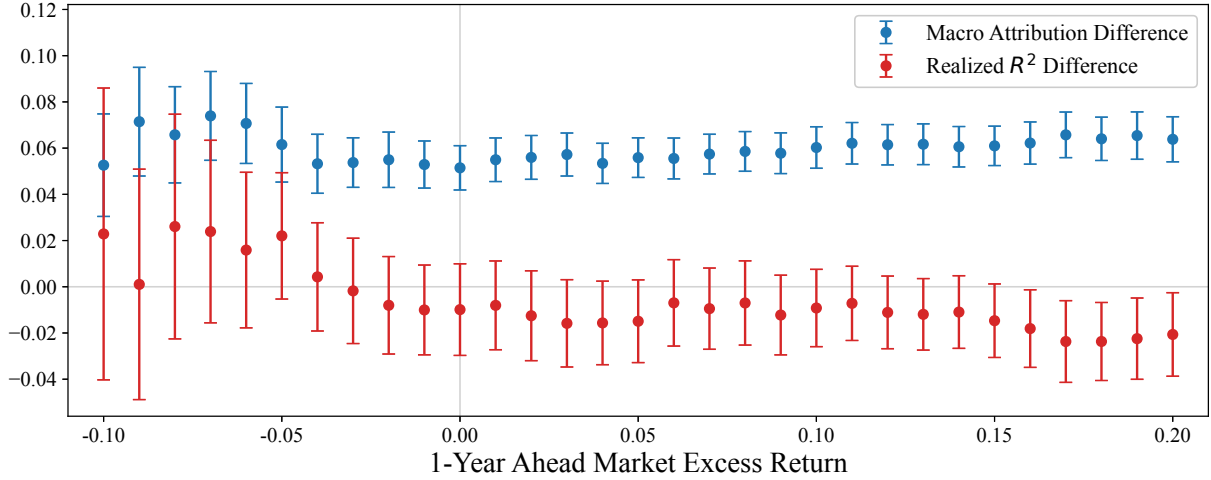


Figure 6 Bear – Base Gaps in Macro Attribution and Realized CAPM R^2 Conditional on Market States.

Note: This figure plots firm-level bear–base differences in macro attribution and realized CAPM R^2 on different market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. We apply the same-firms rule within each bin, requiring firms to have realized-bear, realized-base, and realized-bull reports. Macro attribution is measured as the firm-level average macro attribution within each ex-ante scenario (bear, base, and bull). Realized CAPM R^2 is computed as the firm-level average R^2 over the 252 trading days following the report release for reports ex post classified as realized bear or realized base cases. The plotted values represent the bear-minus-base differences. Error bars show 95% confidence intervals from constant-only regressions.

which tend to co-occur with macro downturns, so bear narratives should naturally emphasize macro factors more. However, the regime-conditional analysis in Figures 5 and 6 directly addresses this. Conditional on the same forward market state, realized CAPM R^2 is similar across bear, base, and bull outcomes, yet the macro-attribution gap remains large and positive. If elevated bear-case macro attribution were a rational response to genuinely higher macro content in bear outcomes, the gap should shrink toward zero once we condition on market state—it does not.

2.6 Unconditional versus Conditional Diagnosticity

The evidence so far shows that the bear–base attribution gap is not warranted conditional on the market state. It does not show that the emphasis analysts place on macro forces is baseless. These are different claims, and the distinction between them is what the model of Section 3 is built on, so we establish it directly.

Pool all dates and market states and ask a simpler question than the one Figure 6 asks: are severe declines, taken as a class across the whole sample period, more macro-driven than ordinary

outcomes? We compare realized CAPM R^2 between reports whose stock ends at or below the bear target and reports whose stock ends between the bear and bull targets. We make the comparison within firm, with firm fixed effects, so that a firm's bad years are compared with its own ordinary years. This comparison does not hold the market environment fixed. Figure 7 reports the results.

Two facts hold at once, and the tension between them is only apparent. Unconditionally, macro forces are more diagnostic of severe declines than of ordinary outcomes. Within firm, realized R^2 is 3.45 percentage points higher in bear outcomes than in base outcomes ($t = 10.27$), against a base-outcome mean of 29.8 percent (Panel A).¹³ Conditionally, the difference disappears. Holding the market state fixed, it is -0.68 percentage points ($t = -2.14$), and it is negative in every market-state bin except years in which the market falls by more than 10 percent. Panel B shows that the entire unconditional gap comes from timing: bear outcomes disproportionately occur in bad market states. Bear outcomes cluster in years when the market falls, and in those years return variation is more systematic for every firm. This timing contributes 4.13 percentage points (standard error 0.17); the within-state difference subtracts 0.68. There is no contradiction. A regularity that holds across the historical record need not hold at any particular date, and the conditional object is what an analyst writing a report in a specific month should be reporting. The macro emphasis in bear narratives therefore has a kernel of truth. What it lacks is conditioning.

Panel C shows that this timing effect becomes stronger as declines become more severe. Define a severe decline as a forward return below $-h$ standard deviations of the firm's pre-report volatility. As h rises from 0.5 to 2.5, the share of severe declines that occur in years when the market falls by more than 10 percent rises from 28 to 59 percent, and the timing component of the bear-base difference rises from 4.7 to 6.7 percentage points (the increase is 2.0 points, $t = 3.67$; Panel C). The total difference does not rise, because within a market state the most severe declines are increasingly firm-specific. Part of this within-state pattern is mechanical: severity is defined over the same 252-day window in which R^2 is measured, so selecting on large firm-specific moves lowers R^2 . Simply put, the more severe the decline, the more likely it is to coincide with a broad market downturn, and the larger the part of the gap that comes from that coincidence.

¹³Pooling reports across firms without firm fixed effects gives a difference of 0.22 percentage points ($t = 0.48$). The cross-firm comparison mixes in which firms fall: volatile firms with low R^2 realize bear outcomes more often. The within-firm comparison removes this composition and keeps all variation across dates. Averaging firm-level means over firms with all three realized outcomes, as in Figures 5 and 6, gives 3.60 percentage points ($t = 9.56$).

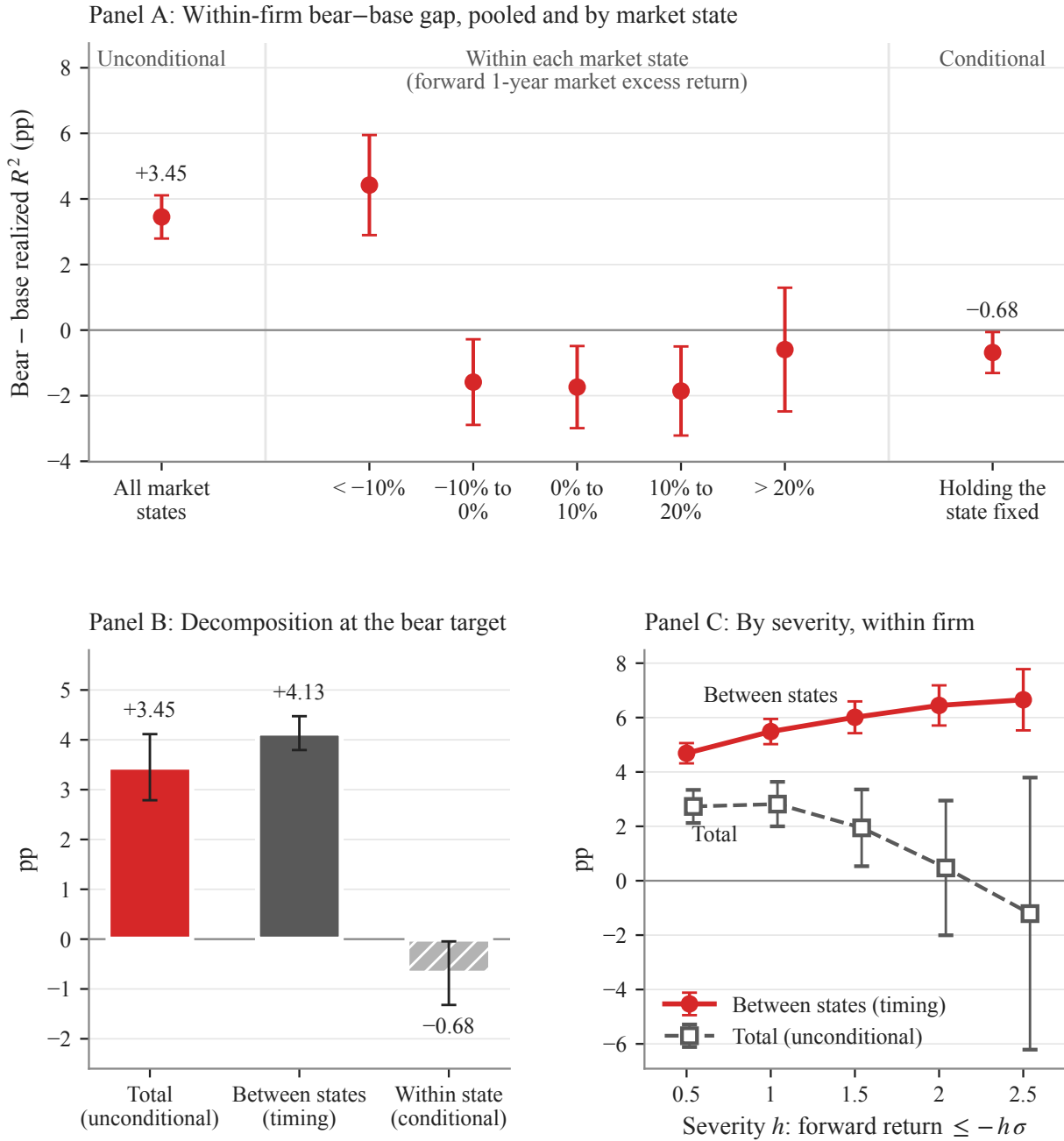


Figure 7 Unconditional versus Conditional Diagnosticity.

Note: This figure reports the difference in realized CAPM R^2 , over the 252 trading days after the report, between reports realized at or below the bear target and reports realized strictly between the bear and bull targets, estimated within firm (firm fixed effects). Panel A pools all market states, estimates the difference within bins of the forward one-year market excess return, and adds market-state-bin fixed effects. Panel B splits the pooled difference into the part due to the timing of bear outcomes across market states and the part within states. Panel C repeats the split for severe declines defined by a forward return below $-h$ standard deviations of pre-report volatility. Error bars are 95% confidence intervals from firm-clustered standard errors (Panel A) or a firm-clustered bootstrap with 1,000 draws (Panels B and C). The sample is unfiltered, as in Figures 5 and 6.

The same data make the point from the other direction. Most severe declines are not market events. Among reports whose stock ends below the bear target, the market component—the firm’s beta times the market excess return—accounts for less than half of the decline in 74 percent of cases, and only 31 percent of these outcomes occur in years when the market falls by more than 10 percent.¹⁴ Severe declines are more often market-driven than ordinary declines, 26 against 16 percent, yet most of them are firm-specific, scattered across firms and dates rather than experienced collectively. Macro-driven collapses are what severe declines are most readily *recalled* to be, not what most of them are.

The regularity documented here also carries a prediction, which we return to once the model is in place. If the bear-case template exaggerates a true feature of severe declines rather than inventing one, then the distortion should be widest exactly where that true feature is strongest. Section 4.2 discusses what the data can and cannot say about this prediction, across firms and over time.

This is as far as the realized data can take us. The comparison between ex ante attribution and ex post macro content establishes that analysts’ emphasis does not move with conditions that should move it, and that the emphasis rests on an unconditional regularity it is being applied beyond. It does not establish that the attribution is wrong relative to the source likelihoods an analyst genuinely faced at the report date, because those likelihoods are not observed. Answering that requires a model of how analysts form the attribution, and of what a correctly formed attribution would look like. That is what we turn to next.

3 A Model of Macro Bear Bias

Section 2 leaves a question the data cannot settle. Bear narratives carry markedly more macroeconomic attribution than base narratives written by the same analyst, about the same firm, on the same date; the gap is close to flat across market states; and the realized data show no matching pattern, with the state-conditional bear–base difference in systematic variation near zero and negative in good states. Why would attribution run at a rate the conditional data do not support, and fail to move when the conditions do?

¹⁴A decline is a negative excess return over the 252 trading days after the report; 6,034 of the 6,740 realized-bear reports qualify. Beta is estimated over the same window. Among declines in realized-base reports, the market component accounts for less than half of the decline in 84 percent of cases.

Our explanation is that analysts do not build the bear case from the current conditional distribution. They recall what severe declines look like, and in memory severe declines are macroeconomic. We formalize this by transposing the representativeness-based theory of stereotypes of [Bordalo et al. \(2016\)](#), hereafter BCGS, built on the recall framework of [Gennaioli and Shleifer \(2010\)](#), to a new object: not the distribution of future outcomes, but the attribution of a conditioning outcome to its causes. Two channels draw on a common memory. Anchoring: the analyst partially substitutes the composition of similar outcomes in memory for the case-specific conditional ([Kahneman and Tversky 1972](#); [Tversky and Kahneman 1973](#)). Representativeness: imagining a severe decline tilts attribution toward whichever source is most diagnostic of that decline, evaluated under the same memory ([Bordalo et al., 2016](#)). Shut both down and the analyst is rational.

One feature of the setting deserves emphasis. In BCGS and throughout this literature the comparison group against which representativeness is evaluated is supplied by the modeler or assigned by the experimenter; inferred from field data it becomes the framework’s least disciplined object.¹⁵ The Risk–Reward protocol supplies it directly: the base case is elicited from the same analyst, about the same firm, on the same date. The comparison group is neither imposed nor estimated. It is reported.

3.1 Setup

A firm i is covered at date t . Let $Y_{i,t+1} \equiv (R_{i,t+1} - \mu_{i,t})/\sigma_{i,t}$ denote its one-year-ahead return, standardized by the analyst’s time- t conditional mean and scale. Standardizing makes a “bear case” mean the same thing for a utility and for a biotech firm, and makes today’s bear case comparable to the crises the analyst remembers. For a bear-severity index $h > 0$ and an upper threshold $u_{i,t} > 0$, the three scenarios partition the outcome space:

$$E_{i,t}^B(h) = \{Y_{i,t+1} \leq -h\}, \quad E_{i,t}^0(h) = \{-h < Y_{i,t+1} < u_{i,t}\}, \quad E_{i,t}^U = \{Y_{i,t+1} \geq u_{i,t}\}. \quad (2)$$

In the data, the lower and upper thresholds correspond to the analyst’s bear and bull target prices, with the base target between them; we index the bear, base, and bull scenarios of Section 2 by

¹⁵[Bordalo et al. \(2019\)](#) estimate the lag of the comparison group at roughly eleven quarters and note that their data cannot distinguish comparing news to past conditions from comparing it to lagged distorted beliefs.

$k \in \{B, 0, U\}$. Raising h moves the lower boundary of the bear and base events together, so both conditionals may depend on it.

The source space is binary, $s \in \{m, n\}$: macroeconomic forces against an aggregate of industry and firm-specific forces.¹⁶ For each realized outcome, nonnegative source contributions $C_m, C_n \geq 0$ with $C_m + C_n > 0$ define the macro contribution share $\omega_m \equiv C_m / (C_m + C_n) \in [0, 1]$. What an analyst should report for scenario k is the average of ω_m over the outcomes that scenario contains, the *rational attribution share*:

$$A_{i,k,t}^*(h) \equiv \mathbb{E}[\omega_m \mid E_{i,t}^k(h), z_t, x_i], \quad k \in \{B, 0\}, \quad (3)$$

with the analogous definition for $k = U$. Here z_t is the information available at the report date and x_i collects firm attributes. Eq. (3) is an ex ante object, so the realized conditional- R^2 benchmark of Section 2 is an ex post proxy for A^* , not the same object: in that proxy, the share of return variance the market explains over the year after a report plays the role of ω_m . Appendix A.7 implements A^* for the special case in which each outcome has a single source, $\omega_m \in \{0, 1\}$. Write $A_{i,k,t}(h)$ for what the analyst actually reports.

3.2 Memory, Representativeness, and Diagnostic Attribution

An analyst building a bear case does not consult the historical record. They consult memory, and memory is a reweighted version of that record.

Let $\mathcal{E}$ be the universe of historical return episodes with probability measure μ^0 ; each episode $e \in \mathcal{E}$ has a standardized outcome $y(e)$ and a macro contribution share $c_m(e) \in [0, 1]$, and we write $e \in E^k(h)$ for $y(e) \in E^k(h)$. Episodes are recalled with rehearsal weights $w_{i,t}(e) \geq 0$: the weight an episode acquires through salience, collective experience, and repetition in market commentary (Tversky and Kahneman, 1973; Bordalo, Gennaioli, and Shleifer, 2020; Bordalo et al., 2023). The average macro share among the remembered episodes falling in scenario k is the rehearsed macro

¹⁶The LLM measures industry and firm sources separately before aggregation. With extracted shares $(a_{i,k,t}^{\text{mac}}, a_{i,k,t}^{\text{ind}}, a_{i,k,t}^{\text{firm}})$ summing to one, the model-facing observed share is $\hat{A}_{i,k,t} = a_{i,k,t}^{\text{mac}}$ and the empirical log odds are $\log[a_{i,k,t}^{\text{mac}} / (a_{i,k,t}^{\text{ind}} + a_{i,k,t}^{\text{firm}})]$ —a deterministic transformation we perform after extraction. The LLM is never asked to infer a likelihood ratio.

share,

$$p_{i,k,t}(h) \equiv \frac{\int_{\mathcal{E}} w_{i,t}(e) c_m(e) \mathbf{1}\{e \in E^k(h)\} d\mu^0(e)}{\int_{\mathcal{E}} w_{i,t}(e) \mathbf{1}\{e \in E^k(h)\} d\mu^0(e)}, \quad k \in \{B, 0, U\}, \quad (4)$$

with $E^U(h)$ read as the bull event, which does not depend on h , and with the objective macro share $p_k^0(h)$ the special case of uniform weights. Two properties of the rehearsal weights do the work below. They are *slow*: they move at the frequency of experience and retelling, not of market realizations. And they are *personal*: an analyst who lived through a macro-driven crash carries weight a colleague who entered later does not (Malmendier and Nagel, 2011; Bordalo, Gennaioli, and Shleifer, 2020). The first generates a gap stable across market states; the second generates cross-analyst heterogeneity in its width.

Assumption 1 (Full support) *For every relevant (i, t, h, k) , the denominator in Eq. (4) is positive and $p_{i,k,t}(h) \in (0, 1)$. Rational attribution shares satisfy $A_{i,k,t}^*(h) \in (0, 1)$.*

A source is diagnostic of an event when it is disproportionately indicative of that event relative to a comparison event. The rehearsed and objective diagnosticity of macro forces for the bear event against the base event are $D_{i,t}(h) \equiv \text{logit } p_{i,B,t}(h) - \text{logit } p_{i,0,t}(h)$ and $D^0(h) \equiv \text{logit } p_B^0(h) - \text{logit } p_0^0(h)$, the log of the relative representativeness of macro forces for the bear event in the likelihood-ratio sense of Tversky and Kahneman (1983) and Definition 1 of Bordalo et al. (2016). The first is diagnosticity as the analyst remembers it; the second is diagnosticity as it is. When firms, sectors, or dates differ in the record available to them, as in Proposition 3, D^0 refers to the record relevant to that firm type and date; we suppress these indices.

Assumption 2 (Kernel of truth) *For the relevant severity range, $D^0(h) > 0$: macro forces are objectively more diagnostic of bear outcomes than of base outcomes.*

The scope of Assumption 2 matters, because at first glance it appears to contradict the paper's headline finding. D^0 is defined over the pooled historical record: across all dates and all market states, severe declines carry a higher macro share than ordinary outcomes. The rational gap $G_{i,t}^*$ below is a different object, conditional on the date and firm through (z_t, x_i) . An unconditional regularity that is positive and a conditional gap that is near zero are not in tension, and the model's

entire content is that the analyst substitutes the former for the latter. Section 2 reports both. The template is not a fabrication; it is a true statement about history applied to a case that does not warrant it.

Assumption 3 (Rehearsal amplification) *Rehearsal preserves and weakly amplifies objective diagnosticity: $D_{i,t}(h) \geq D^0(h)$, with $\partial D_{i,t}/\partial D^0 \geq 1$ across firms, sectors, and dates.*

Crises that are collectively experienced and repeatedly discussed are disproportionately macro-driven, while idiosyncratic disasters—the failed trial, the fraud, the lost contract—are scattered across firms and rarely retold. Memory oversamples the macro tail. The proportionality clause carries the amplification into the cross-section and over time, where rehearsal scales with the truth rather than merely adding to it. Assumption 2 is verifiable from realized outcomes alone; Assumption 3 can be tested only through its implications, so a reader who doubts one need not reject the other.

Two channels carry this memory into what the analyst reports.

Definition 1 (Memory anchoring) *For every scenario $k \in \{B, 0, U\}$, the analyst’s anchored attribution pools rational and rehearsed attribution in log odds:*

$$\text{logit } Q_{i,k,t} = (1 - \lambda) \text{logit } A_{i,k,t}^* + \lambda \text{logit } p_{i,k,t}, \quad \lambda \in [0, 1]. \quad (5)$$

Anchoring is attribute substitution: asked a hard question about this firm at this date, the analyst answers an easier one about what firms like this look like in general (Kahneman and Tversky 1972; Tversky and Kahneman 1974). Eq. (5) is a tractable log-opinion-pooling form for that substitution rather than an implication of BCGS, and it applies to every scenario: at $\lambda > 0$, no reported attribution responds fully to case-specific information.

Definition 2 (Diagnostic tilt) *The reported bear attribution additionally receives a diagnostic tilt—an additive log-odds distortion toward the diagnostic source:*

$$\text{logit } A_{i,B,t} = \text{logit } Q_{i,B,t} + \theta D_{i,t}(h), \quad A_{i,k,t} = Q_{i,k,t} \text{ for } k \in \{0, U\}, \quad \theta \geq 0. \quad (6)$$

The tilt is the stereotype. Having recalled that severe declines are macro events, the analyst does not merely report that composition but overweights it, in exactly the exponential way BCGS propose; Lemma A.1 shows their Definition 2 reduces in the binary case to the additive log-odds shift in Eq. (6). What we add is the anchored baseline it applies to and the rehearsed measure under which representativeness is evaluated.

Two consequences of the binary structure are worth recording. First, the distorted share lies in $(0, 1)$ for any finite $\theta D_{i,t}(h)$, because the tilt is additive in log odds. Bounding the distortion is not free in this literature—continuous-state applications of the same weighting form require Gaussian conjugacy, rank truncation, or a smoothing constant—and here Assumption 1 is all that is needed. The cost is that with two sources there is no distribution over types whose mean or variance could be exaggerated, so BCGS’s higher-moment results have no counterpart; what survives is a single log-odds gap, which is what our data measure. Second, that the tilt hits the bear case alone is an assumption about *cues*. Being asked to imagine a severe decline invites the question of what makes this scenario different; being asked for the modal or favorable case invites no such comparison. The assumption is substantive and Proposition 2 states its implication. At $\lambda = \theta = 0$ the analyst is rational in every scenario.

The model’s central object is the bear–base contrast, which strips out everything common to the two scenarios—the firm’s narrative orientation, the analyst’s writing habits, any scenario-invariant part of anchored attribution—and leaves only what downside framing itself does. Define $G_{i,t}(h) \equiv \text{logit } A_{i,B,t}(h) - \text{logit } A_{i,0,t}(h)$ and $G_{i,t}^*(h) \equiv \text{logit } A_{i,B,t}^*(h) - \text{logit } A_{i,0,t}^*(h)$. Differencing Eqs. (5)–(6) gives an exact decomposition (Appendix A.4):

$$G_{i,t}(h) = (1 - \lambda)G_{i,t}^*(h) + (\lambda + \theta)D_{i,t}(h). \quad (7)$$

The first term is the case-specific rational gap, attenuated: only a fraction $1 - \lambda$ of any variation in rational conditions reaches the narrative. The second is what the gap inherits from memory. Holding the memory state and scenario thresholds fixed, the second term does not move with current conditions at all, so a gap that is large and stubbornly flat across market states is one in which it dominates.

Definition 3 (Macro Bear Bias and MacroBearGap) *The bias component of the observed gap is*

$\mathcal{B}_{i,t}(h) \equiv G_{i,t}(h) - G_{i,t}^*(h) = (\lambda + \theta)D_{i,t}(h) - \lambda G_{i,t}^*(h)$. Macro Bear Bias is the property $\mathcal{B}_{i,t} > 0$. The share-space counterpart of $G_{i,t}$ is $\text{MacroBearGap}_{i,t} \equiv A_{i,B,t} - A_{i,0,t}$, the percentage-point attribution gap.¹⁷

Each channel earns its place because it generates what the other cannot. If the tilt operated alone, $\lambda = 0$, reported attribution would respond one-for-one to case-specific information and the observed gap would track the rational gap shifted by a constant—not what the data show. If anchoring operated alone, $\theta = 0$, the coefficient on rehearsed diagnosticity would be $\lambda \leq 1$, so the analyst could at most report the remembered composition of severe declines and never exceed it. Only the tilt lets the reported gap run past the analyst’s own memory, which is the precise sense in which a stereotype exaggerates.

3.3 Two Implications

Two results follow from Eq. (7) and the definitions alone. The model’s remaining predictions concern how the gap varies with memory and objective diagnosticity; we state each in Sections 4 and 5, where we take it to the data or explain why the field cannot. Table A.1 maps them.

Proposition 1 (Gap decomposition, the rational null, and uniform attenuation) *Under Assumption 1, holding fixed the memory state and scenario definitions:*

- (i) *If $\lambda = \theta = 0$, then $G_{i,t} = G_{i,t}^*$ and reported attribution tracks rational attribution one-for-one in every scenario.*
- (ii) *If $\lambda > 0$, then $\partial \logit A_{i,k,t} / \partial \logit A_{i,k,t}^* = 1 - \lambda$ for every scenario $k \in \{B, 0, U\}$, with the same coefficient in each, and $\partial G_{i,t} / \partial G_{i,t}^* = 1 - \lambda < 1$.*
- (iii) *$G_{i,t} > 0$ if and only if $(\lambda + \theta)D_{i,t} > -(1 - \lambda)G_{i,t}^*$.*
- (iv) *At $\lambda = 1$, $G_{i,t} = (1 + \theta)D_{i,t}$, independent of $G_{i,t}^*$.*

Part (i) is the null Section 2 rejects under the realized- R^2 proxy: a rational analyst’s gap equals the rational gap and inherits its state dependence. Part (ii) says the attenuation is uniform, and carries a restriction easy to overlook. Because λ in Eq. (5) does not depend on k , the same coefficient governs bear, base, and bull; estimating the slope separately by scenario therefore yields

¹⁷ $\text{MacroBearGap}_{i,t}$ is linked to $G_{i,t}$ exactly by $A_{i,B,t} - A_{i,0,t} = A_{i,0,t}(1 - A_{i,0,t})(e^{G_{i,t}} - 1) / (1 - A_{i,0,t} + A_{i,0,t}e^{G_{i,t}})$ (Appendix A.6). A stable log-odds gap does not mechanically imply a stable percentage-point gap.

three coefficients the model requires to be equal.¹⁸ Accounts in which analysts apply a bear-specific template, or in which incentives distort the downside narrative in particular, give no reason for base-case and bull-case attribution to be equally sluggish. The restriction does not separate memory from generic narrative inertia, such as text carried over from one report to the next, which would make attribution equally sluggish in every scenario. The realized evidence offers only an informal check: base-case and bull-case attribution are as flat across market states as bear-case attribution (Panel B of Figure 5). Part (iii) gives the sign, so a MacroBearGap that stays positive across environments is a statement about relative magnitudes rather than a theorem. Part (iv) is the limit in which reported gaps carry no information about current conditions.

Together these say what the data can check. Rational conditions move with the aggregate environment; the memory state does not, or not at that speed. Over horizons across which rehearsal is stable, $dG/dz = (1 - \lambda) dG^*/dz + (\lambda + \theta) dD/dz$ collapses to its attenuated first term. Note what the model does not claim: memory is slow, not frozen, so the gap should track current conditions weakly rather than not at all, and the object worth measuring is the gap and a rational benchmark together.

Proposition 2 (Scenario asymmetry) Define $G_{i,t}^U \equiv \text{logit } A_{i,U,t} - \text{logit } A_{i,0,t}$, $G_{i,t}^{*U}$ analogously, and $D_{i,t}^U \equiv \text{logit } p_{i,U,t} - \text{logit } p_{i,0,t}$. Under Definitions 1–2,

$$G_{i,t}^U = (1 - \lambda)G_{i,t}^{*U} + \lambda D_{i,t}^U, \quad (8)$$

which carries no θ term, so $\mathcal{B}_{i,t}^U = \lambda(D_{i,t}^U - G_{i,t}^{*U})$. If rehearsal does not differentially favor macro forces among upside episodes, $D_{i,t}^U \approx 0$ and $\mathcal{B}_{i,t}^U \approx -\lambda G_{i,t}^{*U}$, which vanishes wherever the rational bull–base gap is small.

The asymmetry is a prediction, not a robustness check, and a demanding one. Every account in which analysts write formulaically, or in which macro language is a stylistic default, predicts the tilt whenever a non-modal scenario is described; the cue assumption predicts it on the downside

¹⁸The test requires an ex ante measure of the rational conditional $A_{i,k,t}^*$ for all three scenarios of each report, built from information available before the report date. The realized benchmark of Section 2 cannot serve this role: it is measured after the fact, and only for the scenario that materializes. Any ex ante measure is a noisy proxy for $A_{i,k,t}^*$, so the estimated slope is biased toward zero and understates $1 - \lambda$. The equality test remains valid when the proxy is equally noisy across the three scenarios.

only. Two cautions. Eq. (8) retains an anchoring term, so the prediction is that the bull-side *excess* is small, not that bull attribution is rational. And the prediction is joint with $D^U \approx 0$, itself an empirical claim about memory: exceptional upside episodes are firm-specific in a way that crashes are not.

3.4 Calibration and Discussion

Quantitative illustration. To connect the model to the magnitude of Macro Bear Bias, it is useful to rewrite Eq. (7) as

$$G = G^* + \lambda(D - G^*) + \theta D. \quad (9)$$

The three terms have a direct interpretation. G^* is the bear-base gap warranted by current firm and market information. The anchoring term $\lambda(D - G^*)$ pulls the current assessment toward the remembered composition of severe declines. The diagnostic-tilt term θD further overweights the source that is representative of the bear case.

We pick illustrative values for memory and match the size of the observed gap. Suppose severe declines are remembered as 40% macroeconomic and ordinary outcomes as 25% macroeconomic. This implies $D = \text{logit}(0.40) - \text{logit}(0.25) = 0.69$. In the return-matched sample the observed narrative gap is about 0.39 in log odds (23.3 against 17.0 percent; Table 2). Treating the rational gap as approximately zero, Eq. (7) implies $\lambda + \theta \approx \frac{0.39}{0.69} = 0.57$.

Thus, the observed magnitude of Macro Bear Bias can be generated by a moderate combined weight on memory and representativeness. The field moments identify the sum $\lambda + \theta$, not its decomposition. To visualize the two mechanisms separately, we use the illustrative split $\lambda = 0.45$ and $\theta = 0.12$, which preserves $\lambda + \theta = 0.57$. The particular split is therefore not estimated; it is used only to illustrate how the two channels combine.¹⁹

Figure 8 shows why the reported gap can remain large as current conditions improve. As G^* declines, the anchoring distortion becomes larger because the distance between current evidence and the remembered crash template widens. Representativeness contributes an additional bear-

¹⁹Measured instead at the market state where the realized bear-base R^2 gap is zero (Figure 5), the narrative gap is 0.34 in log odds (21.5 against 16.4 percent), which implies $\lambda + \theta \approx 0.49$. The illustration keeps 0.57.

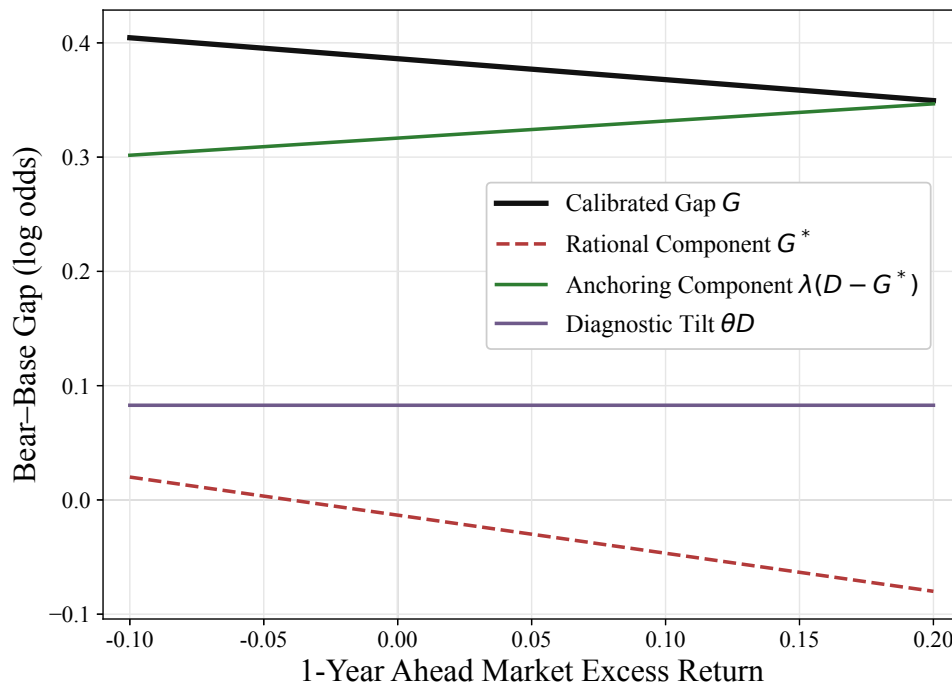


Figure 8 Decomposition of the Model-Implied MacroBearGap.

Note: This figure illustrates Eq. (9). The rational component G^* captures the bear-base gap warranted by current information. The anchoring component $\lambda(D - G^*)$ pulls attribution toward the remembered composition of severe declines, while the diagnostic-tilt component θD captures the additional bear-specific distortion from representativeness. The illustration uses $D = 0.69$, $\lambda = 0.45$, and $\theta = 0.12$. The data discipline $\lambda + \theta = 0.57$; the split between λ and θ is illustrative.

specific macro tilt. Together, the two forces offset much of the decline in the rational benchmark: with $\lambda = 0.45$, the reported gap moves only $1 - \lambda = 0.55$ as much as G^* .

The more relevant question is whether the same calibration reproduces the central empirical pattern. Figure 9 compares the model-implied MacroBearGap with the empirical bear-base macro-attribution gap and the realized R^2 gap across market states.

The exercise highlights the model's central mechanism. A rational attribution gap inherits the state dependence of current systematic risk, while memory moves more slowly. Anchoring therefore pulls attribution toward a slow-moving historical template, and representativeness further tilts the bear case toward macro explanations. The resulting gap can remain positive and nearly state-invariant even when the current conditional benchmark is close to zero or negative. This exercise does not separately identify λ and θ ; Section 4 tests the restrictions the field can reach, none of which separates the two channels.

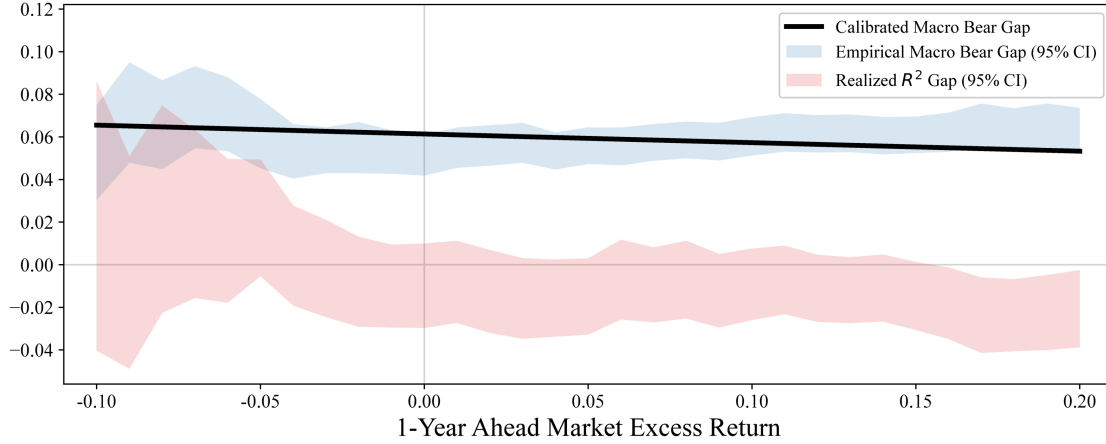


Figure 9 Model-Implied and Empirical MacroBearGap.

Note: The black line plots the model-implied bear-base macro-attribution gap in share space under the illustrative calibration. The blue band reports the empirical bear-base macro-attribution gap with its 95% confidence interval, and the red band reports the corresponding realized CAPM R^2 gap with its 95% confidence interval. The realized R^2 gap is an ex post proxy for the rational conditional benchmark rather than G^* itself. The model reproduces the central pattern: the narrative gap remains positive and nearly flat as the realized systematic gap falls toward zero and becomes negative in stronger market states.

Relation to other accounts. The nearest rival keeps the tilt but evaluates it under current conditions rather than memory; it amplifies the rational gap without breaking its state dependence, and Section 6.2 shows why the data reject it. The distortion here operates on a different object than in [Bordalo et al. \(2019\)](#) and [Bordalo et al. \(2024\)](#): diagnostic expectations distort the distribution of future outcomes, this model the attribution of a conditioning outcome to its causes. The mapping to BCGS is exact rather than analogical—their group is our scenario event, their comparison group our base event, their types our sources, their representativeness ratio our e^D —and what the setting adds is that the comparison group is reported rather than assumed. Rational attention allocation, by contrast, predicts that macro attention concentrates when aggregate states are volatile ([Kacperczyk, Van Nieuwerburgh, and Veldkamp, 2016](#)); here the slope on current information is $1 - \lambda$ and memory enters through λ and θ instead.

In short, Macro Bear Bias is what happens when a true fact about tail risk is stored in memory, exaggerated on recall, and then substituted for a calculation the analyst does not perform.

4 What Moves the Attribution Gap

Section 3 states two results and defers the rest to the sections that take them to the data. This section takes to the analyst data every prediction the field can bear, and it is organized by which object in the decomposition

$$G_{i,t} = (1 - \lambda)G_{i,t}^* + (\lambda + \theta)D_{i,t}$$

each test moves. Section 4.1 varies the cue rather than any parameter, and shows that the tilt fires on downside framing alone. Section 4.2 varies the objective diagnosticity D^0 and asks whether the distortion scales with the truth it exaggerates, across firms and over time. Sections 4.3 and 4.4 then turn to the rehearsed measure D itself: what memory contains, and what happens to the reported gap when that memory moves.

What the field cannot do is separate the anchoring parameter λ from the diagnosticity parameter θ where memory moves. There the field identifies only the combined coefficient $\lambda + \theta$, and we flag this where it binds. Table A.1 maps the results to the sections that take them up.

4.1 Scenario Asymmetry

Proposition 2 is the model's most demanding field prediction, because it follows from the cue structure rather than from the size of any parameter. The diagnostic tilt is triggered by being asked to imagine a severe decline; being asked for the modal or the favorable case invites no such comparison. The bull–base gap therefore carries no θ term, and its bias component reduces to $\mathcal{B}_{i,t}^U = \lambda(D_{i,t}^U - G_{i,t}^{*U})$.

The comparison is already visible in the evidence of Section 2, where it appeared as part of the argument that the pattern is not a crash forecast; the model promotes it to a test. Panel B of Figure 5 shows bull-case macro attribution running close to base-case attribution across market states, and the bull–base gap in Table 1 is -0.40 percentage points against a bear–base gap of 6.47 —an order of magnitude smaller, and of the opposite sign. Every account in which analysts write formulaically, or in which macro language is a stylistic default for any non-modal scenario, predicts a tilt on both sides. The data show it on one.

Two qualifications belong with the result. The bull–base gap retains an anchoring term, so the model does not require the bull-side excess over its own rational benchmark to vanish everywhere. With $D_{i,t}^U \approx 0$, Proposition 2 gives $\mathcal{B}_{i,t}^U \approx -\lambda G_{i,t}^{*U}$: the bull–base attribution gap should stay flat near zero, and its excess should move opposite to the rational bull–base gap, vanishing where that gap is small. The bear side differs in kind. It carries the tilt, so its excess should be positive in every state. The prediction is also joint with $D_{i,t}^U \approx 0$, which is an empirical claim about memory rather than an assumption of convenience: exceptional upside episodes—a successful drug, a platform breakout, a takeover—are firm-specific in a way that crashes are not, so rehearsal has no macro tail to oversample on the upside.

Figure 10 applies the construction of Figure 6 to the bull side, and the evidence is consistent with both parts of the prediction. The bull–base attribution gap stays within one percentage point of zero in every market-state window, between -0.7 and 1.0 points, while the bull–base gap in realized R^2 ranges from -3.2 to 1.3 points (Panel A). Bull-side attribution does not follow the realized gap, as anchoring predicts. The bull-side excess is small: it lies between -1.3 and 2.6 percentage points, and its 95% confidence interval covers zero in 26 of the 31 windows. The five exceptions sit at moderate market returns, where the realized bull–base gap is most negative—the pattern $-\lambda G_{i,t}^{*U}$ implies. The bear-side excess lies between 3.0 and 8.9 points and is positive in 29 of the 31 windows (Panel B). The tilt shows up on the downside alone, suggesting that the pattern is specific to downside framing rather than a general feature of scenario writing.

4.2 Diagnosticity Scaling

Section 2.6 established the first half of the model’s kernel-of-truth claim directly from realized outcomes: pooled across market states, macro forces really are more diagnostic of severe declines than of ordinary ones, so $D^0 > 0$. This subsection takes up the second half. Assumption 3 says rehearsal does not merely add a constant to that regularity but scales with it, and Definition 3 then implies that the bias component $\mathcal{B}_{i,t}$ should be widest exactly where the underlying regularity is strongest. The two subsections are a matched pair: one verifies that the template has a basis, the other asks whether the distortion tracks that basis.

The prediction has two forms. Across firms, objective diagnosticity varies with how systematic a firm’s downside risk genuinely is, so the gap should be wider for cyclical and financials than for

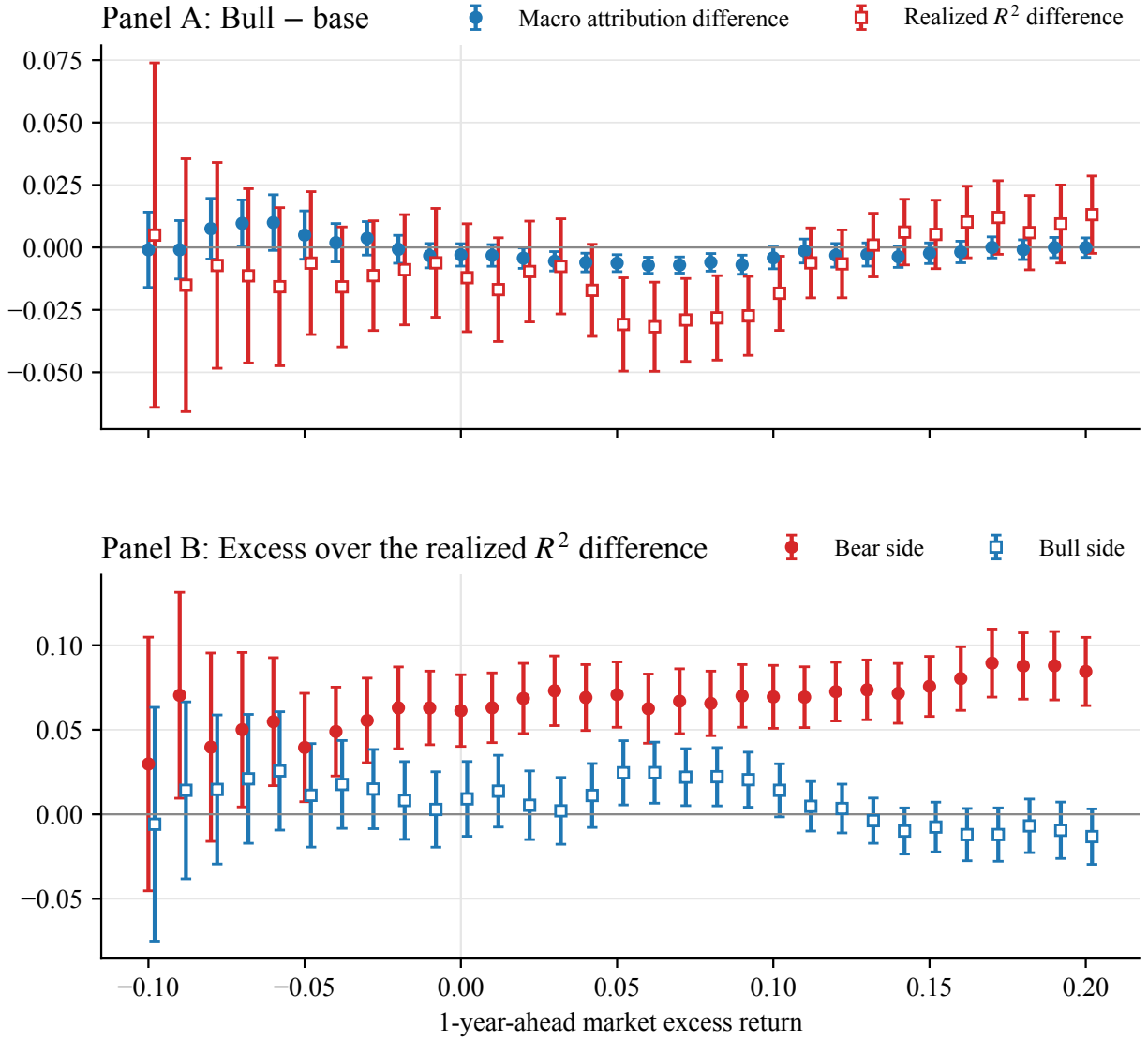


Figure 10 Bull – Base Gaps and the Excess over Realized R^2 , Conditional on Market States.

Note: Panel A applies the construction of Figure 6 to the bull side: firm-level bull–base differences in macro attribution (divided by 100) and in realized CAPM R^2 . Panel B plots, firm by firm, the excess of the attribution difference over the realized R^2 difference, for the bear side (bear minus base) and the bull side (bull minus base). For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release, and apply the same-firms rule within each bin. Error bars show 95% confidence intervals from constant-only regressions.

firms whose tail risk is a failed trial or a court ruling. Over time, it varies with the record of crises an analyst has to draw on, so the gap should widen when that record makes severe declines look more macro-driven.

Proposition 3 (Diagnosticity scaling) *Compare firms, sectors, or dates whose available record*

differs in objective diagnosticity D^0 . Under Assumption 3,

$$\frac{\partial \mathcal{B}_{i,t}}{\partial D^0} = (\lambda + \theta) \frac{\partial D_{i,t}}{\partial D^0} - \lambda \frac{\partial G_{i,t}^*}{\partial D^0},$$

which, because $\partial D_{i,t}/\partial D^0 \geq 1$ under Assumption 3, is strictly positive whenever $\lambda \partial G_{i,t}^*/\partial D^0 < \lambda + \theta$. If in addition $\partial G_{i,t}^*/\partial D^0 \geq 0$, the observed gap is also increasing in objective diagnosticity.

Where macro forces genuinely are more diagnostic of a firm's severe declines—cyclicals, financials, firms with high downside betas—the reported gap should be wider, and so should its excess over the rational benchmark. Where downside risk is genuinely idiosyncratic—a single-asset biotech, a firm whose tail risk is litigation or a binary regulatory event—both should be narrower. This would be the sharpest available separation from a template account: a fixed narrative template predicts a uniform macro tilt with no cross-sectional gradient in objective diagnosticity, and an incentive account predicts a gradient in visibility or coverage intensity, not in the comovement structure of a firm's downside.

Testing the proposition requires firms that differ in objective diagnosticity, and Section 2.6 suggests why such variation is hard to find. The regularity behind $D^0 > 0$ is mainly a matter of timing: severe declines cluster in years when the market falls, and those years are shared across firms. Proxies for cross-firm differences in objective diagnosticity built from pre-report information—the downside-beta spread, and the realized bear-base gap among a firm's or sector's earlier reports—do not separate firms by the realized diagnosticity of their subsequent outcomes, so portfolio sorts on them cannot test the proposition.²⁰

The time dimension carries the variation, and Figure 11 shows how the attribution gap responds to it. For each report year, the figure plots mean MacroBearGap and the within-firm bear-base difference in realized R^2 among that year's reports. The realized difference swings with the market. It is large in 2007 and in 2019–2022 and near zero in most other years, with a standard deviation across years of 2.32 percentage points. MacroBearGap barely moves: its standard deviation across years is 0.76 percentage points, a third as large, and it is uncorrelated with the realized difference (slope -0.002 , $t = -0.03$). Because firms within a year share most of their market environment, the realized difference here is close to the conditional benchmark. Simply put, the truth moves

²⁰Appendix D.1 reports five such sorts, with their first stages, and a time-series version of the test.

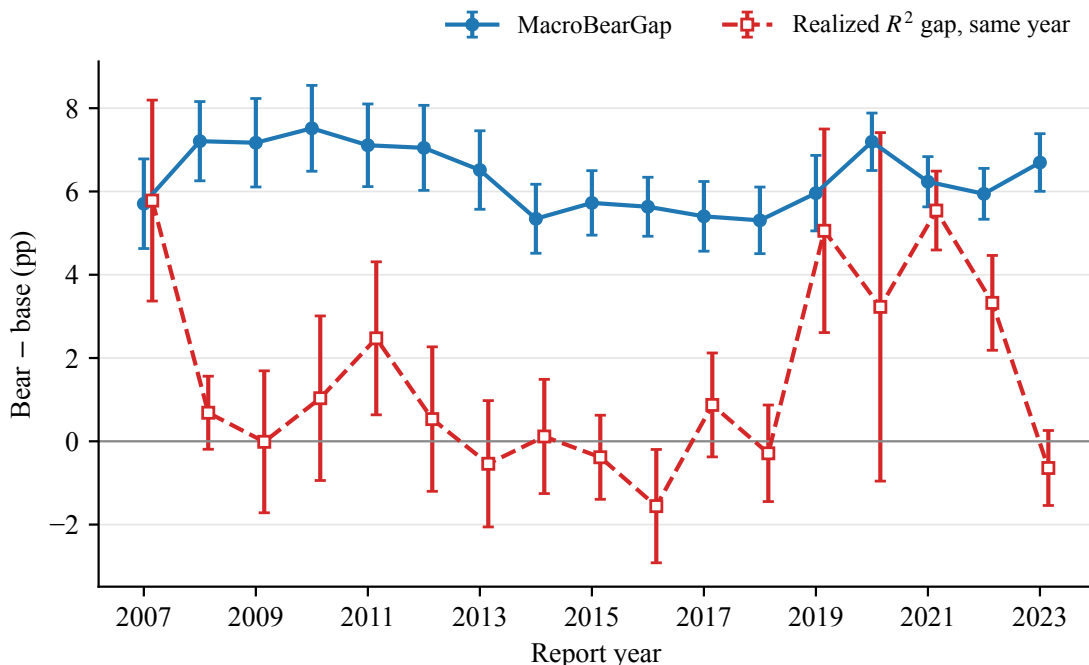


Figure 11 MacroBearGap and the Realized Bear–Base R^2 Difference by Report Year.

Note: For each report year, this figure plots mean MacroBearGap (bear minus base macro attribution) and the within-firm difference in realized CAPM R^2 , over the 252 trading days after the report, between realized-bear and realized-base reports dated in that year (firm fixed effects). Both are in percentage points. Error bars show 95% confidence intervals from standard errors clustered by firm. Unfiltered return-matched sample, 2007–2023.

from year to year, and the attribution gap does not—the central fact of Section 2, seen in a new dimension.

4.3 Bear-Case Macro Topics

The exemplar result stated in Section 4.4 (Proposition 4) concerns the rehearsed macro share $p_{i,B,t}$: what moves the reported gap is which episodes are available when the analyst imagines a severe decline. That object is not directly observed, but its content leaves a trace in the narratives themselves. This subsection reads that trace, asking which macroeconomic topics actually populate bear cases and whether their weight tracks current news or persists after the originating events recede. The answer characterizes the memory the next subsection then perturbs.

To identify topic-level narrative content, we apply a second LLM extraction step that classifies each macro-risk mention extracted from the bear-case narrative into one of 52 macro topic categories drawn from Zhou (2025), excluding 7 corporate finance topics from their original

Table 3 Bear Analyst Attention and WSJ Attention Across Key Topics

Note: This table reports macro topics in analysts’ bear-case narratives and corresponding attention to the same topics in Wall Street Journal articles. Bear analyst attention for a given topic is measured as the share of topic mentions in the bear-case narrative associated with this topic across analyst reports. WSJ attention for a given topic is measured as the average monthly share of Wall Street Journal articles associated with that topic over the sample period. Panel A shows the top five topics ranked by average monthly bear analyst attention. Panel B shows the top five topics with the highest correlation between monthly bear analyst attention and monthly WSJ attention. Bear Rank and WSJ Rank report the rankings of each topic by average attention within analyst bear narratives and the Wall Street Journal, respectively.

Panel A: Top Five Topics by Average Monthly Bear Analyst Attention					
Topic	Bear Avg.	Bear Rank	WSJ Avg.	WSJ Rank	Corr.
Economic Growth and Recovery Outlook	10.66%	1	2.36%	12	0.49
COVID-19 Pandemic and Vaccine Developments	5.36%	2	5.34%	2	0.76
Market Activity and Financial Performance	4.75%	3	5.44%	1	-0.01
Consumer Price Index and Inflation Trends	2.40%	4	0.86%	27	0.36
International Trade Relations and Economic Policies	2.31%	5	1.60%	16	0.32

Panel B: Top Five Topics by Correlation Between Bear Analyst Attention and WSJ Attention					
Topic	Bear Avg.	Bear Rank	WSJ Avg.	WSJ Rank	Corr.
COVID-19 Pandemic and Vaccine Developments	5.36%	2	5.34%	2	0.76
Economic Growth and Recovery Outlook	10.66%	1	2.36%	12	0.49
Consumer Behavior Amid Economic Strains	1.91%	7	0.24%	49	0.38
Geopolitical Tensions and Economic Impacts	0.46%	20	4.40%	3	0.38
Economic Stimulus and Government Interventions	0.40%	25	0.46%	40	0.37

59-topic taxonomy. A single bear-case narrative may therefore contribute multiple topic labels, one per mention. Appendix C.2 describes the prompt and Table A.2 reports each topic’s description and summary statistics. Following Zhou (2025), bear-case analyst attention to a topic in a given month is the share of mentions associated with that topic across all bear-case narratives released in that month. We benchmark this against Wall Street Journal (WSJ) coverage. WSJ attention to a topic is the monthly share of WSJ articles tagged with that topic. The two series therefore use different denominators—mention shares on the analyst side and article shares on the WSJ side—so the comparison in Figure 12 and Table 3 is best read as comovement in standardized time-series variation rather than a level-for-level match.

Table 3 reports average topic-level attention and time-series co-movement with the WSJ. Panel A ranks topics by average monthly bear-case analyst attention. *Economic Growth and Recovery Outlook* is the dominant bear-narrative topic, accounting for 10.66% of bear-case attention on average—more than four times its WSJ share of 2.36% (WSJ rank: 12). This disproportion suggests that macroeconomic growth narratives function as a canonical bear-case template: analysts invoke them persistently, at far higher rates than contemporaneous news coverage would predict. *COVID-19 Pandemic and Vaccine Developments* ranks second at 5.36%. *Market Activity and Financial Performance* ranks third (4.75%), despite being the single most-covered topic in the WSJ (rank 1, 5.44%), suggesting analysts allocate less narrative emphasis to general market conditions than the news cycle does. Inflation (*Consumer Price Index and Inflation Trends*, rank 4) and trade policy (*International Trade Relations and Economic Policies*, rank 5) round out the top five, each receiving materially more analyst attention than their WSJ ranks imply.

The reverse asymmetry is equally informative. The previous paragraph shows that analysts devote unusually high bear-case attention to recession-and-growth topics even when those topics are not among the most prominent in the WSJ. The opposite pattern appears for geopolitics: *Geopolitical Tensions and Economic Impacts* is the third most-covered WSJ topic, with a 4.40% share, but it ranks only 20th in bear-case analyst attention, with a 0.46% share. Thus, analysts are not simply importing the most visible macro topics from the news into their bear narratives. If availability operated through generic media exposure, geopolitical risk should be much more prominent in bear cases. Instead, analysts selectively emphasize a narrower set of recession-and-growth topics while largely excluding geopolitical risk despite its media salience. This pattern suggests that the relevant form of availability is not “whatever is currently in the news,” but a familiar macro-crisis template for explaining downside risk. Put differently, analysts appear to reach for the kind of macro story that resembles past market-wide crises. Section 4.4 tests this interpretation directly: MacroBearGap rose more after the GFC in high-systematic-risk sectors, and it is wider among analysts with Morgan Stanley-specific GFC experience, although that estimate is marginal ($t = 1.65$).

Panel B ranks topics by the time-series correlation between monthly bear-case analyst attention and WSJ attention. COVID-19 leads (0.76), followed by economic growth (0.49). High correlations indicate that analysts respond to the news environment in their narrative topic choices. The critical

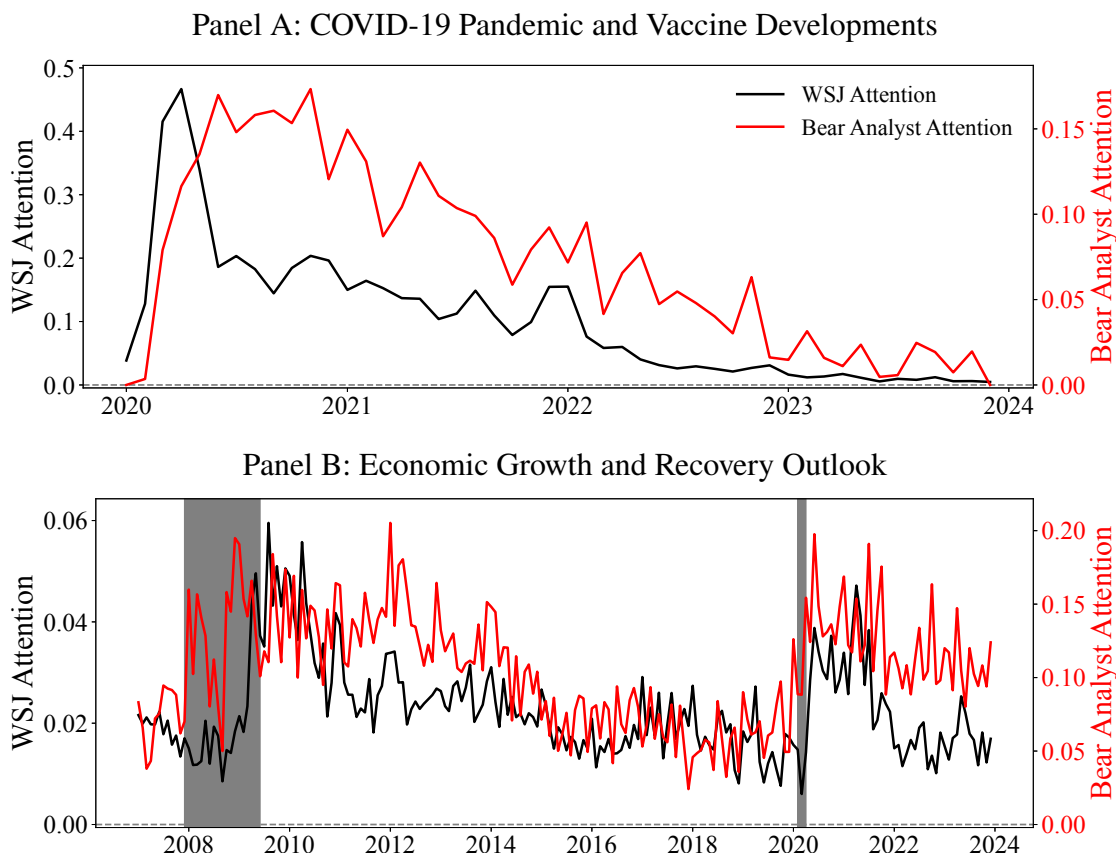


Figure 12 Bear Analyst Attention and WSJ Attention Over Time for Selected Topics.

Note: This figure plots monthly bear analyst attention (red, right axis) alongside monthly Wall Street Journal (WSJ) attention (black, left axis) for two representative macro topics. Bear analyst attention for a given month is the share of bear-case narrative mentions across all reports in that month associated with the topic. WSJ attention is the monthly share of WSJ articles associated with the topic. Panel A covers *COVID-19 Pandemic and Vaccine Developments*. Panel B covers *Economic Growth and Recovery Outlook*; gray bands indicate the Global Financial Crisis recession period (2008-2009) and the COVID-19 recession period (2020).

question, however, is whether that response is symmetric: do analysts reduce topic emphasis as quickly as they raise it, or do bear-case templates persist long after the originating events have receded?

Figure 12 examines this asymmetry directly. Each panel overlays monthly bear-case analyst attention (red, right axis) against WSJ attention (black, left axis) for the two highest-correlation topics. Panel A, covering COVID-19, shows a sharp joint spike in early 2020. WSJ attention subsequently declines steeply, reaching near zero by 2023; bear-case analyst attention remains materially elevated for roughly two additional years. Panel B, covering economic growth, replicates this pattern across both the Global Financial Crisis and the COVID-19 period: analyst attention

risers alongside WSJ coverage during each crisis but reverts far more slowly, remaining elevated long after media attention has subsided. This asymmetric response is consistent with template-based updating: salient crises make particular macro explanations easy to recall, and analysts keep deploying them as default bear-case narratives long after the originating shock has passed. Bear narratives therefore do not simply mirror contemporaneous news salience; salient macro episodes leave durable templates that continue to shape how analysts describe downside risk. A formal test supports the visual impression, though only marginally. Across the 52 macro topics, a one-percentage-point increase in WSJ coverage raises bear-case attention to the topic by 0.215 percentage points over the following six months ($t = 1.87$), while a one-point decrease lowers it by only 0.062 points ($t = 1.38$); the difference is 0.153 points ($t = 1.81$).²¹

4.4 Exemplar Rehearsal and Crisis Experience

The comparative static below moves memory itself, holding the world fixed. We test it with variation the field supplies.

Proposition 4 (Exemplar sensitivity) *Fix h , hold rational attribution fixed, and suppose $\lambda + \theta > 0$. Increasing the rehearsal weights on a set of episodes $S \subseteq E^B(h)$ with $\mu^0(S) > 0$ and $c_m(e) > p_{i,B,t}(h)$ for every $e \in S$ strictly raises $p_{i,B,t}(h)$, $D_{i,t}(h)$, and $G_{i,t}(h)$, provided Assumption 1 is preserved. Doing so on a set on which $c_m(e) < p_{i,B,t}(h)$ strictly lowers them.*

Rehearsal alone moves the reported gap while the world stands still. Make a macro-heavy crisis more available and bear-case macro attribution rises; make an idiosyncratic disaster more available and it falls. This is the model’s account of why the gap widens after the Global Financial Crisis.

Because rehearsal weights are personal as well as slow, the same comparative static has a cross-analyst counterpart requiring no manipulation at all.

Corollary 1 (Experience heterogeneity) *Let analysts a and a' cover the same firm at the same date, and suppose $w_{i,t}^a$ places greater weight than $w_{i,t}^{a'}$ on macro-driven episodes in $E^B(h)$, with equal weight elsewhere. Then $D_{i,t}^a(h) > D_{i,t}^{a'}(h)$ and, by Eq. (7), $G_{i,t}^a(h) > G_{i,t}^{a'}(h)$.*

²¹We regress the monthly change in bear-case attention to a topic on the current and five lagged monthly changes in its WSJ attention, split into increases and decreases, with month and topic fixed effects, and sum the six coefficients on each side. Standard errors are clustered by topic. The sample covers 52 topics from 2007 to 2023, 9,880 topic-months with the six lags available; the levels regression in Table 4 uses a different sample.

Analysts whose careers span more macro-driven downside episodes should write wider gaps than colleagues covering the same firm on the same date. In the ideal test, firm and date are held fixed, so the comparison survives the fixed effects that absorb any explanation operating at the firm or market level. It connects the model to evidence on experience effects in beliefs (Malmendier and Nagel, 2011), and it is the analogue of the finding in Bordalo, Gennaioli, and Shleifer (2020) that the strength of a cue depends on what it competes with in a given person's database.

Neither Proposition 4 nor Corollary 1 can be tested by manipulation here, so we use the two sources of variation the field does supply: aggregate shocks to which crises are rehearsed (topic news coverage and the GFC by sector), and cross-analyst differences in crisis experience (the GFC at Morgan Stanley). Morgan Stanley rarely assigns two analysts to the same firm at the same time: only 0.3 percent of firm-quarters in the analyst-matched sample have two, while 44 percent of firms are covered by different analysts over the sample. The experience test therefore compares a firm's analysts across years, with firm and year fixed effects, rather than two analysts covering the same firm on the same date.

The psychological basis for the rehearsal weights is the availability heuristic. The availability heuristic (Tversky and Kahneman, 1973, 1974) holds that people assess the probability of an event by the ease with which instances come to mind: events that are vivid, widely shared, or recently experienced are recalled more readily, inflating their judged likelihood. Our setting brings these mechanisms into a high-stakes professional context.

Crisis rehearsal in the field. Three tests take Proposition 4 and Corollary 1 to the data. Two vary which crises are rehearsed: news coverage that makes a macro topic salient (Table 4), and the Global Financial Crisis, whose contagion template applied most directly to some sectors (Table 5, column 1). The third varies who is writing: analysts who lived through the crisis at Morgan Stanley (columns 2–3). Column 4, a check on peer comovement, is discussed in Appendix D.2.

Topic salience. News coverage is a rehearsal cue. A burst of coverage of a macro topic raises the rehearsal weight of the episodes it recalls, so Proposition 4 predicts that analysts draw on the topic more readily when they construct bear cases.

We test this prediction at the topic-month level by regressing bear-case analyst attention to

topic k at time t on the WSJ topic shocks, its interaction with WSJ sentiment, controlling for WSJ sentiment and three lags of bear-case attention:

$$\begin{aligned} \text{BearAttention}_{t,k} = & \beta_1 \hat{\varepsilon}_{t-\tau,k}^{\text{WSJAttention}} + \beta_2 \hat{\varepsilon}_{t-\tau,k}^{\text{WSJAttention}} \times \text{WSJSentiment}_{t-\tau,k} \\ & + \beta_3 \text{WSJSentiment}_{t-\tau,k} + \sum_{j=1}^3 \rho_j \text{BearAttention}_{t-j,k} + \alpha_t + u_{t,k}, \end{aligned} \quad (10)$$

where $\hat{\varepsilon}_{t-\tau,k}^{\text{WSJAttention}}$ denotes the autoregressive (AR) residual of WSJ attention to topic k at lag τ , capturing the unexpected component of topic attention.²² $\text{WSJSentiment}_{t-\tau,k}$ is the average WSJ news sentiment for topic k at lag τ , and α_t is the time fixed effect. The three lags of $\text{BearAttention}_{t-j,k}$ control for persistence in analysts' own topic attention, and $\hat{\varepsilon}_{t-\tau,k}^{\text{WSJAttention}}$ isolates only the transient, unexpected increase in attention. The specification nests two predictions: a level effect ($\beta_1 > 0$), whereby coverage surprises raise analyst bear-case attention, and an asymmetry effect ($\beta_2 < 0$), whereby negative-sentiment coverage amplifies this response.

Table 4 reports results for $\tau \in \{1, 2, 3, 4\}$; Appendix Table A.4 reports the same regressions for bull-case attention. The level effect is positive and marginally significant at short lags: a one-percentage-point attention shock raises bear-case attention to the topic by 0.064 percentage points at $\tau = 1$ ($t = 1.85$) and 0.052 points at $\tau = 2$ ($t = 1.72$), declining to an insignificant 0.042 and 0.021 at $\tau = 3$ and $\tau = 4$. The interaction coefficient is negative, significant at $\tau = 2$ and 3 and marginally so at $\tau = 1$: $\hat{\beta}_2 = -0.188, -0.237$, and -0.186 at $\tau = 1, 2$, and 3 (t -statistics $-1.81, -2.43, -2.10$), attenuating to an insignificant -0.038 at $\tau = 4$. Baseline sentiment alone is economically negligible and statistically insignificant across all columns, indicating that the direction of coverage matters only in amplifying surprises, not independently.

These results are consistent with coverage acting as a rehearsal cue. Unexpected increases in macro news coverage shift bear-case analyst attention toward the newly salient topic, and this shift is amplified when coverage is negative—precisely the threatening, crisis-type signal identified by the availability literature as most cognitively accessible (Tversky and Kahneman, 1973). The response to a coverage surprise is transient: the level effect fades by $\tau = 3$ and the interaction by $\tau = 4$. That is a statement about the news shock, not about the template. Section 4.3 suggests that attention

²²We use an AR(3) model for each topic, since the Akaike information criterion (AIC) selects AR(3) for 40 of the 52 topics.

Table 4 Bear-Case Macro Attention and Topic Salience

Note: This table reports panel regressions of bear-case analyst attention to topic k at time t on the autoregressive (AR) residual of WSJ news coverage (attention shock), its interaction with WSJ news sentiment, the sentiment level, and three lags of the dependent variable (analyst attention to the topic); see Equation (10). Columns correspond to lag $\tau \in \{1, 2, 3, 4\}$. All specifications include time fixed effects. Heteroskedasticity-robust (HC1) t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	$\tau = 1$	$\tau = 2$	$\tau = 3$	$\tau = 4$
WSJ Attention Shock $_{t-\tau,k}$	0.0642* (1.85)	0.0524* (1.72)	0.0417 (1.54)	0.0207 (1.50)
WSJ Attention Shock $_{t-\tau,k} \times$ WSJ Sentiment $_{t-\tau,k}$	-0.1876* (-1.81)	-0.2374** (-2.43)	-0.1855** (-2.10)	-0.0377 (-0.55)
WSJ Sentiment $_{t-\tau,k}$	0.0001 (0.72)	0.0000 (0.24)	0.0001 (0.56)	0.0001 (0.64)
WSJ Attention $_{t-1,k}$	0.3085*** (11.02)	0.2998*** (10.87)	0.3004*** (10.48)	0.3011*** (10.14)
WSJ Attention $_{t-2,k}$	0.2835*** (10.33)	0.2868*** (10.44)	0.2862*** (10.20)	0.2867*** (10.16)
WSJ Attention $_{t-3,k}$	0.3826*** (11.94)	0.3876*** (12.58)	0.3873*** (12.12)	0.3865*** (11.68)
Time FE	Yes	Yes	Yes	Yes
Observations	11,440	11,388	11,336	11,284
R^2	88.90%	88.96%	88.95%	88.93%

to a crisis topic stays elevated after coverage subsides, so a transient response to surprises and a persistent template are two parts of the same pattern.

Industry-crisis salience. Proposition 4 also predicts where a crisis should widen the gap most: among firms whose downside the crisis episode most resembles, because there the newly rehearsed episode is most relevant. The 2008 Global Financial Crisis was widely framed as a systematic-risk event centered on financial contagion, so the availability of macro-downside narratives should have increased most for firms in high-systematic-exposure sectors (e.g., financials) relative to defensive sectors such as utilities. We test this by interacting a post-GFC indicator with a high-systematic-industry dummy, which equals one for firms in the GICS financials, energy, and materials sectors and zero otherwise. The indicator is a sector category, not a measured similarity to the crisis: it groups financials, where the GFC was the sector's own crash, with energy and materials, whose

defining downturns are commodity collapses. The test therefore captures exposure to the systematic downturn the crisis made salient rather than resemblance to the GFC itself.²³

Column (1) of Table 5 reports the results. The interaction is economically and statistically significant: after the GFC, MacroBearGap increases by 4.9 percentage points more for firms in high-systematic sectors than for other firms ($t = 2.61$). Combining the main effect and the interaction, high-systematic sectors had on average -8.9 percentage-point *lower* MacroBearGap than other sectors before the GFC; post-GFC, the differential narrows by 4.9 points to about -4.0 percentage points before any common post-GFC effect is added. The crisis therefore did not flip high-systematic sectors to higher MacroBearGap; it narrowed a pre-existing negative gap, which is the within-firm pattern Proposition 4 predicts when the crisis raises the rehearsal weight of a macro-driven episode most for these sectors.²⁴

Analyst crisis experience. The crisis experience tests require an analyst-level identifier and therefore restrict the sample to the *analyst-matched subsample*. We match each Morgan Stanley report to IBES via the analyst’s name (cleaned of titles and suffixes) and the covered firm’s name within a 90-day window around the report date; a report enters the matched subsample if at least one IBES analyst record satisfies the join, and we retain the longest-tenured matching analyst when multiple candidates appear. The matched subsample contains 17,257 report observations, a 45% reduction from the full panel of 31,445 observations used in column (1).²⁵

Corollary 1 predicts that analysts whose careers include the GFC place more weight on macro-driven episodes and so write wider gaps. We construct two GFC-experience indicators: one for analysts with a record at Morgan Stanley before March 2009, and one for analysts with a record at *any* brokerage before that date, each proxied by the analyst’s earliest report date in the respective

²³We use the GICS classification rather than a continuous measure of systematic risk exposure (e.g., beta) because crisis templates are rehearsed as categories rather than along continuous dimensions. The GICS classification is widely used in financial media and industry practice, making it a natural proxy for the salient categories that drive narrative construction.

²⁴We also test whether the post-COVID increase in MacroBearGap is concentrated in COVID-sensitive industries (airlines, hotels, retail). It is not: the COVID interaction with a COVID-sensitive industry indicator is essentially zero (-0.06 , $t = -0.06$). This contrast suggests that COVID activated a universal “shutdown” narrative across all sectors, whereas the GFC activated a sector-specific “financial contagion” template.

²⁵The reduction is driven primarily by IBES coverage gaps (about 32%) and ambiguous name matches that the join rule discards conservatively (about 13%). Matched and unmatched reports differ on observable dimensions of analyst tenure and firm coverage: matched reports skew toward larger-cap, longer-tenured analyst coverage, which is exactly the segment in which the GFC-experience and IB-relationship tests have the most variation.

database. This pair of indicators distinguishes two channels: the MS-specific measure captures exposure to the firm’s crisis-era narrative conventions, while the profession-wide measure captures generic crisis memory.

Columns (2) and (3) of Table 5 report the results. Analysts who experienced the GFC at Morgan Stanley write gaps 1.2 points wider, an estimate significant only at the 10 percent level ($t = 1.65$); analysts with GFC experience at any brokerage show no discernible effect. The divergence is consistent with rehearsal through firm-specific narrative templates: the pattern is more consistent with exposure to a particular institution’s crisis-era framing conventions than with a generic crisis-memory proxy, though it does not fully separate institutional template exposure from selection into Morgan Stanley or differences in the matched analyst populations.²⁶

Bear-case attention thus responds to negative coverage surprises, MacroBearGap widened after the GFC in the sectors the crisis template fit most directly, and it is wider for analysts who lived through the crisis at Morgan Stanley, although that last estimate is marginal. The industry and experience results hold the firm and the year fixed.

In summary, the field evidence supports the parts of the model it can reach. The excess attribution sits on the downside alone (Figure 10); the gap does not follow the year-to-year variation in realized diagnosticity (Figure 11); and it moves when the set of rehearsed crises moves, holding the firm and the year fixed (Section 4.4). What the field cannot do is say how much of the movement is anchoring and how much is exaggeration, since Proposition 4 and Corollary 1 both operate through the combined coefficient $\lambda + \theta$. Section 5 asks whether the distortion matters for outcomes.

5 From Narrative to Forecast Errors

The preceding sections establish that Macro Bear Bias exists and that it behaves as a memory-based distortion should. They do not establish that it matters. A biased bear narrative would be a curiosity if it stayed inside the bear narrative, and nothing in Section 3 requires that it escape: the attribution

²⁶We also examine whether the MS-specific GFC effect reflects generic MS tenure rather than crisis-specific experience. In a horse-race regression including both the GFC-experience indicator and years of MS tenure, the GFC coefficient remains positive (+1.27) while the tenure coefficient is essentially zero (-0.02 , $t = -0.14$). The point estimate is directionally consistent but imprecisely estimated in the horse-race specification; we therefore read the standalone result as suggestive rather than definitive.

Table 5 Crisis Rehearsal: Determinants of MacroBearGap

Note: This table reports panel regressions of MacroBearGap (bear minus base macro attribution, 0–100 scale) on proxies for crisis rehearsal. Column (1) interacts a post-GFC indicator with a high-systematic-industry dummy (financials, energy, materials). Columns (2)–(3) test whether analysts who experienced the GFC at Morgan Stanley (col 2) or at any brokerage (col 3) exhibit larger MacroBearGap. Column (4) relates MacroBearGap to the leave-one-out sector–quarter average of MacroBearGap, a measure of sector comovement that may reflect herding or common shocks (Appendix D.2). Columns (2)–(3) are restricted to the analyst-matched subsample. All specifications include firm and year fixed effects. Standard errors are clustered by firm. *t*-statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	MacroBearGap			
	(1)	(2)	(3)	(4)
Post-GFC $\times$ High Systematic	4.9229*** (2.61)			
High Systematic	−8.9346*** (−4.30)			
GFC Experienced (MS)		1.1544* (1.65)		
GFC Experienced (Prof.)			0.1594 (0.25)	
LOO Sector MacroBearGap				0.0828* (1.75)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	31,445	17,257	17,257	30,979
Within R^2	0.0008	0.0007	0.0000	0.0010

model governs how analysts describe scenarios, not how they forecast. This section asks whether the distortion travels, and it does so by adding one assumption to the model rather than by deriving a further implication of it.

That assumption, Assumption 4 below, is that excess macro emphasis in the bear case depresses the base-case forecast in proportion to the firm’s systematic exposure. Corollary 2 then delivers the testable object: MacroBearGap should predict subsequent base-case forecast errors, and more strongly for high-beta firms.

One caution belongs at the outset, because it determines what the evidence below can and cannot show. The *existence* of a forecast-error relation is not evidence of bias. Set $\lambda = \theta = 0$, so that reported attribution is fully rational, and Assumption 4 still delivers a positive coefficient: an analyst who correctly perceives high systematic exposure and marks the base case down accordingly generates exactly the same correlation. What discriminates is the magnitude of the response relative

to the rational gap—whose realized proxy Section 2 shows to be near zero—together with the cross-sectional gradient in β_i , which undifferentiated pessimism does not predict and which our specification does not yet separately identify. The tests below are therefore tests of consequence, not of the model’s belief mechanism.

Throughout this section, $\text{MacroBearGap}_{i,t}$ denotes the bear–base macro-attribution gap in percentage points, as in Definition 3; $\text{MacroBullGap}_{i,t}$ and $\text{FirmBearGap}_{i,t}$ denote the corresponding bull–base and firm-source contrasts used in the specifications below.

5.1 Transmission and the Testable Prediction

Prices are anchored on the base case, and the base case is written by an analyst who has just talked themselves through a macro-driven collapse. Absent bias, the base-case forecast reflects expected market conditions,

$$R_{i,t+1}^{\text{base}} = \alpha_{i,t} + \beta_i \mu_t, \quad (11)$$

where $\mu_t = \mathbb{E}_t[M_{t+1}]$ is the analyst’s expectation of the market excess return and β_i the firm’s systematic exposure. We assume that an inflated bear–base gap depresses this forecast through the firm’s beta: a macro-heavy downside story raises the aggregate exposure the analyst perceives, and perceived aggregate exposure is priced at the rate β_i .

Assumption 4 (Base-case spillover) *Excess macro emphasis in the bear narrative shifts the base-case forecast downward in proportion to systematic exposure and the attribution gap:*

$$R_{i,t+1}^{\text{base}} = \alpha_{i,t} + \beta_i \mu_t - \phi \beta_i \text{MacroBearGap}_{i,t}, \quad \phi > 0, \quad (12)$$

with $\text{MacroBearGap}_{i,t}$ as in Definition 3.

Corollary 2 (Forecast-error predictability) *Let $\text{Forecast Error}_{i,t+1} = R_{i,t+1}^{\text{realized}} - R_{i,t+1}^{\text{base}}$. Under Assumption 4, $\text{Forecast Error}_{i,t+1} = \beta_i(M_{t+1} - \mu_t) + \phi \beta_i \text{MacroBearGap}_{i,t} + \varepsilon_{i,t+1}$, so that $\partial \text{Forecast Error}_{i,t+1} / \partial \text{MacroBearGap}_{i,t} = \phi \beta_i > 0$: a larger attribution gap predicts a more positive subsequent base-case forecast error, and more strongly so for firms with higher systematic exposure.*

Here returns are measured relative to the report-date price, $R_{i,t+1}^{\text{realized}} = P_{i,t+1}^{\text{realized}}/P_{i,t} - 1$ and $R_{i,t+1}^{\text{base}} = P_{i,t+1}^{\text{base}}/P_{i,t} - 1$, so a positive forecast error indicates that the analyst’s base-case forecast is too pessimistic relative to the subsequently realized return.

Assumption 4 is a reduced-form transmission assumption rather than an implication of the attribution model. Nothing rests on its linear form: the testable content is the sign of ϕ and the interaction with β_i . That interaction is what gives the assumption content beyond a claim that gloomy narratives accompany gloomy forecasts, since undifferentiated pessimism predicts no cross-sectional gradient in systematic exposure. It is also, together with the near-zero rational gap documented in Section 2, what separates the model from the rational benchmark that generates the same unconditional correlation.

Assumption 4 asserts the spillover; the discount-rate channel below supplies its microfoundation and explains why the spillover should scale with β_i rather than entering additively.

The primary channel operates through discount rates. A bear case described as predominantly macro-driven leads the analyst to perceive elevated systematic risk, implicitly applying a higher discount rate to base-case cash flows. Let the base-case price target be $P^{\text{base}} = \sum_{\tau} CF_{\tau}/(1 + r_{\tau})^{\tau}$, where $r_{\tau} = r_f + \tilde{\beta}_{i,t} \cdot \mu_t$. Macro Bear Bias inflates the analyst’s perceived beta *multiplicatively*, so that the distortion scales with the firm’s true systematic exposure:

$$\tilde{\beta}_{i,t} = \beta_{i,t} (1 + \psi \cdot \text{MacroBearGap}_{i,t}), \quad \psi > 0. \quad (13)$$

Under this specification, the incremental discount-rate wedge is $\beta_{i,t} \psi \cdot \text{MacroBearGap}_{i,t} \cdot \mu_t$: a high-beta firm is hit by a larger absolute wedge than a low-beta firm at the same MacroBearGap. The wedge raises the discount rate and depresses P^{base} even if point estimates of base-case cash flows are unchanged, and the scaling by $\beta_{i,t}$ generates the $\beta_i \times \text{MacroBearGap}_{i,t}$ interaction tested in the empirical specifications below.²⁷ This microfoundation maps directly onto the reduced-form

²⁷A natural refinement would decompose $\beta_{i,t}$ into cash-flow and discount-rate components (Campbell and Vuolteenaho, 2004) and ask which component carries the forecast-error response: our spillover specification operates through the discount rate, while the content of bear narratives—recessions, contagion, regime shifts—describes persistent cash-flow shocks. We do not pursue this decomposition because the news terms it requires are recovered from a VAR whose output is sensitive to the choice of state variables (Chen and Zhao, 2009; Engsted, Pedersen, and Tanggaard, 2012), and subsequent work has favored direct measurement of cash-flow fundamentals over VAR-implied news (Lochstoer and Tetlock, 2020). Our reduced-form specification in Eq. (12) is agnostic about whether the perceived-risk distortion loads on cash-flow or discount-rate news: the sign prediction of Corollary 2 holds under either channel, so the forecast-error tests do not depend on resolving this decomposition.

spillover parameter ϕ of Assumption 4 and Eq. (12): ϕ captures the product of ψ and the sensitivity of the price target to the discount rate.^{28,29}

A second, complementary route runs through narrative substitution rather than the discount rate: an analyst whose base case already embeds firm-specific pessimism has less need of a macro bear narrative, so base-case position relative to consensus should covary with the macro content of the bear case. Section 6.3.2 shows that base-case position relative to consensus predicts the macro content of the bear case, symmetrically above and below consensus, and discusses what that result can and cannot separate.

The flat gap documented in Section 2.5 also has state-dependent consequences, which is why a linear specification understates the predictive content of Macro Bear Bias: in strong markets the same macro-heavy bear narrative diverges more sharply from realized conditions and generates larger forecast errors, while in weak markets the template is closer to correct. Section 5.3 takes this up.

We now take Corollary 2 to the data. Its forecast-error decomposition motivates a panel regression of forecast errors on MacroBearGap with firm fixed effects and a market-state control. Because α_i cancels in the forecast error, the role of firm fixed effects is to absorb any residual persistent firm-level forecast-error component (e.g., time-invariant analyst optimism on a given firm) that is not modeled in the framework but could correlate with MacroBearGap. Because β_i enters the slope multiplicatively, firm fixed effects do not remove cross-sectional variation in β_i ; the pooled OLS slope on MacroBearGap instead recovers a variance-weighted average of the firm-specific products $\phi\beta_i$. The forward-looking market surprise $\beta_i(M_{t+1} - \mu_t)$ enters the residual, so identification requires that MacroBearGap does not predict future market returns. We examine

²⁸Some DCF implementations weight scenarios explicitly: $P^{\text{reported}} = w_{\text{bear}}P^{\text{bear}} + w_{\text{base}}P^{\text{base}} + w_{\text{bull}}P^{\text{bull}}$. If availability inflates w_{bear} or macro narratives compress P^{bear} , the composite target shifts down. This is a reduced-form version of the discount-rate channel.

²⁹A numerical example anchors the economic magnitude. At the sample-mean values $\hat{\beta}_{\text{CAPM}} = 1.13$ (Table 2), a six-percent equity risk premium, and a bias coefficient $\psi = 0.005$, a MacroBearGap at its return-matched sample mean of 6.3 percentage points (Table 2) raises the perceived beta from 1.13 to $1.13 \times (1 + 0.005 \times 6.3) = 1.166$. The incremental discount-rate wedge is therefore $\hat{\beta}_{\text{CAPM}} \psi \text{MacroBearGap} \cdot \mu = 1.13 \times 0.005 \times 6.3 \times 6\% \approx 21$ bp. At an effective valuation duration of ten years, modified duration is $10/(1+r) \approx 9.4$, so the implied compression of the base-case target price is $-9.4 \times 21 \text{ bp} \approx -2.0\%$. The same calculation under $\beta = 2.0$ (high-beta firm) yields a wedge of $2.0 \times 0.005 \times 6.3 \times 6\% \approx 38$ bp and a target-price compression of about -3.5% ; under $\beta = 0.5$ (low-beta firm) the wedge is about 9 bp and the compression is about -0.9% . The wedge therefore scales roughly four-to-one between extreme high- and low-beta firms in the sample, which delivers the cross-sectional content of the $\beta_i \times \text{MacroBearGap}_{i,t}$ interaction in Corollary 2: high-beta firms should display the largest forecast-error response to MacroBearGap.

Corollary 2 in two steps. Section 5.2 uses a linear panel regression; Section 5.3 uses a flexible nonlinear specification that allows the effect to vary across market states.

5.2 Empirical Evidence: Forecast Error Prediction and Portfolio Performance

Guided by the framework in Section 5.1, our central empirical prediction is that stronger Macro Bear Bias should be associated with more positive subsequent base-case forecast errors. Intuitively, when analysts place excessive macro emphasis on downside scenarios, they embed too much downside macro risk into the central forecast. As a result, the base-case forecast becomes too pessimistic relative to subsequently realized returns.

To test this prediction, we estimate the following specification:

$$\text{Forecast Error}_{i,t+1} = a + b_1 \hat{\beta}_{i,t} \times \text{MacroBearGap}_{i,t} + b_2 M_t + \mathbf{b}' X_{i,t} + FE_i + u_{i,t+1}, \quad (14)$$

where $\text{MacroBearGap}_{i,t}$ is the bear–base gap in macro attribution, $\hat{\beta}_{i,t}$ is the past 252-day CAPM beta, M_t is the past 252-day market excess return prior to report release, $X_{i,t}$ is a vector of time-varying firm characteristics including the stock’s own past 252-day return $R_{i,t}$ and CAPM beta $\hat{\beta}_{i,t}$ before report release, and γ_i denotes firm fixed effects. We include M_t to control for the market state prevailing when the analyst issues the report. The framework predicts $b_1 > 0$: firms with stronger Macro Bear Bias should exhibit larger positive subsequent forecast errors.

For the empirical analysis, we go beyond the bear–base macro-attribution gap and also examine the bull–base macro-attribution gap, as well as the corresponding gaps in firm-specific attribution as placebo tests:

$$\text{Forecast Error}_{i,t+1} = a + b_1 \hat{\beta}_{i,t} \times \text{AttributionGap}_{i,t} + b_2 M_t + b_3 \hat{\beta}_{i,t} + b_4 R_{i,t} + FE_i + u_{i,t+1}, \quad (15)$$

where $\text{AttributionGap}_{i,t}$ denotes the narrative attribution-gap variable. This design helps distinguish Macro Bear Bias from more general differences in narrative intensity across scenarios. If forecast errors are specifically driven by Macro Bear Bias, then the relevant predictor should be the bear–base gap in macro attribution, rather than scenario differences in upside narratives or in firm-level risk attribution.

The regression results are reported in Table 6. Across columns (1), (5), and (6), the coefficient on the bear–base macro-attribution gap is positive and statistically significant, with t -statistics of 2.5, 2.3, and 3.2, respectively. These results indicate that firms for which analysts place relatively greater macro emphasis in the bear case than in the base case subsequently exhibit larger base-case forecast errors. Under our definition of forecast error, this means that analysts set base-case forecasts that are too low for these firms. The evidence is therefore consistent with Macro Bear Bias distorting analysts’ base-case expectations in a systematically pessimistic direction.³⁰ Additionally, because the predictor enters multiplicatively as $\hat{\beta}_{i,t} \times \text{MacroBearGap}_{i,t}$, the positive and significant first coefficient is consistent with the cross-sectional content of Corollary 2. Because the gap enters only through the interaction, the specification does not separately identify a gradient in β_i .

By contrast, the bull–base macro-attribution gap is not statistically significant, which is consistent with the view that the bias is specific to downside macro narratives rather than to scenario differentiation more broadly. The firm-specific gap is insignificant on its own (column 3) but significant alongside the macro gap in column 6 ($t = 2.16$), with a coefficient close to the macro coefficient. Because the three attribution shares sum to one, the two bear–base gaps together mirror the industry gap, so column 6 does not isolate the macro component; the univariate evidence in columns 1–3 points to it, but the joint specification cannot rule out a role for the bear-case shift away from industry attribution.

The univariate-to-multivariate comparison reveals a suppression effect. The Macro Bear–Base coefficient rises from 0.0011 ($t = 2.5$) in column (1) to 0.0022 ($t = 3.2$) in column (6). Because the macro and firm bear–base gaps are mechanically negatively correlated—the compositional constraint $A^{\text{macro}} + A^{\text{firm}} + A^{\text{industry}} = 100$ means more macro bear-case attribution implies less firm-specific attribution—the univariate estimate is attenuated by the omitted Firm Bear–Base gap. Column (6) recovers the partial effect and is our preferred linear specification. In economic terms, a one-standard-deviation increase in the beta-weighted bear–base macro-attribution gap predicts a 2 to 4% increase in the base-case forecast error (based on the coefficients in Columns 1 and 6, respectively). This magnitude is economically meaningful given that the absolute mean forecast error in this sample is 5.5%.

³⁰We also conduct a robustness check in which we regress forecast errors directly on attribution gaps, rather than on the interaction between $\hat{\beta}_{i,t}$ and attribution gaps. The results, reported in Table A.3, are very similar to those in Table 6. This is expected because the vast majority of estimates satisfy $\hat{\beta}_{i,t} > 0$, as shown in Table 2.

Table 6 Forecast Error Regressions

Note: This table reports empirical results from the following forecast-error prediction regression (15):

$$\text{Forecast Error}_{i,t+1} = a + b_1 \hat{\beta}_{i,t} \times \text{AttributionGap}_{i,t} + b_2 M_t + b_3 \hat{\beta}_{i,t} + b_4 R_{i,t} + FE_i + u_{i,t+1},$$

where each narrative attribution-gap variable is interacted with stock i 's CAPM beta, $\hat{\beta}_{i,t}$, estimated using daily returns over the 252 trading days prior to the report release. M_t and $R_{i,t}$ denote the market excess return and stock i 's excess return over the same 252-day pre-release window. Columns (1)–(4) include each beta-weighted attribution gap individually. Column (5) includes the two macro beta-weighted gaps jointly. Column (6) includes all four beta-weighted attribution-gap variables simultaneously. Firm fixed effects are included in all specifications. Standard errors are clustered by firm and month. t-statistics are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
Attribution Gaps						
$\hat{\beta}_{i,t} \times (\text{Macro Bear} - \text{Base})$	0.0011** (2.49)				0.0011** (2.29)	0.0022*** (3.21)
$\hat{\beta}_{i,t} \times (\text{Macro Bull} - \text{Base})$		0.0005 (0.75)			0.0001 (0.16)	-0.0002 (-0.25)
$\hat{\beta}_{i,t} \times (\text{Firm Bear} - \text{Base})$			0.0006 (1.06)			0.0018** (2.16)
$\hat{\beta}_{i,t} \times (\text{Firm Bull} - \text{Base})$				-0.0001 (-0.16)		-0.0005 (-0.60)
Control Variables						
M_t	-0.1449 (-1.51)	-0.1457 (-1.52)	-0.1461 (-1.52)	-0.1456 (-1.51)	-0.1449 (-1.51)	-0.1452 (-1.52)
$\hat{\beta}_{i,t}$	-0.0392 (-0.68)	-0.0316 (-0.55)	-0.0280 (-0.49)	-0.0315 (-0.55)	-0.0390 (-0.68)	-0.0351 (-0.61)
$R_{i,t}$	-2.5075 (-0.15)	-2.4404 (-0.15)	-2.2543 (-0.13)	-2.3862 (-0.14)	-2.5196 (-0.15)	-2.2828 (-0.14)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	43,306	43,306	43,306	43,306	43,306	43,306
Within R^2	0.0005	0.0004	0.0004	0.0003	0.0005	0.0007

One may object that these estimates identify something other than pessimism: our framework treats $R_{i,t+1}^{\text{base}}$ as the analyst's subjective mean, but the base case may be read as the mode—the single most likely case. The modal reading, however, can only strengthen the interpretation of the results in Table 6. If analysts' subjective return distributions are left-skewed—as the typically greater distance of the bear target from the base target suggests—the mode lies above the mean, so a mode-based target is too high relative to the subjective mean and embeds a *negative* wedge in the forecast error defined in Corollary 2. This wedge pushes measured forecast errors downward—toward apparent optimism—and therefore cannot manufacture the positive forecast errors that

MacroBearGap predicts. The wedge could distort the slope estimates only through its covariance with MacroBearGap, and the plausible direction of that covariance again works against us: reports with macro-heavier bear cases set bear targets further below the base, so subjective left skew is largest where MacroBearGap is largest, attenuating the estimated ϕ in Corollary 2 toward zero. That the coefficients in Table 6 are positive and significant despite this attenuation is therefore evidence that a large MacroBearGap marks base-case targets that are too pessimistic. Even under the modal interpretation, the estimates are a lower bound on the bias.

We next examine whether correcting for Macro Bear Bias improves the out-of-sample cross-sectional ranking of stock returns. The key idea is that the analyst’s base-case forecast may contain useful firm-level information but may also be distorted by the predictable pessimism induced by macro-heavy bear narratives. Rather than replacing the analyst forecast with a purely statistical return forecast, we therefore estimate the component of the analyst’s forecast error that is predictable from Macro Bear Bias and use it to adjust the original base-case forecast. This exercise asks whether the narrative-based bias documented above contains return-relevant information beyond the analyst’s raw valuation signal.

To implement this test, we use the specification in Column (6) of Table 6, but exclude the time-varying firm characteristics $R_{i,t}$ and $\hat{\beta}_{i,t}$. This restriction ensures that out-of-sample predictability comes from the interaction between firms’ systematic exposure and the narrative attribution gaps, rather than from standalone sorts on past returns or beta. We estimate the following regression separately for each firm:

$$\begin{aligned} \text{Forecast Error}_{i,t+1} = & a_i + b_{i,1} \hat{\beta}_{i,t} \times \text{MacroBearGap}_{i,t} + b_{i,2} \hat{\beta}_{i,t} \times \text{MacroBullGap}_{i,t} \\ & + b_{i,3} \hat{\beta}_{i,t} \times \text{FirmBearGap}_{i,t} + b_{i,4} \hat{\beta}_{i,t} \times \text{FirmBullGap}_{i,t} \\ & + b_{i,5} M_t + u_{i,t+1}, \end{aligned} \tag{16}$$

where $\text{MacroBearGap}_{i,t}$ is the bear–base macro-attribution gap, $\text{MacroBullGap}_{i,t}$ is the bull–base macro-attribution gap, $\text{FirmBearGap}_{i,t}$ is the bear–base firm-attribution gap, and $\text{FirmBullGap}_{i,t}$ is the bull–base firm-attribution gap. Because (16) is estimated for each firm individually rather than as a panel regression, all slope coefficients carry a firm index i in contrast to (15).

We construct firm-level out-of-sample forecasts using a rolling training window of 60

observations. At each forecast date, the training sample consists of the most recent 60 reports for the same firm whose outcomes are observable, meaning that the reports were released at least one year before the test date.³¹ Each firm is therefore forecasted using only its own historical relation between beta-weighted attribution gaps, the aggregate market state, and subsequent forecast errors.

We then evaluate portfolio performance using two sorting signals. The first signal is the Bias-adjusted Forecast, defined as the analyst's base-case forecast return plus the out-of-sample forecast error predicted from (16). This adjusted measure corrects the analyst's base-case forecast for the predictable bias component associated with Macro Bear Bias. The second signal is the analyst's raw base-case forecast return, labeled Base Forecast, which serves as the benchmark for evaluating portfolio performance before correcting for Macro Bear Bias.

For each signal, we form long-short portfolios using an annual rebalancing strategy implemented separately at the end of each calendar month. Specifically, for each calendar month, we construct a portfolio that is rebalanced annually in that month, yielding 12 distinct annually rebalanced strategies. To focus on the part of the cross-section where the signal is strongest, we go long stocks in the top decile of the sorting signal and short stocks in the bottom decile. We construct both value-weighted (VW) and equal-weighted (EW) portfolios. We then average performance across the 12 calendar-month implementations and report the annualized Sharpe ratio, average monthly return, and CAPM alpha.

Table 7 reports portfolio performance for the two sorting signals. The Base Forecast portfolio, which sorts stocks solely on analysts' raw base-case forecast returns, does not deliver significant performance. The VW long-short portfolio has a negative average return and a negative CAPM alpha, and the EW portfolio is similarly weak. Thus, analysts' raw base-case forecasts do not provide meaningful out-of-sample cross-sectional return predictability.

By contrast, the Bias-adjusted Forecast delivers economically large and statistically significant abnormal returns in the VW portfolio. The VW long-short portfolio earns an average monthly return of 1.52% ($t = 2.31$) and a CAPM alpha of 1.36% ($t = 2.00$), with an annualized Sharpe ratio of 0.83. The EW portfolio also earns positive returns and positive CAPM alpha in contrast to

³¹Because the forecast error is defined as the difference between the realized one-year-ahead return and the analyst's base-case forecast return, we use only training observations released at least one year before the test report. This timing convention avoids using realized returns that would not have been observable at the forecast date. For example, to predict the forecast error for a report released on June 1, 2021, the training sample consists of reports released before June 1, 2020.

Table 7 Portfolio Performance of Linear Forecast Error Prediction

Note: This table reports monthly long-short portfolio performance for Bias-adjusted Forecast and Base Forecast. For each signal, the portfolio goes long stocks in the top decile and short stocks in the bottom decile. Bias-adjusted Forecast sorts on the analyst’s base-case forecast return plus the out-of-sample forecast error predicted from (16). Base Forecast sorts on the analyst’s raw base-case forecast return and serves as the benchmark before bias adjustment. Portfolios are rebalanced annually across 12 calendar-month implementations. Annual SR is the annualized Sharpe ratio. Avg Return is the average monthly return of the long-short portfolio. CAPM α is the intercept from a CAPM time-series regression. VW and EW denote value-weighted and equal-weighted portfolios, respectively. t -statistics are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Return Forecast	VW			EW		
	Annual SR	Avg Return	CAPM α	Annual SR	Avg Return	CAPM α
Bias-adjusted	0.8348	0.0152** (2.31)	0.0136** (2.00)	0.4864	0.0080 (1.35)	0.0064 (1.05)
Base	-0.1330	-0.0019 (-0.37)	-0.0043 (-0.86)	-0.0424	-0.0005 (-0.12)	-0.0034 (-0.79)

the Base Forecast. These results indicate that the predictable forecast errors associated with Macro Bear Bias contain substantial return-relevant information. The stronger VW results further suggest that this return-relevant bias is concentrated among larger, more investable firms, where analyst reports are more likely to be read by investors and incorporated into prices.

These findings provide a direct economic interpretation of Macro Bear Bias. The bias is not merely a textual regularity in analyst narratives. It predicts systematic errors in analysts’ numerical base-case forecasts, and those forecast errors contain information about subsequent stock returns. Stocks whose base-case forecasts are too pessimistic after accounting for macro-heavy bear narratives subsequently outperform stocks whose base-case forecasts are less favorably adjusted. Thus, the bias-adjusted signal recovers return-relevant information obscured in raw analyst forecasts, suggesting that narrative misattribution can spill over into valuation and affect the cross-section of expected returns.

5.3 Nonlinear Forecast Error Prediction and Portfolio Performance

The linear correction in Section 5.2 generates portfolio returns in the predicted direction, but they are smaller than under the nonlinear correction and insignificant in the equal-weighted portfolio. This pattern suggests that a single linear slope captures only part of the return-relevant information

contained in Macro Bear Bias. The conceptual framework explains why: Macro Bear Bias reflects a failure to condition appropriately on aggregate market states. The same bear–base macro-attribution gap may therefore imply different forecast distortions depending on whether market conditions are weak or strong. In weak market states, a given amount of downside macro emphasis may translate into one degree of pessimism in the base-case forecast; in stronger market states, the same narrative gap may imply a different distortion. More generally, the relationship between subjective belief objects and realized outcomes should be expected to be nonlinear by construction. [Dew-Becker, Giglio, and Molavi \(2025\)](#) establish this point formally, showing that under Bayesian filtering—including under behavioral departures such as misspecified priors—belief dynamics are generically nonlinear and state-dependent. The effect of Macro Bear Bias on forecast errors is therefore likely governed by nonlinear interactions between narrative-based attribution measures and market conditions, rather than by their separate linear effects.

To capture these nonlinearities, we follow [Kelly, Malamud, and Zhou \(2024, 2022\)](#) and use the Random Fourier Features (RFF) approach of [Rahimi and Recht \(2007, 2008\)](#) to predict out-of-sample forecast error. The raw predictors are the macro bear–base gap, macro bull–base gap, firm bear–base gap, firm bull–base gap, and the cumulative market excess return over the 252 trading days prior to the report release. Let $X_{i,t}$ denote the 5×1 predictor vector for a given firm-date observation:

$$X_{i,t} = [\text{MacroBearGap}_{i,t}, \text{MacroBullGap}_{i,t}, \text{FirmBearGap}_{i,t}, \text{FirmBullGap}_{i,t}, M_t] . \quad (17)$$

Compared to the linear specification in (16), we exclude the CAPM beta $\hat{\beta}_{i,t}$ from $X_{i,t}$. This choice makes the nonlinear specification deliberately conservative: firm-specific information enters only through the analyst’s own narrative attribution measures.

The RFF procedure maps $X_{i,t}$ into nonlinear features,

$$S_{i,t,p} = [\sin(\gamma \omega_p' X_{i,t}), \cos(\gamma \omega_p' X_{i,t})]' , \quad \omega_p \sim i.i.d. N(0, I), \quad (18)$$

where each ω_p defines a random projection of the original predictors, which is then transformed using sine and cosine functions. This transformation generates a rich set of nonlinear combinations of attribution gaps and market states. As emphasized by [Kelly, Malamud, and Zhou \(2024\)](#), RFF

can be interpreted as a wide two-layer neural network with random first-layer weights and estimated second-layer coefficients. Its advantage in our setting is that it introduces flexible nonlinearities while remaining computationally tractable and straightforward to evaluate out of sample. We set the bandwidth parameter to $\gamma = 2$ and draw up to $P_{\max} = 180,000$ random projections for each firm. To study the role of model complexity, we estimate forecasts over a grid of model sizes P ranging from 6 to 180,000, using nested subsets of these projections and holding the ridge penalty fixed at $z = 1,000$.

The corresponding linear benchmark of the RFF regression is

$$\begin{aligned} \text{Forecast Error}_{i,t+1} = & a_i + b_{i,1}\text{MacroBearGap}_{i,t} + b_{i,2}\text{MacroBullGap}_{i,t} \\ & + b_{i,3}\text{FirmBearGap}_{i,t} + b_{i,4}\text{FirmBullGap}_{i,t} \\ & + b_{i,5} M_t + u_{i,t+1}, \end{aligned} \quad (19)$$

which feeds the same five predictors $X_{i,t}$ from Eq. (17) into a linear specification, and therefore omits the $\hat{\beta}_{i,t}$ interaction used in the linear out-of-sample model of Eq. (16). Holding the predictor set fixed at $X_{i,t}$, the comparison between this linear benchmark and the RFF forecast isolates the incremental contribution of nonlinearity.

Following the out-of-sample design in Section 5.2, we estimate firm-level RFF forecasts and the corresponding linear benchmark in (19) using a 60-observation rolling window. To avoid look-ahead bias, each training observation must have been released at least one year before the test date. Forecasts are estimated separately by firm, using the firm's attribution-gap variables and the aggregate market state.

We then examine whether the nonlinear forecast-error correction improves the cross-sectional portfolio performance more than the linear forecast-error correction. In addition to the analyst's raw base-case forecast return (denoted as the Base Forecast), we construct a Nonlinearity-adjusted Forecast, defined as the base-case forecast return plus the out-of-sample forecast error predicted by the RFF model. To compare the performance improvement contributed by nonlinearity, we also construct the portfolio by sorting on the corresponding Linearly-adjusted Forecast, which is the base-case forecast return plus the out-of-sample forecast error predicted by (19).

Portfolio construction follows the same procedure as in Section 5.2. For each signal, we form

Table 8 Portfolio Performance of Nonlinear Forecast Error Prediction

Note: This table reports monthly long-short portfolio performance for Nonlinearity-adjusted Forecast, Linearly-adjusted Forecast, and Base Forecast. Each portfolio goes long the top decile and short the bottom decile. Nonlinearity-adjusted Forecast (Nonlinear) sorts on the analyst's base-case forecast return plus the RFF-predicted forecast-error correction. Linearly-adjusted Forecast (Linear) sorts on the analyst's base-case forecast return plus the out-of-sample forecast error predicted from (19). Base Forecast sorts on the analyst's raw base-case forecast return and serves as the benchmark before bias adjustment. RFF specification uses bandwidth $\gamma = 2$, shrinkage $z = 1,000$, and complexity $c = 3,000$, the largest value on the grid in Figure 13. Portfolios are rebalanced annually across 12 calendar-month implementations. Annual SR is the annualized Sharpe ratio. The "vs. Linear" columns report the CAPM alpha of the Nonlinear minus the Linear long-short portfolio. Avg Return is the average monthly return of the long-short portfolio. CAPM α is the intercept from a CAPM time-series regression. VW and EW denote value-weighted and equal-weighted portfolios, respectively. t -statistics are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Return Forecast	VW				EW			
	Annual SR	Avg Return	CAPM α	vs. Linear	Annual SR	Avg Return	CAPM α	vs. Linear
Nonlinear	1.0985	0.0202*** (3.04)	0.0191*** (2.84)	0.0111** (2.51)	0.9901	0.0180*** (2.74)	0.0174** (2.55)	0.0110** (2.46)
Linear	0.6518	0.0125* (1.80)	0.0107 (1.53)		0.5168	0.0094 (1.43)	0.0075 (1.13)	
Base	-0.0663	-0.0010 (-0.18)	-0.0030 (-0.61)		-0.0198	-0.0003 (-0.05)	-0.0028 (-0.63)	

top-minus-bottom decile long-short portfolios, construct both equal-weighted (EW) and value-weighted (VW) versions, implement the annual rebalancing strategy separately across the 12 calendar months, and report average performance across these implementations using the annualized Sharpe ratio, average monthly return, and CAPM alpha.

Table 8 shows a sharp contrast across the three signals. The Base Forecast signal, which sorts on the analyst's base-case forecast return alone, generates no abnormal return.³² By contrast, the Nonlinearity-adjusted Forecast delivers economically large and statistically significant abnormal returns. The VW long-short portfolio earns an average monthly return of 2.02% ($t = 3.04$) and a CAPM alpha of 1.91% ($t = 2.84$), with an annualized Sharpe ratio of 1.10. The EW portfolio earns 1.80% per month ($t = 2.74$) with a CAPM alpha of 1.74% ($t = 2.55$) and an annualized Sharpe ratio of 0.99. In comparison, the Linearly-adjusted Forecast produces substantially weaker performance: the average monthly return is 1.25% ($t = 1.80$) in the VW portfolio and 0.94%

³²The numbers are slightly different from those for the Base Forecast portfolio performance in Table 7 because Table 7 removes observations with no $\hat{\beta}_{i,t}$.

($t = 1.43$) in the EW portfolio, while the corresponding CAPM alpha is statistically insignificant in both cases.

The nonlinear adjustment also delivers economically meaningful gains relative to the linear adjustment. The nonlinearity-adjusted portfolio outperforms the linearly-adjusted portfolio significantly, with a monthly alpha of 1.11% and t -statistics of 2.51 in the VW portfolio and a monthly alpha of 1.10% and t -statistics of 2.46 in the EW portfolio.³³ These results indicate that the return-relevant component of Macro Bear Bias is not fully captured by a linear correction. Allowing forecast-error predictions to vary nonlinearly with narrative gaps and market conditions substantially improves out-of-sample portfolio performance.

Figure 13 shows that portfolio performance improves as model complexity increases. Model complexity c is defined as the number of RFF features divided by the 60-observation training window, so higher values of c correspond to a richer and more flexible nonlinear feature space. Consistent with Kelly, Malamud, and Zhou (2024, 2022), the RFF Predicted Forecast Error signal, shown in the right column, becomes more powerful as c rises: the annualized Sharpe ratio increases steadily, and both the CAPM alpha and the corresponding t -statistics display a similar upward pattern. The Nonlinearity-adjusted Forecast, shown in the left column, exhibits the same pattern. Its annualized Sharpe ratio and CAPM alpha are lower at low levels of c , indicating that a sparse RFF feature space is unable to capture the relevant interactions among narrative gaps and market conditions. As c increases, however, the RFF approximation becomes richer and recovers more of the state-dependent predictive content embedded in analyst narratives. Near $c = 3,000$, the largest complexity on our grid and the value used in Table 8, the VW portfolio reaches an annualized Sharpe ratio of approximately 1.1, and the CAPM alpha has a t -statistic above 2.7.

Taken together, these results show that Macro Bear Bias has economically meaningful predictive power for both forecast errors and cross-sectional returns, but that this predictive content is recovered more effectively through a nonlinear specification. A linear correction compresses state-dependent forecast distortions into an average slope and therefore captures only limited portfolio performance. By contrast, the RFF specification allows the forecast-error correction to vary flexibly with the interactions among macro-heavy bear narratives, firm-level narrative content, and market

³³The nonlinearity-adjusted portfolio also outperforms the bias-adjusted portfolio in Table 7. The VW monthly alpha is 0.9% with a t -statistic of 1.92, significant only at the 10 percent level. The EW monthly alpha is 1.15% with a t -statistic of 2.53.

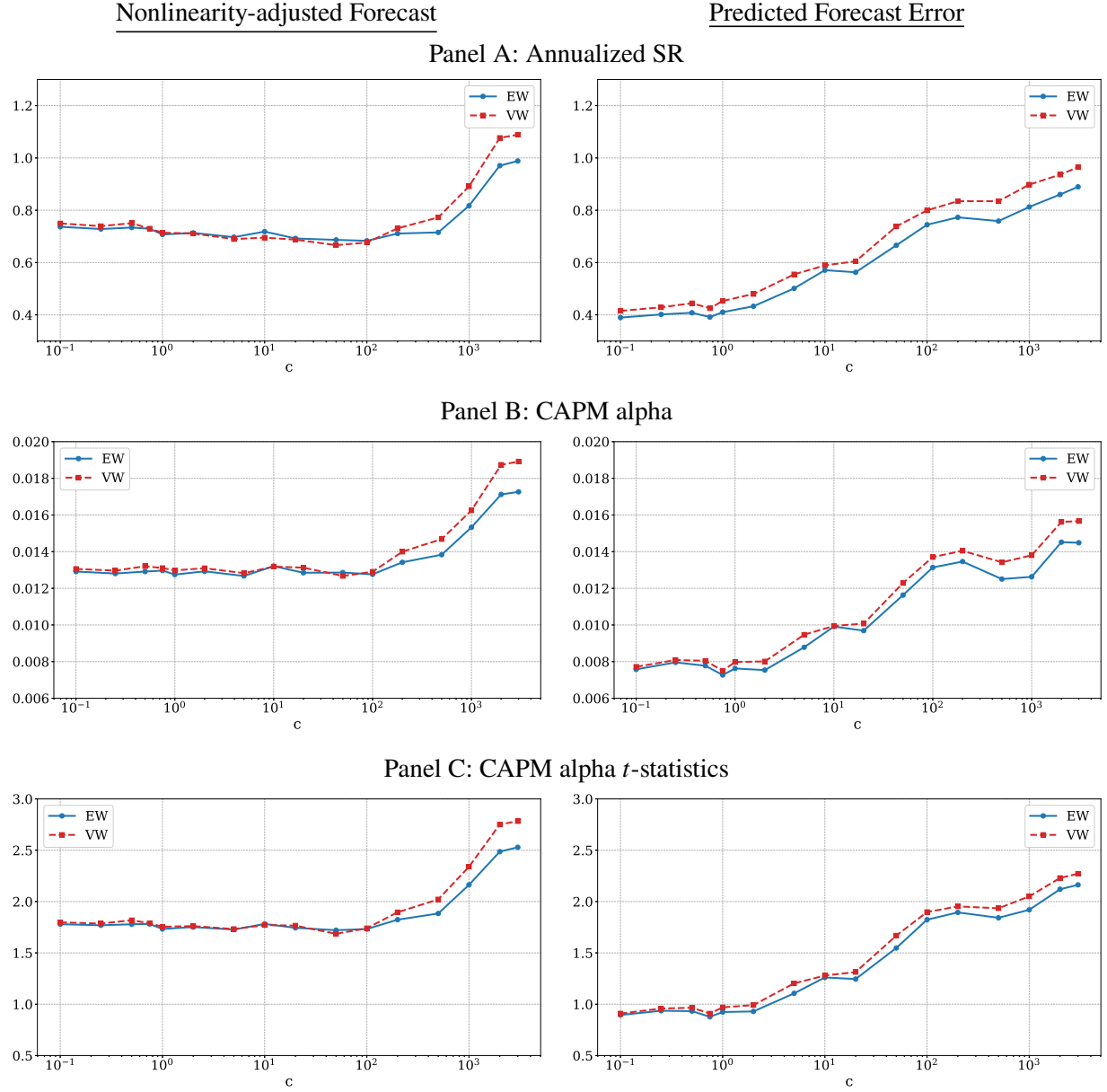


Figure 13 Out-of-sample Long-short Portfolio Performance.

Note: This figure shows the out-of-sample long-short portfolio performance when the model complexity c increases. The left column sorts on the Nonlinearity-adjusted Forecast of Table 8; the right column sorts on the RFF-predicted forecast error alone. The training window is 60 observations for each firm, and RFF count P ranges from 6 to 180,000 with bandwidth $\gamma = 2$ and shrinkage $z = 1000$.

conditions. The resulting improvement in out-of-sample performance indicates that the pricing content of Macro Bear Bias depends not only on whether analysts over-attribute downside risk to macro forces, but also on the states in which this misattribution occurs.

6 Alternative Explanations

Three accounts could produce a persistent, downside-specific macro tilt without the mechanism of Section 3, and they differ in what they would require of the data. The first is that the tilt is not a distortion at all: analysts may be reporting the correct source likelihoods, and our benchmark may simply be the wrong one. The second keeps representativeness but relocates it, evaluating diagnosticity under current conditions rather than under memory. The third replaces the psychology entirely: analysts may load macro explanations into the downside because doing so serves them, whether through the reputational incentives of the sell-side or through the more general human tendency to attribute bad outcomes to external forces. We take these in turn.

The third deserves a word at the outset, because it is the rival this paper takes most seriously. An analyst who writes a macro-heavy bear case has, whether or not they intend to, constructed an alibi: if the call goes wrong, the fault lies with the economy rather than with the stock selection. This is a genuine competing psychology, not a weaker version of ours, and we do not claim to eliminate it. What we do claim is narrower and, we think, more defensible. Two features of the data—the response of the gap to which crises are rehearsed, and its dependence on an individual analyst’s crisis experience holding the firm and the year fixed—have no motivated counterpart. We therefore read memory as necessary, though the second feature rests on a marginal estimate. A motivated channel may well operate alongside it, supplying a reason to reach for a template that memory has already stocked, and nothing in our evidence rules that out.

6.1 Rational Source Likelihoods

The first possibility is that there is no bias to explain. If macro forces genuinely are more likely to drive severe declines than ordinary outcomes at the moment each report is written, then a bear-base attribution gap is what a well-calibrated analyst should produce, and the realized- R^2 comparison of Section 2 misstates the benchmark rather than revealing a distortion.

Proposition 1(i) makes this account precise and, in doing so, testable. Under rational attribution the reported gap equals $G_{i,t}^*$ exactly, which has two consequences. Its level is pinned to the true conditional gap, and—more restrictively—it inherits whatever state dependence the true gap has. The second is the binding restriction. Whatever one believes about the level of G^* , a rational

analyst's gap must move when the conditional macro content of downside outcomes moves, and Section 2 shows that the realized bear–base difference in systematic variation falls to zero and turns negative as markets strengthen while the attribution gap does not follow.

A defender of the rational account is therefore committed to a specific claim: that the true ex ante source likelihoods diverge from their ex post realizations in exactly the pattern needed to justify a flat gap. That is not impossible, but it is a strong assumption. Confronting it directly requires estimating G^* from information available at the report date rather than proxying for it with realized outcomes.

6.2 A State-Conditioned Tilt

The second account concedes that attribution is distorted but denies that memory is doing the work. Suppose the representativeness tilt operates exactly as in Definition 2, but diagnosticity is evaluated under the current conditional distribution rather than under the rehearsed one. Lemma A.1 then delivers a shift of $\theta G_{i,t}^*$ rather than $\theta D_{i,t}$, and the reported gap becomes

$$G_{i,t} = (1 + \theta)G_{i,t}^*. \quad (20)$$

This is an attractive rival. It requires no memory, no rehearsal weights, and no anchoring parameter, and it produces a bear–base gap that exceeds the rational benchmark—which is the paper's headline fact.

It fails on the second arm of the puzzle rather than the first. Eq. (20) amplifies the rational gap but preserves its sign and its state dependence: the reported gap must collapse toward zero wherever $G_{i,t}^*$ does, and must turn negative wherever $G_{i,t}^*$ is negative. Section 2 shows neither. The realized bear–base difference in systematic variation is near zero in weak markets and negative in strong ones, and across that entire range the attribution gap stays large and positive. No value of θ reconciles a multiplicative amplification with a gap that is flat in a benchmark that is not.

The comparison identifies precisely which ingredient of the model is load-bearing. Exaggeration alone cannot generate a distortion that survives when the thing being exaggerated has gone to zero. What the model adds is a comparison set that moves slowly, which is what allows a regularity

that holds across the historical record to be applied to a date at which it does not. Appendix [A.8](#) develops the comparison formally.

6.3 Motivated Attribution

The two subsections above dispute how the attribution is formed while granting that something cognitive forms it. The third rival disputes the psychology itself. On this account the downside-specific macro tilt is not a failure of recall but a service the analyst’s narrative performs for the analyst, and the mechanism has both an institutional and a psychological form.

The institutional form is career concerns. Sell-side analysts face asymmetric reputational costs: firm-specific errors are falsifiable and attributable, while consensus-correlated errors provide reputational cover ([Hong and Kubik, 2003](#)). This generates a “cover your asset” heuristic in which macro-based bear narratives are attractive because they are difficult to falsify, relationally neutral toward management, and consensus-aligned.

The psychological form is more general and, we think, the more serious rival. Self-serving attribution—the tendency to assign favorable outcomes to one’s own judgment and unfavorable ones to external circumstance—is among the most robust findings in social psychology and has an established place in finance through models of biased self-attribution and overconfidence ([Daniel, Hirshleifer, and Subrahmanyam, 1998](#); [Gervais and Odean, 2001](#)). On this reading the macro-heavy bear case is an excuse constructed in advance, whether or not any strategic calculation takes place. The account requires no incentive contract and no conscious deliberation, which is what makes it a peer of the mechanism in Section [3](#).

Motivated attribution is also, unlike the two rivals above, an account that our headline facts do not discriminate against. It predicts a tilt that appears on the downside and not the upside, since there is nothing to excuse in a favorable outcome; it therefore reproduces Proposition [2](#) exactly. What it does not predict is any dependence on *which* crises an analyst happens to have available, on *who* the analyst is, holding the firm and the year fixed, or on the objective comovement structure of a firm’s downside. Those are the margins on which the two accounts part company; our data reach the first two, and we take them up after reporting what the incentive proxies show.

6.3.1 Incentive proxies

We begin with the institutional form, which our proxies address directly.

If career incentives drive macro emphasis, three predictions follow that our proxies can test: (i) junior analysts, who face greater reputational risk from differentiated misses, should exhibit larger MacroBearGap; (ii) firms with investment-banking relationships with Morgan Stanley should show larger MacroBearGap, since firm-specific bear narratives are relationally costly; and (iii) analyst identity should explain a substantial share of MacroBearGap variation. The data provide little support for (i) and (ii) and only partial support for (iii).

Appendix Table A.6 reports regressions of MacroBearGap on analyst experience, measured both as Morgan Stanley tenure and as years of industry experience across all brokerages (using the full IBES price-target detail file to identify each analyst's career start at any firm). Career concerns predict that junior analysts exhibit larger MacroBearGap; the data contradict this. The MS-tenure junior indicator enters with a *negative* coefficient (-0.69 , $t = -1.66$): analysts new to Morgan Stanley show *less* macro emphasis in bear narratives, not more. The profession-wide junior indicator is statistically indistinguishable from zero. The negative MS-tenure result, though marginal, is consistent with the rehearsal channel of Section 3, whose psychological basis is availability, with the caveat that availability in our setting operates through both personal recall of crisis episodes and repeated exposure to firm-specific narrative templates. Analysts new to Morgan Stanley have not yet been steeped in the firm's macro-heavy bear-case template; as they accumulate tenure, the template becomes cognitively more accessible. This is the institutional-repetition leg of availability, not personal-memory availability in the original Tversky–Kahneman sense. Our tests cannot discriminate between the two within a single-brokerage sample, and we do not claim to: the GFC-at-MS finding in Table 5 is consistent with both personal crisis memory and institution-specific socialization into the firm's crisis-era framing. Continuous tenure itself does not predict the gap, so we read both legs only as consistent with rehearsal, and discuss the limits of this identification in Section 7.

If career concerns operate through relational incentives, then analysts covering firms with active Morgan Stanley investment-banking relationships should show larger MacroBearGap, since firm-specific bear narratives risk antagonizing management during the advisory relationship. We

construct an IB-relationship indicator equal to one if Morgan Stanley served as bookrunner or lead manager on a new issuance for the covered firm within a specified lookback window, using the LSEG SDC New Issues database. Table 9, columns (1)–(4), reports the results across four lookback windows (6, 12, 24, and 36 months). All coefficients are positive, consistent with the predicted direction, but none approaches statistical significance (t -statistics range from 0.46 to 1.00).³⁴

Columns (5)–(7) of Table 9 test three additional proxies motivated by the career-concerns literature. If analysts who cover more firms rely on generic templates, coverage breadth should predict higher MacroBearGap (column 5: -0.04 , $t = -0.80$). If analysts who write more reports entrench house templates, report frequency should predict higher MacroBearGap (column 6: -0.01 , $t = -0.52$). If analysts with poor historical forecast accuracy use macro framing as reputational insurance, average forecast error should predict higher MacroBearGap (column 7: 0.48 , $t = 0.78$). None of these proxies is statistically significant, and the coverage and frequency measures carry signs opposite to the career-concerns prediction.³⁵

On balance, none of the career-concern proxies—analyst experience, IB relationships, coverage breadth, report frequency, forecast accuracy—significantly predicts MacroBearGap in the direction the channel implies. The only marginally significant coefficient, junior MS tenure, carries the wrong sign. We read this as little support for the career-concerns channel through the proxies available to us; we do not claim the channel is conclusively ruled out, since institutional-client demand, sector-head review, internal compliance, and management-access pressure are not captured by these measures and could in principle generate Macro Bear Bias through routes our data do not observe.³⁶

³⁴The SDC data match 34.5% of deals to our panel via CUSIP-to-PERMNO mapping, yielding IB-relationship rates of 12% (6-month window) to 39% (36-month window). The null result is unlikely to be driven by insufficient variation in the IB indicator.

³⁵We also examine whether analyst identity explains a meaningful share of MacroBearGap variation by adding 232 analyst fixed effects to a specification with firm and year fixed effects (Appendix Table A.7). The R^2 of the firm-demeaned regression, which includes year dummies, increases by 2.05 percentage points (from 0.28% to 2.33%), indicating persistent heterogeneity in macro-narrative style across analysts. This is consistent with either career incentives or cognitive differences in scenario construction, and does not uniquely identify career concerns.

³⁶The proxies in Table 9 explain almost none of the within-firm variation in MacroBearGap. A formal power analysis would bound the size of career-concern effect that the IB-relationship estimates can rule out.

Table 9 Career Concern Proxies: IB Relationships and Analyst Characteristics

Note: This table reports panel regressions of MacroBearGap (bear minus base macro attribution, 0–100 scale) on career-concern proxies. Columns (1)–(4) test whether Morgan Stanley investment-banking relationships predict MacroBearGap, using lookback windows of 6, 12, 24, and 36 months. IB relationship equals one if MS served as bookrunner or lead manager on a new issuance for the covered firm within the indicated window (source: SDC New Issues). Coverage equals the number of firms the analyst covers in the same calendar year. Reports equals the analyst’s annual report count. Accuracy equals the analyst’s mean absolute forecast error. Columns (5)–(7) are restricted to the analyst-matched subsample. All specifications include firm and year fixed effects. Standard errors are clustered by firm. *t*-statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	MacroBearGap						
	IB 6m	IB 12m	IB 24m	IB 36m	Coverage	Reports	Accuracy
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
MS IB Relationship	0.1484 (0.46)	0.3363 (1.00)	0.2481 (0.59)	0.3474 (0.70)			
Firms Covered					−0.0351 (−0.80)		
Reports / Year						−0.0055 (−0.52)	
Avg. Forecast Error							0.4776 (0.78)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	31,445	31,445	31,445	31,445	17,257	17,257	17,255
Within R^2	0.0000	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000

6.3.2 What separates the two accounts

The incentive proxies bear only on the institutional form of the rival. The psychological form makes no prediction about investment-banking relationships or coverage breadth, so the nulls above leave it untouched. Two results in Section 4 do bear on it, and they are the ones that carry the argument.

First, the gap responds to which crises are available. MacroBearGap widened more after the Global Financial Crisis in financials, energy, and materials, the sectors most exposed to the systematic downturn the crisis made salient (Section 4.4), and the topic-level evidence indicates, marginally, that attention lingers after the originating events leave the news cycle (Section 4.3). An alibi is useful in proportion to the reputational exposure it covers, which does not change when a crisis recedes from memory; a template is invoked in proportion to how readily it comes to mind, which does.

Second, the gap depends on who is writing. Holding the firm and the year fixed, analysts who lived through the GFC at Morgan Stanley write wider gaps, profession-wide GFC experience does not, and years of Morgan Stanley tenure, measured continuously, do not predict the gap (Table 5, columns 2–3; Appendix Table A.6). An analyst’s reputational exposure on a given stock does not depend on where they spent 2008; the set of crises they remember does. The estimate is marginal ($t = 1.65$), so this leg is suggestive rather than decisive.

The model makes a third distinguishing prediction—that the gap scales with the objective diagnosticity of a firm’s downside, where a motivated account predicts a gradient only in what the analyst has to lose—but our proxies for cross-firm diagnosticity do not separate firms, so the data cannot test it (Section 4.2).

One result is consistent with the rival but does not discriminate, and we report it as such. The psychological form of the rival predicts that the bear narrative depends on what the analyst has already committed to in the base case: an analyst whose base-case target sits above the consensus has made a bold firm-specific call, and a macro-heavy bear case is the natural excuse should it fail. Matching each report to the most recent IBES consensus target for the firm within 30 days (26,858 reports), we find that MacroBearGap is 1.05 percentage points higher when the base case is above consensus ($t = 4.47$) and 1.09 points lower when it is below ($t = -4.66$), with firm and year fixed effects (Table A.8); because the two indicators are close to complements, these are essentially one contrast estimated twice. The signed and absolute deviations from consensus have no detectable slope, so the pattern is a contrast between regimes rather than a gradient in boldness. Narrative coherence predicts the same contrast: a base case that already embeds firm-specific problems supplies idiosyncratic downside, and one that does not leaves the analyst to look elsewhere. The result therefore does not separate the accounts, and we do not read it as evidence for memory.

Our conclusion is deliberately limited. The incentive proxies available to us do not support the institutional form of the rival, and two features of the data have no counterpart in the psychological form, which is why we read memory as necessary to account for what we observe, though the second rests on a marginal estimate. It is not obviously sufficient. Motivated attribution and memory-based representativeness are not mutually exclusive: a template that memory has stocked is also a template that is convenient to reach for, and the two channels would reinforce each other in exactly the settings where we find the largest gaps. Distinguishing them cleanly requires variation

in an analyst's stake that is orthogonal to what they remember, which our setting does not supply and which we discuss as a design question in Section 7.

7 Conclusion

We document a Macro Bear Bias in sell-side analyst research: analysts systematically over-attribute a firm's downside risk to macroeconomic forces. Bear-case narratives place substantially more weight on macro forces than base-case narratives for the same firm at the same date, while bull-case narratives show no comparable gap. This bear-base gap is remarkably insensitive to market conditions. Yet, conditional on the same market state, realized bear outcomes are no more market-driven than realized base outcomes.

We explain this pattern with a memory-based attribution mechanism. The mechanism begins with a kernel of truth: historically, severe declines are more macro-driven on average because they disproportionately occur during broad market downturns. Analysts overgeneralize this remembered regularity to firm-specific downside scenarios under current market conditions, where it need not hold. Anchoring leads them to substitute remembered severe declines for the case-specific conditional distribution, while representativeness tilts attribution toward the macro feature most associated with those declines.

The evidence is consistent with this interpretation. The macro tilt is specific to downside framing rather than to scenario writing in general, and the gap stays sticky even as realized diagnosticity moves substantially over time. Bear-case narratives draw disproportionately on recession- and crisis-oriented themes rather than simply mirroring contemporaneous news. Finally, the gap shifts when crisis memories become more salient: it widened more after the Global Financial Crisis in financials, energy, and materials, the sectors most exposed to the crisis, and it is marginally wider among analysts who experienced the crisis at Morgan Stanley. Memory need not act alone. Analysts may also attribute self-servingly, writing a macro-heavy bear case as a ready excuse should the call fail, so we read memory as necessary to explain the data but not as sufficient; the two may well reinforce each other. Whatever its precise source, the distortion matters economically. A larger bear-base gap predicts more pessimistic base-case forecasts, and hence more positive

forecast errors over the following year. Adjusting forecasts for the predictable component of the bias with a nonlinear correction generates substantial return predictability.

More broadly, our findings suggest that belief distortions can arise not only in what economic agents expect about outcomes but also in how they attribute those outcomes to their causes. This distinction matters because narratives often serve as inputs into valuation, forecasting, and decision-making. A historical regularity can therefore be both true and misleading: true on average, yet inappropriate for the conditional decision problem at hand. In this sense, narrative errors need not come from inventing false relationships; they can arise from applying the right relationship in the wrong context.

This perspective opens several directions for future research. One is attribution biases in other professional settings in which agents explain adverse outcomes, including portfolio management, credit analysis, corporate forecasting, and policy analysis. Another is how remembered crises shape subsequent risk attribution, how long those effects persist, and what types of new information lead professionals to update or abandon those templates. More generally, it remains open when narrative attribution improves judgment and when it substitutes for state-contingent reasoning. Scenario-based analyst reports provide one setting in which these mechanisms are observable, but similar forces may operate wherever professionals rely on remembered stories to interpret risk.

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INTERNET APPENDIX

A Model Derivations and Additional Details

Table A.1 From results to tests

Result	Prediction	Stated and tested in	What it would rule out
Prop. 1(i)	Reported gap equals the rational gap and inherits its state dependence	§2	Nothing—this is the null
Prop. 1(ii)	Attenuation coefficient $1 - \lambda$ identical across bear, base, and bull	§3.3; requires an ex ante benchmark	Bear-specific templates; incentive accounts
Prop. 2	Bull-side excess small; bull attribution flat across market states	§4.1	Symmetric distortion; stylistic default
Prop. 3	Gap and bias component scale with objective diagnosticity across firms, sectors, and dates	§4.2, App. D.1; untestable across firms, underpowered over time	Fixed templates; visibility-based incentives
Prop. 4	Gap responds to rehearsed crises, firm and year held fixed	§4.4	Purely rational updating
Cor. 1	Analyst crisis experience predicts gap width, within firm across analysts over time	§4.4	Firm- and time-level explanations
Cor. 2	Gap predicts base-case forecast errors, scaling in β_i	§5	Undifferentiated pessimism (gradient not yet identified)

A.1 The BCGS Tilt in the Binary Case

Lemma A.1 *Let a binary distribution $(q, 1 - q)$ over sources $\{m, n\}$ be distorted by the exponential weighting of Definition 2 in [Bordalo et al. \(2016\)](#), with representativeness ratios*

$$R_m = \frac{p_B(h)}{p_0(h)}, \quad R_n = \frac{1 - p_B(h)}{1 - p_0(h)}.$$

Then

$$A = \frac{qR_m^\theta}{qR_m^\theta + (1-q)R_n^\theta}$$

satisfies

$$\text{logit } A = \text{logit } q + \theta \log \frac{R_m}{R_n} = \text{logit } q + \theta D(h).$$

Proof. Directly,

$$\text{logit } A = \log \frac{qR_m^\theta}{(1-q)R_n^\theta} = \text{logit } q + \theta \log \frac{R_m}{R_n}.$$

Moreover,

$$\log \frac{R_m}{R_n} = \log \frac{p_B(h)/(1-p_B(h))}{p_0(h)/(1-p_0(h))} = D(h).$$

□

The BCGS weighting form carries over exactly. Our two extensions are the anchored baseline of Eq. (5), which replaces q , and the rehearsed measure under which D is constructed. Boundedness is immediate from the display: $\text{logit } A$ is finite whenever $\text{logit } q$ and $D(h)$ are, and Assumption 1 guarantees both. No renormalization, truncation, or smoothing constant is required, in contrast to the continuous-state applications of the same weighting form.

A.2 Proof of Proposition 2

By Definition 2, $A_{i,U,t} = Q_{i,U,t}$ and $A_{i,0,t} = Q_{i,0,t}$, so no tilt term enters either object. Applying Definition 1 to each and differencing,

$$G_{i,t}^U = \text{logit } Q_{i,U,t} - \text{logit } Q_{i,0,t} = (1-\lambda)(\text{logit } A_{i,U,t}^* - \text{logit } A_{i,0,t}^*) + \lambda(\text{logit } p_{i,U,t} - \text{logit } p_{i,0,t}),$$

which is Eq. (8). The bias component follows by subtracting $G_{i,t}^{*U}$:

$$\mathcal{B}_{i,t}^U = G_{i,t}^U - G_{i,t}^{*U} = \lambda(D_{i,t}^U - G_{i,t}^{*U}).$$

Setting $D_{i,t}^U = 0$ gives $\mathcal{B}_{i,t}^U = -\lambda G_{i,t}^{*U}$. Comparing with Definition 3, the bear-side bias component contains the additional term $\theta D_{i,t}$, which is strictly positive under Assumptions 2–3 whenever $\theta > 0$. The difference is $\mathcal{B}_{i,t} - \mathcal{B}_{i,t}^U = \theta D_{i,t} + \lambda[(D_{i,t} - D_{i,t}^U) - (G_{i,t}^* - G_{i,t}^{*U})]$: the tilt term has no bull-side counterpart, and the remainder comes from anchoring.

A.3 Proof of Proposition 3

From Definition 3, $\mathcal{B}_{i,t} = (\lambda + \theta)D_{i,t} - \lambda G_{i,t}^*$. Differentiating with respect to D^0 , holding (λ, θ) fixed,

$$\frac{\partial \mathcal{B}_{i,t}}{\partial D^0} = (\lambda + \theta) \frac{\partial D_{i,t}}{\partial D^0} - \lambda \frac{\partial G_{i,t}^*}{\partial D^0}.$$

Under Assumption 3, $\partial D_{i,t} / \partial D^0 \geq 1$, so the first term is at least $\lambda + \theta$. The expression is therefore strictly positive whenever $\lambda \partial G_{i,t}^* / \partial D^0 < \lambda + \theta$, which is the stated condition. The condition is weak: for $\lambda > 0$ it fails only if the rational conditional gap responds to objective diagnosticity more than $(\lambda + \theta) / \lambda \geq 1$ times as strongly as memory does, and for $\lambda = 0$ it reduces to $\theta > 0$. The corresponding derivative of the observed gap is $\partial G_{i,t} / \partial D^0 = (1 - \lambda) \partial G_{i,t}^* / \partial D^0 + (\lambda + \theta) \partial D_{i,t} / \partial D^0 > 0$ whenever in addition $\partial G_{i,t}^* / \partial D^0 \geq 0$: the first term is then nonnegative, and the second is at least $\lambda + \theta$, which the stated condition makes positive.

A.4 Proof of Proposition 1

Part (i) is immediate. At $\lambda = 0$ Eq. (5) gives $\text{logit } Q_{i,k,t} = \text{logit } A_{i,k,t}^*$ for every k , and at $\theta = 0$ Eq. (6) leaves the bear case undistorted, so $A_{i,k,t} = A_{i,k,t}^*$ in all three scenarios; differencing bear and base gives $G_{i,t} = G_{i,t}^*$ and unit slope.

Using Definitions 1 and 2,

$$\begin{aligned}
G_{i,t} &= \text{logit } A_{i,B,t} - \text{logit } A_{i,0,t} \\
&= \left[(1 - \lambda) \text{logit } A_{i,B,t}^* + \lambda \text{logit } p_{i,B,t} + \theta D_{i,t} \right] \\
&\quad - \left[(1 - \lambda) \text{logit } A_{i,0,t}^* + \lambda \text{logit } p_{i,0,t} \right] \\
&= (1 - \lambda) G_{i,t}^* + (\lambda + \theta) D_{i,t}.
\end{aligned}$$

All objects are finite by Assumption 1. Differentiating with respect to G^* while holding D fixed gives $1 - \lambda$. The positivity condition follows by rearrangement. Setting $\lambda = 1$ yields $G = (1 + \theta)D$. The same pooling equation gives the scenario-level derivative in part (ii). $\square$

A.5 Proof of Proposition 4

Fix h . Because $E^B(h)$ and $E^0(h)$ are disjoint, changing the weights on a set of episodes in $E^B(h)$ leaves $p_{i,0,t}(h)$ unchanged. Write

$$p_{i,B,t}(h) = \frac{\int_{E^B(h)} w_{i,t}(e) c_m(e) d\mu^0(e)}{\int_{E^B(h)} w_{i,t}(e) d\mu^0(e)}.$$

Increasing the weights on a set $S \subseteq E^B(h)$ with $\mu^0(S) > 0$ on which c_m exceeds the current weighted mean raises that mean strictly. Hence $\text{logit } p_{i,B,t}$ and $D_{i,t}$ rise strictly. The gap then rises by $(\lambda + \theta)\Delta D > 0$. The reverse argument applies when c_m is below the current weighted mean on S . $\square$

A.6 Exact Mapping from Log-Odds Gaps to Share Gaps

Let $G = \text{logit } A_B - \text{logit } A_0$. Then

$$A_B = \frac{A_0 e^G}{1 - A_0 + A_0 e^G},$$

so

$$A_B - A_0 = \frac{A_0(1 - A_0)(e^G - 1)}{1 - A_0 + A_0 e^G}.$$

The mapping depends on the base share A_0 as well as on G , so a stable log-odds gap yields an approximately stable percentage-point gap only when the logistic Jacobian varies little over the observed range of A_0 . We therefore treat G as the model-facing outcome and report MacroBearGap alongside it as the economic magnitude.

If the LLM returns shares $(a^{\text{mac}}, a^{\text{ind}}, a^{\text{firm}})$ summing to one, the exact empirical objects are

$$\widehat{A} = a^{\text{mac}}, \quad \text{logit } \widehat{A} = \log \frac{a^{\text{mac}}}{a^{\text{ind}} + a^{\text{firm}}},$$

and

$$\widehat{G} = \log \frac{a_B^{\text{mac}}}{a_B^{\text{ind}} + a_B^{\text{firm}}} - \log \frac{a_0^{\text{mac}}}{a_0^{\text{ind}} + a_0^{\text{firm}}}.$$

Observations at the boundary require a pre-specified smoothing or clipping rule. We stress again that this transformation is performed after extraction; the LLM reports shares.

A.7 A Rational Source-Likelihood Benchmark

The rational attribution share can be implemented through a binary source process. Let

$$\pi_{m,i,t} = \Pr(S = m \mid z_t, x_i)$$

be the ex ante macro-source prior, and let

$$L_{m,i,t}^k = \Pr(E_{i,t}^k \mid S = m, z_t, x_i), \quad L_{n,i,t}^k = \Pr(E_{i,t}^k \mid S = n, z_t, x_i)$$

be source-conditioned scenario likelihoods. Bayes' rule gives

$$A_{i,k,t}^* = \frac{\pi_{m,i,t} L_{m,i,t}^k}{\pi_{m,i,t} L_{m,i,t}^k + (1 - \pi_{m,i,t}) L_{n,i,t}^k}. \quad (\text{A.1})$$

When the same prior applies across scenarios,

$$G_{i,t}^* = \log \frac{L_{m,i,t}^B / L_{n,i,t}^B}{L_{m,i,t}^0 / L_{n,i,t}^0}. \quad (\text{A.2})$$

In the analyst data these likelihoods can be estimated from rolling or cross-fitted source-conditioned return distributions, using only information available before the report date; realized conditional R^2 remains an ex post measure.

A.8 The State-Conditioned Alternative

Suppose representativeness is evaluated under the current rational conditional rather than the memory measure:

$$R_m^z = \frac{A_{i,B,t}^*}{A_{i,0,t}^*}, \quad R_n^z = \frac{1 - A_{i,B,t}^*}{1 - A_{i,0,t}^*}.$$

Lemma A.1 implies a diagnostic shift

$$\theta \log \frac{R_m^z}{R_n^z} = \theta G_{i,t}^*.$$

Without memory anchoring,

$$G_{i,t} = (1 + \theta)G_{i,t}^*.$$

This distortion amplifies the rational gap, but it preserves the sign and the state dependence of that gap. When the rational benchmark is small or negative, as it is in the data, a state-conditioned tilt cannot deliver a wide and stable positive gap. That is the work the anchored, memory-evaluated component does.

B Morgan Stanley Risk-Reward Framework Example

Gartner Inc. (IT.N), April 26, 2024

Risk Reward – Gartner Inc. (IT.N)

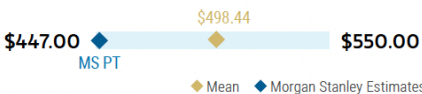
Strong Growth Story, but Balanced Risk/Reward

PRICE TARGET \$447.00

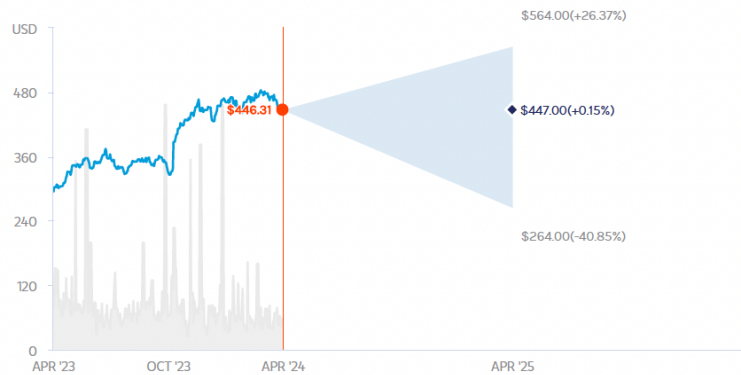
Derived from our DCF model. Our DCF includes a 8.6% WACC, 1.06 beta, and a 3.0% terminal growth rate.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



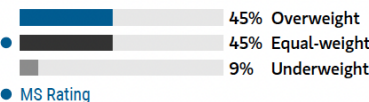
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- Gartner is the market leader in technology research, and is expanding into new functional areas (marketing, supply chain, finance, legal, sales, and HR)
- IT tends to have one of the highest top line growth rates in the Info Services peer group
- Global Business Sales' (GBS) growth has been strong and we expect it to continue due to its large TAM
- Appetite for technology research and secular growth in IT continue to provide solid backdrop
- We find valuation fair at current levels vs. peers, and would be more constructive if there were signs of approaching a positive CV inflection.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: Positive
Pricing Power: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

\$564.00

27.0x Bull Case '25e EV/MS Adj. EBITDA
Salesperson productivity accelerates, driving contract value growth and adj. EBITDA margin expansion. Revenue and adj. EBITDA CAGRs of 12% and 13% through '27; adj. EBITDA margins expand from 16% in '19 to 26% in '27.

BASE CASE

\$447.00

23.8x Base Case '25e EV/MS Adj. EBITDA
Sales force expansion drives top-line growth and modest margin expansion. Revenue and adj. EBITDA CAGRs of 9% through '27; adj. EBITDA margins expand to 25% in '27.

BEAR CASE

\$264.00

17.8x Bear Case '25e EV/MS Adj. EBITDA
Revenue growth decelerates as macro environment becomes more challenged and margin expansion is limited. Revenue and adj. EBITDA grow at 7% and 3% CAGRs through '27; adj. EBITDA margins of 22% in '27.

C LLM Prompts

C.1 Subjective Risks Identification Prompt

You are a financial analyst trained in equity valuation and empirical asset pricing. You will analyze the Bull, Base, and Bear case texts from a Morgan Stanley equity research report. Your task is to extract reasoning, classify it as firm-level, industry-level, or macro-level, identify the economic channel, and quantify attention shares across cases. Write in a concise, research-style tone used in equity research reports.

Return your full answer ****strictly in JSON format****, with one JSON object per report (suitable for .jsonl storage). Each JSON object must include keys: "filename", "bull_case", "base_case", "bear_case", and "cross_case_summary". Do not include explanations outside the JSON.

Tasks:

1. Price Targets: List the bull case, base case, and bear case price targets stated in the Risk Reward section.
2. Reason Extraction: Identify all distinct reasons for the bull, base, and bear cases. Requirements:
 - 2.1 Include any explicitly stated upside risks as part of the bull-case reasons.
 - 2.2 Include any explicitly stated downside risks as part of the bear-case reasons.
 - 2.3 Preserve the original wording as closely as possible (avoid paraphrasing).
 - 2.4 Number each reason sequentially (e.g., 1., 2., 3.).
 - 2.5 Label each reason as one of the following categories: Firm, Industry, Macro, or Mixed. If labeled Mixed, specify which component (Firm, Industry, or Macro) is dominant.
 - 2.6 Assign each reason to a single economic channel that best represents its underlying mechanism, such as expected cash flow, discount rate, risk aversion or asset demand.
3. Attention Quantification: Quantify the analyst's relative attention across Firm, Industry, and Macro news as: (Firm %, Industry %, Macro %), ensuring $\text{Firm \%} + \text{Industry \%} + \text{Macro \%} \leq 100 \%$.

- 4.Attention Evolution Summary: Summarize how the analyst’s attention distribution across Firm, Industry, and Macro news (from Step 2.5) evolves across the bull, base, and bear cases.
- 5.Channel Evolution Summary: Summarize how the dominant economic channels (from Step 2.6) evolve across the bull, base, and bear cases.

Output format (strict JSON):

```
{
  "filename": "<report_name>",
  "bull_case": {
    "bull_price_target": "<numerical price target or range>",
    "reasons": [
      {
        "id": 1,
        "text": "<verbatim reason>",
        "category": "Firm | Industry | Macro | Mixed (dominant: Firm/Industry/Macro)",
        "economic_channel": "Expected Cash Flow | Discount Rate | Risk Aversion | Asset Demand"
      }
    ],
    "attention_distribution": { "Firm": X, "Industry": Y, "Macro": Z },
    "analytical_summary": "[one concise sentence summarizing key focus of bull case]"
  },
  "base_case": {
    "base_price_target": "<numerical price target or range>",
    "reasons": [
      {
        "id": 1,
        "text": "<verbatim reason>",
        "category": "Firm | Industry | Macro | Mixed (dominant: ...)",
        "economic_channel": "Expected Cash Flow | Discount Rate | Risk Aversion | Asset Demand"
      }
    ]
  }
}
```

```

    ],
    "attention_distribution": { "Firm": X, "Industry": Y, "Macro": Z },
    "analytical_summary": "[one concise sentence summarizing key focus of base
        case]"
    },

    "bear_case": {
        "bear_price_target": "<numerical price target or range>",
        "reasons": [
            {
                "id": 1,
                "text": "<verbatim reason>",
                "category": "Firm | Industry | Macro | Mixed (dominant: ...)",
                "economic_channel": "Expected Cash Flow | Discount Rate | Risk
                    Aversion | Asset Demand"
            }
        ],
        "attention_distribution": { "Firm": X, "Industry": Y, "Macro": Z },
        "analytical_summary": "[one concise sentence summarizing key focus of bear
            case]"
    },

    "cross_case_summary": {
        "attention_evolution": "[summary of how Firm/Industry/Macro attention
            shifts across bull -> base -> bear]",
        "channel_evolution": "[summary of how dominant economic channels shift
            across bull -> base -> bear]"
    }
}

```

Rules:

- All percentages must be integers (no decimals).
- Maintain an analytical and academic tone throughout.
- Do NOT include any text outside the JSON structure.
- Ensure structural consistency and valid JSON formatting.

C.2 Topic Classification Prompt

You are a financial research assistant.

The input summaries already contain ONLY macro-related content.

Do NOT extract or filter anything.

Tasks:

1. Classify each scenario (Bull, Base, Bear) using the taxonomy.
 - Multiple topics allowed.
 - If broad -> "Mixed"
 - If not covered -> "Other: <specific macro theme>"
 - If empty -> "No explicit macro content"
 - Also assign a percentage weight to each topic based on its importance in the summary:
 - The percentages must sum to 100 for each scenario
 - Use integers (e.g., 20, 30, 50)
 - The order of percentages must match the order of topics exactly

1. Compare macro differences:

Bear vs Base:

- If both Bear and Base are empty -> "No explicit macro content"
- If Base is empty but Bear is not:
 - > return ALL Bear topics
- If Bear is empty but Base is not:
 - > "No meaningful difference"
- Otherwise:
 - Identify ONLY macro topics that appear in Bear but NOT in Base
 - Do NOT include topics already in Base
 - If no additional topics -> "No meaningful difference"

- For any returned topics:
 - Assign percentage weights that sum to 100
 - Reflect relative importance of differences
 - Order must match topics exactly

Bull vs Base:

- If both Bull and Base are empty -> "No explicit macro content"
- If Base is empty but Bull is not:
 - > return ALL Bull topics
- If Bull is empty but Base is not:
 - > "No meaningful difference"
- Otherwise:
 - Identify ONLY macro topics that appear in Bull but NOT in Base
 - Do NOT include topics already in Base
 - If no additional topics -> "No meaningful difference"
- For any returned topics:
 - Assign percentage weights that sum to 100
 - Reflect relative importance of differences
 - Order must match topics exactly

Rules:

- Use only provided text
- No inference beyond text
- Topics must come from taxonomy (+ Mixed / Other)
- "Other" must specify a real macro concept
- Output valid JSON only

Taxonomy:

```
{TOPIC_TAXONOMY}
```

Input:

```
[BULL]
{bull_summary}
```

```
[BASE]
{base_summary}
```

```
[BEAR]
{bear_summary}
```

Output JSON:

```
{
  "bull": {
    "topics": ["..."],
    "percentages": []
  },
  "base": {
    "topics": ["..."],
    "percentages": []
  },
  "bear": {
    "topics": ["..."],
    "percentages": []
  },
  "bear_vs_base": {
    "topics": ["..."],
    "percentages": []
  },
  "bull_vs_base": {
    "topics": ["..."],
    "percentages": []
  }
}
```

}

C.3 Systematic vs. Idiosyncratic Prompt

You are a financial research analyst.

Your task is to classify each statement based only on its text content.

Determine whether the primary driver described in the statement is:

- Systematic
- Idiosyncratic
- Mixed

Judge only from the text itself.

DEFINITIONS

Systematic:

Primarily driven by broad common factors,
affecting multiple firms or the overall market,
and generally not diversifiable.

Idiosyncratic:

Primarily driven by firm-specific factors,
mainly affecting one firm or a small number of firms,
and generally diversifiable.

Mixed:

Use only when both systematic and idiosyncratic factors are explicitly stated and equally important.

CRITICAL RULES

1. Primary Driver Rule

Classify based on the main economic driver.

2. Mixed Rule

Use Mixed only when both types are explicitly present and neither is clearly dominant.

3. Forced Choice Rule

Every statement must be classified as:

- Systematic
- Idiosyncratic
- Mixed

4. Background vs Driver Rule

If broad market or environmental conditions are only background context, but the company itself drives the outcome, classify as Idiosyncratic.

INSTRUCTIONS

For each item:

1. Read the statement carefully.
2. Determine the true primary driver.
3. Return one final label.

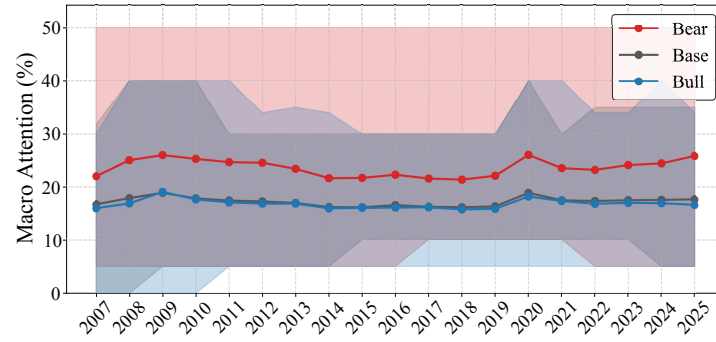
OUTPUT FORMAT

Return ONLY valid JSON:

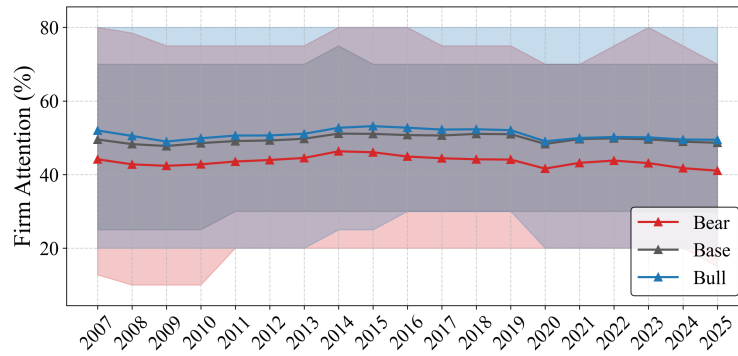
```
{
  "items": [
    {
      "scenario": "bull/base/bear",
      "segment_number": "1",
      "original_text": "...",
      "correct_label": "Systematic",
      "explanation": "short explanation"
    }
  ]
}
```

D Additional Exhibits

Panel A: Macro Attention



Panel B: Firm Attention



Panel C: Industry Attention

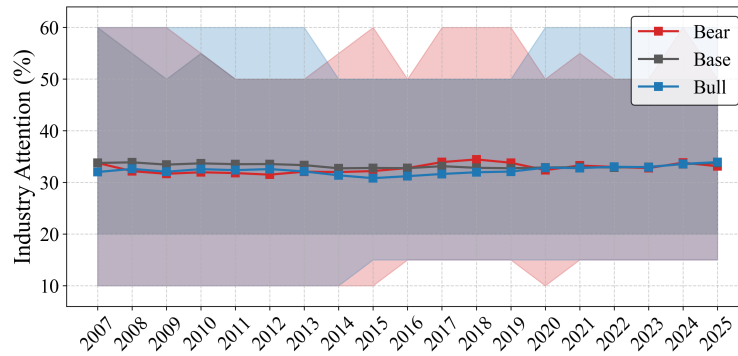


Figure A.1 Time Series of Attribution to Macro, Firm, and Industry News.

Note: Time series of annual average attribution shares to firm-, industry-, and macro-level news. The solid lines represent yearly means. The shaded areas correspond to the empirical 5th to 95th percentile range of the cross-sectional distribution in each year.

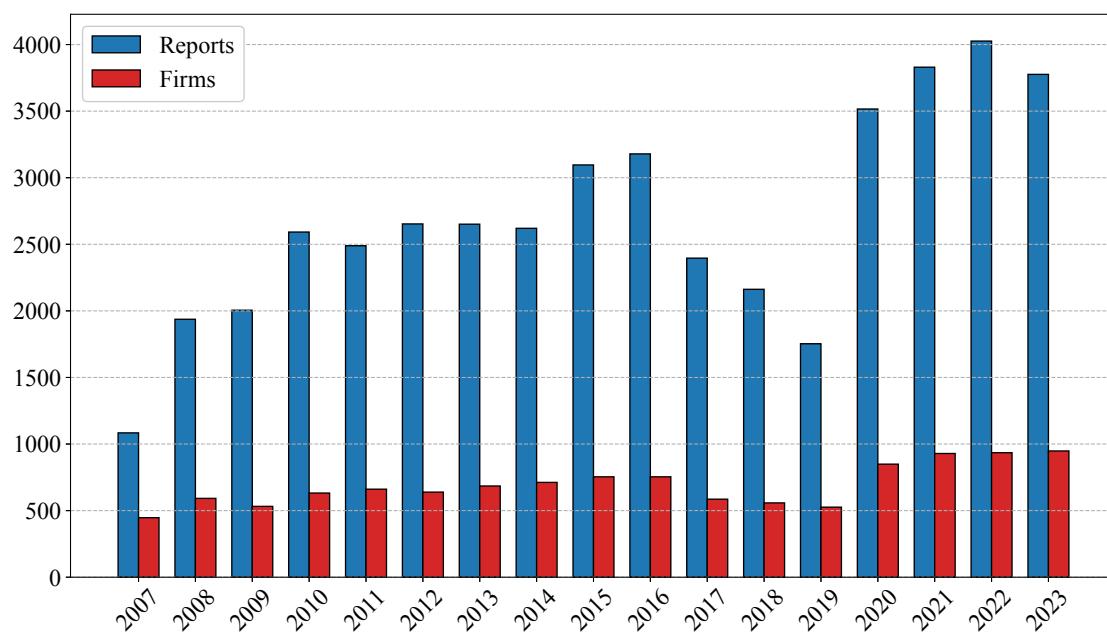


Figure A.2 Yearly Distribution of Reports and Firms after Matching with Returns.

Note: This figure shows the annual number of Morgan Stanley Risk-Reward analyst reports and the number of covered firms after matching the reports with one-year-ahead firm returns.

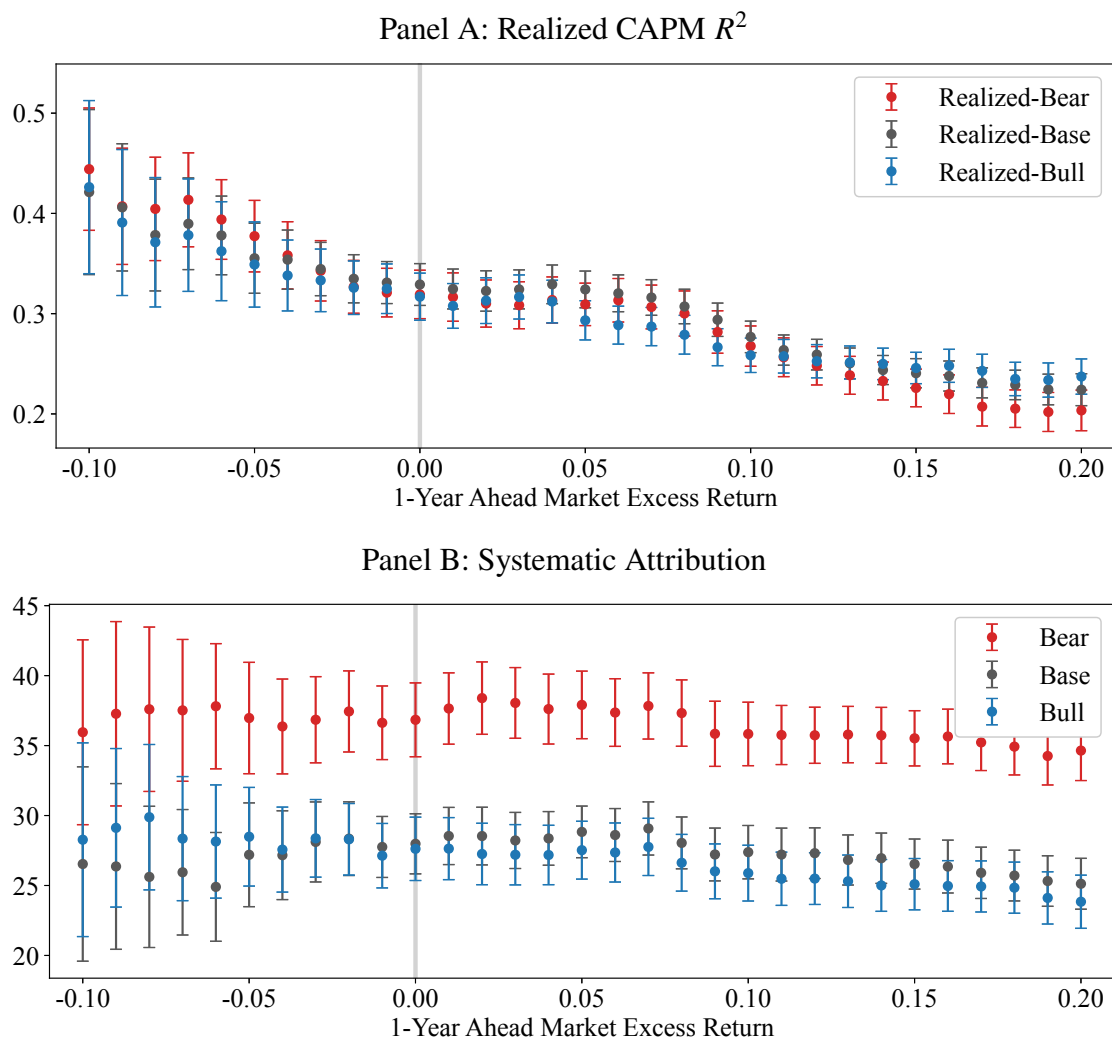


Figure A.3 Realized CAPM R^2 and Systematic Attribution Conditional on Market States.

Note: This figure shows the realized R^2 and systematic attribution conditional on different market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. We apply the same-firms rule within each bin, requiring firms to have realized-bear, realized-base, and realized-bull reports. Systematic attribution is measured as the firm-level average systematic attribution within each ex-ante scenario (bear, base, and bull). Realized CAPM R^2 is computed as the firm-level average R^2 over the 252 trading days following the report release for reports ex post classified as realized bear/base/bull cases. Error bars show 95% confidence intervals from constant-only regressions.

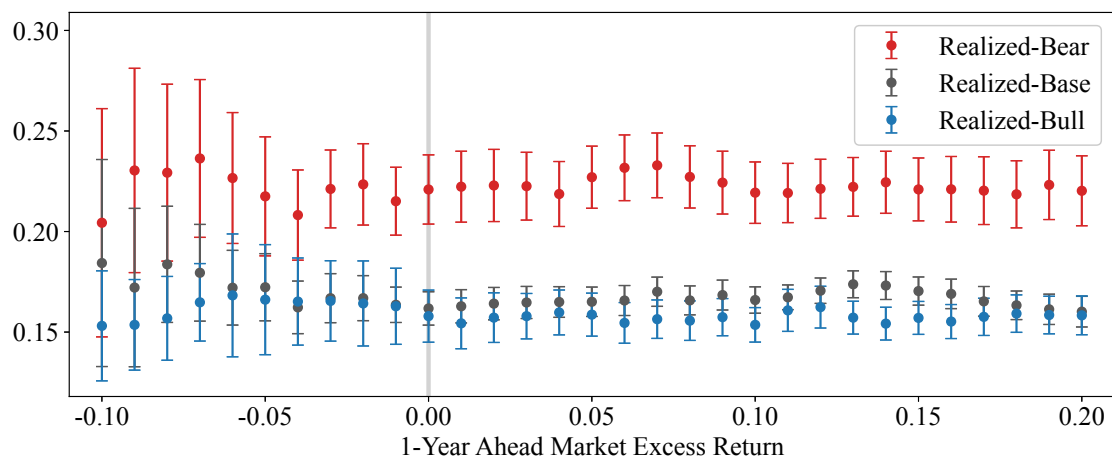


Figure A.4 Realized Macro Attribution Conditional on Market States.

Note: This figure shows the realized macro attribution conditional on different market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. We apply the same-firms rule within each bin, requiring firms to have realized-bear, realized-base, and realized-bull reports. Macro attribution is measured as the firm-level average macro attribution within each realized scenario (bear, base, and bull). For example, Macro Bear attribution includes a report's Macro Bear attribution only if the report's outcome is realized as a bear case. Error bars show 95% confidence intervals from constant-only regressions.

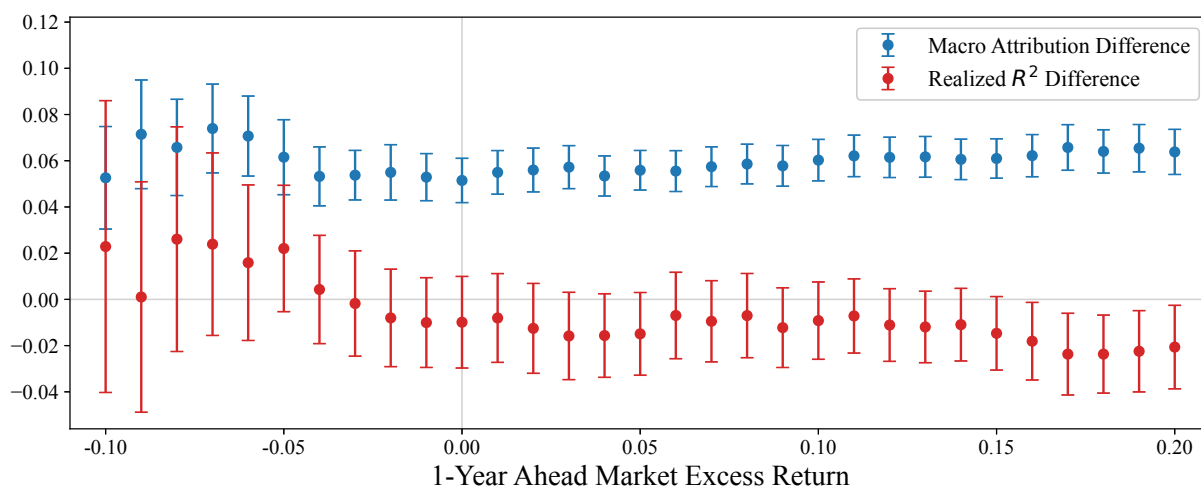


Figure A.5 Bear – Base Gaps in Systematic Attribution and Realized CAPM R^2 Conditional on Market States.

Note: This figure plots firm-level bear–base differences in systematic attribution and realized CAPM R^2 on different market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. We apply the same-firms rule within each bin, requiring firms to have realized-bear, realized-base, and realized-bull reports. Systematic attribution is measured as the firm-level average systematic attribution in the bear and base cases. Realized CAPM R^2 is computed as the firm-level average R^2 over the 252 trading days following the report release for reports ex post classified as realized bear or realized base cases. The plotted values represent the bear-minus-base differences. Error bars show 95% confidence intervals from constant-only regressions.

Table A.2 Descriptions of Key Topics Used in Attention and Correlation Analysis

Note: Bear Attn. is the average monthly share of bear-case macro-risk mentions assigned to the topic. News Attn. is the average monthly share of WSJ articles tagged with the topic. News Sent. is the average WSJ news sentiment for the topic. Section 4.3 describes the construction.

Topic	Description	Bear Attn. (%)	News Attn. (%)	News Sent.
Economic Growth and Recovery Outlook	Indicators suggesting potential economic recovery emerge from consumer confidence, job growth, and corporate earnings. Positive signals improve investor outlook, although inflation and labor market weakness may remain concerns.	10.66	2.36	0.48
COVID-19 Pandemic and Vaccine Developments	The COVID-19 pandemic significantly affected global health and economic conditions. Vaccine development, distribution logistics, variants, and public health responses shaped market expectations and business conditions.	5.36	5.34	0.41
Market Activity and Financial Performance	Stock market activity reacts to earnings releases, firm performance, and sector-wide developments. Strong results support rallies, while weak performance increases caution and volatility.	4.75	5.44	0.39
Consumer Price Index and Inflation Trends	Rising consumer prices and inflation pressures influence monetary policy, investor sentiment, and broader economic strategies. Central bank responses are especially important for markets.	2.40	0.86	0.08
International Trade Relations and Economic Policies	Trade tensions, tariff policy, and negotiations among major economies affect global commerce, supply chains, and investor confidence.	2.31	1.60	0.12
Consumer Behavior Amid Economic Strains	Economic pressures change household spending patterns, especially for discretionary purchases. Firms adjust pricing and marketing strategies amid inflation and fuel cost concerns.	1.91	0.24	-0.05
Geopolitical Tensions and Economic Impacts	Political instability, wars, and international conflicts affect stock prices, commodity markets, and investor sentiment through heightened uncertainty.	0.46	4.40	-0.11
Economic Stimulus and Government Interventions	Fiscal stimulus packages, government support programs, and intervention policies are used to stabilize growth, employment, and financial markets during downturns.	0.40	0.46	0.07

Table A.3 Forecast Error Regressions on Attribution Gaps

Note: This table reports empirical results from the following forecast-error prediction regression:

$$\text{Forecast Error}_{i,t+1} = a + b_1 \text{AttributionGap}_{i,t} + b_2 M_t + b_3 \hat{\beta}_{i,t} + b_4 R_{i,t} + FE_i + u_{i,t+1},$$

where each narrative attribution-gap variable is NOT interacted with stock i 's CAPM beta. Stock i 's CAPM beta $\hat{\beta}_{i,t}$ is estimated using daily returns over the 252 trading days prior to the report release. M_t and $R_{i,t}$ denote the market excess return and stock i 's excess return over the same 252-day pre-release window. Columns (1)–(4) include each attribution gap individually. Column (5) includes the two macro attribution gaps jointly. Column (6) includes all four attribution-gap variables simultaneously. Firm fixed effects are included in all specifications. Standard errors are clustered by firm and month. t-statistics are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
Attribution Gaps						
Macro Bear – Base	0.0012** (2.38)				0.0011** (2.09)	0.0023*** (3.26)
Macro Bull – Base		0.0009 (1.03)			0.0005 (0.54)	0.0002 (0.24)
Firm Bear – Base			0.0006 (1.11)			0.0018** (2.28)
Firm Bull – Base				-0.0002 (-0.23)		-0.0004 (-0.55)
Control Variables						
M_t	-0.1451 (-1.55)	-0.1462 (-1.56)	-0.1459 (-1.56)	-0.1458 (-1.55)	-0.1454 (-1.55)	-0.1454 (-1.55)
$\hat{\beta}_{i,t}$	-0.0323 (-0.58)	-0.0317 (-0.56)	-0.0315 (-0.56)	-0.0316 (-0.56)	-0.0323 (-0.57)	-0.0322 (-0.57)
$R_{i,t}$	-2.4847 (-0.15)	-2.4249 (-0.15)	-2.2599 (-0.14)	-2.3870 (-0.15)	-2.5042 (-0.15)	-2.2778 (-0.14)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	43,306	43,306	43,306	43,306	43,306	43,306
Within R^2	0.0004	0.0004	0.0004	0.0003	0.0004	0.0006

Table A.4 Bull-Case Macro Attention and Topic Salience

Note: This table reports panel regressions of bull-case analyst attention to topic k at time t on the autoregressive (AR) residual of WSJ news coverage (attention shock), its interaction with WSJ news sentiment, the sentiment level, and three lags of the dependent variable (analyst attention to the topic); see Equation (10). Columns correspond to lag $\tau \in \{1, 2, 3, 4\}$. All specifications include time fixed effects. Heteroskedasticity-robust (HC1) t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	$\tau = 1$	$\tau = 2$	$\tau = 3$	$\tau = 4$
WSJ Attention Shock $_{t-\tau,k}$	0.0315* (1.85)	0.0250* (1.70)	0.0103 (0.97)	-0.0025 (-0.35)
WSJ Attention Shock $_{t-\tau,k} \times$ WSJ Sentiment $_{t-\tau,k}$	-0.0783 (-1.35)	-0.0876 (-1.52)	-0.0540 (-1.08)	-0.0338 (-0.83)
WSJ Sentiment $_{t-\tau,k}$	0.0003** (2.08)	0.0003* (1.85)	0.0002 (1.29)	0.0003** (1.97)
Bull Attention $_{t-1,k}$	0.2889*** (11.22)	0.2849*** (10.95)	0.2842*** (10.86)	0.2849*** (10.85)
Bull Attention $_{t-2,k}$	0.1844*** (6.65)	0.1864*** (6.70)	0.1818*** (6.51)	0.1804*** (6.45)
Bull Attention $_{t-3,k}$	0.4919*** (16.22)	0.4938*** (16.30)	0.4987*** (16.39)	0.5004*** (16.37)
Time FE	Yes	Yes	Yes	Yes
Observations	11,440	11,388	11,336	11,284
R^2	85.05%	85.10%	85.16%	85.27%

D.1 Proxies for Objective Diagnosticity

Table A.5 reports five portfolio sorts on proxies for cross-firm differences in objective downside diagnosticity, each built from information available at the report date. None has a first stage in the predicted direction. In no panel does the within-firm realized bear-base R^2 gap of the quintile's subsequent outcomes rise significantly across quintiles (column 3), and in Panel E it falls significantly (high minus low -3.84 , standard error 1.03); across panels the high-minus-low difference ranges from -3.84 to 1.30 percentage points. Without a first stage, the sorts cannot test Proposition 3. We report them for completeness: MacroBearGap does not rise across quintiles in any panel, and its high-minus-low point estimate is negative in all five (column 2). A continuous regression of MacroBearGap on the firm-mean downside-beta spread, with month fixed effects, gives the same sign (-1.46 , $t = -2.08$, standard errors clustered by firm); within firm, the slope on a report's own spread is zero (0.01 , $t = 0.06$). Across firms, the point estimates thus run against the proposition, although the proxies do not separate realized diagnosticity.

Figure A.6 reports a time-series version of the test. Memory is built from the record available at the report date, so years in which the recent record makes severe declines look more macro-driven should have wider gaps. For each report year y from 2011 to 2023, the figure plots mean MacroBearGap against the within-firm realized bear-base R^2 gap among reports dated $y - 4$ to $y - 2$, whose one-year windows closed before year y . The slope is 0.077 ($t = 0.88$, Newey–West standard errors, 13 years), and the slope for the bias component is 0.048 ($t = 0.20$). The sign is the one the model predicts, but thirteen annual observations leave the test without power.

D.2 Additional Evidence on Mechanisms

Analysts covering the same sector may draw on a common pool of rehearsed crises—through shared media exposure, industry conferences, and peer discussions—so their MacroBearGap should comove within sector–quarters, even after absorbing firm fixed effects. We construct a leave-one-out (LOO) sector–quarter average of MacroBearGap and test whether an analyst's own MacroBearGap covaries with her peer average. The leave-one-out construction removes the focal observation, so the comovement is not mechanical; it does not, however, isolate peer influence from common sector–quarter shocks.

Column (4) of Table 5 is consistent with this prediction, though only marginally. The leave-one-out sector MacroBearGap positively predicts the analyst's own MacroBearGap (0.083 , $t = 1.75$), suggesting that analysts covering firms in the same sector–quarter converge on similar levels of macro emphasis beyond what firm fixed effects explain.

Table A.5 MacroBearGap and Realized Diagnosticity across Portfolio Sorts

Note: This table sorts reports into quintiles of five proxies for objective downside diagnosticity, all built from information available at the report date. Panels A–C use the downside-beta spread $\beta^- - \beta^+$ from trailing 252-day regressions on positive and negative market returns. Panels D–E use the realized bear–base R^2 gap among the firm’s (sector’s) earlier reports whose one-year windows had closed before the report date, requiring at least two earlier realized-bear and two realized-base outcomes. Column (1) reports the mean of the sort variable. Column (2) reports mean MacroBearGap. Column (3) reports the within-firm (firm fixed effects) difference in realized CAPM R^2 between realized-bear and realized-base reports in the quintile, the first stage: whether the sort separates objective diagnosticity. Column (4) adds market-state-bin fixed effects. Column (5) is column (2) minus column (4). All differences are in percentage points. Firm-clustered bootstrap standard errors (1,000 draws) in parentheses. Unfiltered sample.

	(1) Sort var.	(2) MacroBearGap	(3) Realized gap (first stage)	(4) Realized gap, state held fixed	(5) Bias
<i>Panel A: Firms sorted by mean pre-report downside-beta spread, $\beta^- - \beta^+$</i>					
Q1	−0.33	5.29 (0.47)	1.82 (0.69)	−1.86 (0.58)	7.15 (0.73)
Q2	−0.08	6.60 (0.42)	5.09 (0.84)	0.15 (0.81)	6.46 (0.90)
Q3	0.01	7.51 (0.41)	4.14 (0.67)	−0.22 (0.66)	7.73 (0.77)
Q4	0.14	5.98 (0.44)	2.57 (0.81)	−1.36 (0.76)	7.34 (0.83)
Q5	0.44	5.21 (0.44)	2.79 (0.73)	−0.53 (0.59)	5.74 (0.72)
High – low		−0.09 (0.65)	0.96 (1.02)	1.33 (0.84)	−1.41 (1.01)
<i>Panel B: Reports sorted by own pre-report downside-beta spread, within year</i>					
Q1	−0.56	6.09 (0.30)	2.79 (0.63)	−0.46 (0.60)	6.55 (0.66)
Q2	−0.17	6.63 (0.31)	3.33 (0.78)	−0.13 (0.71)	6.75 (0.79)
Q3	0.01	7.09 (0.28)	3.90 (0.84)	−0.25 (0.75)	7.35 (0.78)
Q4	0.19	6.81 (0.27)	3.09 (0.76)	−1.82 (0.73)	8.62 (0.75)
Q5	0.63	5.75 (0.25)	3.90 (0.65)	−0.04 (0.58)	5.79 (0.62)
High – low		−0.34 (0.32)	1.11 (0.91)	0.42 (0.83)	−0.76 (0.87)
<i>Panel C: Sectors (two-digit NAICS) sorted by mean downside-beta spread</i>					
Q1	−0.02	7.25 (0.44)	3.45 (0.80)	−1.16 (0.78)	8.40 (0.91)
Q2	0.02	5.79 (0.29)	2.66 (0.59)	−1.24 (0.56)	7.02 (0.59)
Q3	0.03	6.37 (0.55)	4.75 (0.82)	−0.85 (0.71)	7.23 (0.91)
Q4	0.05	7.84 (0.56)	5.15 (1.00)	1.95 (0.92)	5.89 (1.07)
Q5	0.12	5.89 (0.49)	1.77 (1.16)	−1.51 (0.95)	7.40 (0.96)
High – low		−1.36 (0.67)	−1.67 (1.38)	−0.35 (1.21)	−1.00 (1.29)
<i>Panel D: Reports sorted by the firm’s own past realized R^2 gap, within year</i>					
Q1	−10.74	7.19 (0.81)	2.12 (1.38)	−0.91 (1.24)	8.10 (1.27)
Q2	−2.46	6.73 (0.58)	1.41 (1.12)	−0.99 (1.02)	7.72 (1.06)
Q3	2.17	7.14 (0.51)	3.25 (1.07)	0.18 (0.97)	6.96 (1.02)
Q4	7.64	6.64 (0.52)	2.67 (1.21)	−0.01 (1.09)	6.65 (1.21)
Q5	18.10	5.77 (0.52)	3.42 (1.49)	0.87 (1.36)	4.90 (1.45)
High – low		−1.42 (0.94)	1.30 (2.04)	1.79 (1.88)	−3.20 (1.96)
<i>Panel E: Reports sorted by the sector’s past realized R^2 gap, within year</i>					
Q1	−4.45	6.41 (0.29)	4.99 (0.67)	0.96 (0.60)	5.44 (0.66)
Q2	−1.56	6.23 (0.30)	2.25 (0.86)	−2.14 (0.84)	8.38 (0.87)
Q3	−0.35	6.81 (0.34)	2.25 (0.72)	−1.69 (0.64)	8.50 (0.70)
Q4	1.76	6.89 (0.32)	3.20 (0.70)	−0.24 (0.64)	7.12 (0.69)
Q5	4.96	5.52 (0.33)	1.16 (0.76)	−1.12 (0.64)	6.65 (0.73)
High – low		−0.88 (0.43)	−3.84 (1.03)	−2.09 (0.90)	1.21 (1.01)

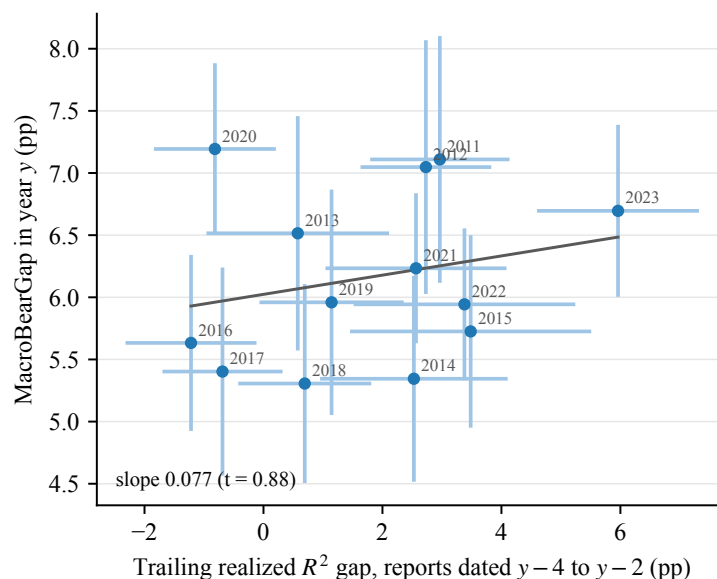


Figure A.6 MacroBearGap and Trailing Realized Diagnosticity.

Note: Each point is a report year y from 2011 to 2023. The horizontal axis is the within-firm difference in realized CAPM R^2 between realized-bear and realized-base reports dated $y-4$ to $y-2$ (firm fixed effects), all of whose one-year windows closed before year y . The vertical axis is mean MacroBearGap of reports dated in year y . Bars show 95% confidence intervals from standard errors clustered by firm; the line is the OLS fit.

Table A.6 reports panel regressions of MacroBearGap on analyst experience, measured as Morgan Stanley tenure (columns 1 and 3), industry-wide experience across all brokerages (columns 2 and 4), and a horse-race specification including both measures (column 5). See Section 6.3 for discussion.

Table A.7 compares the within- R^2 of a baseline specification (firm and year fixed effects) with a specification that adds 232 analyst fixed effects. See Section 6.3 for discussion.

Table A.8 reports the regressions of MacroBearGap on the position of the base-case target relative to the IBES consensus discussed in Section 6.3.2.

Table A.6 Analyst Experience and Macro Bear Bias

Note: This table reports panel regressions of MacroBearGap (bear minus base macro attribution, 0–100 scale) on analyst experience measures. Junior (MS) equals one if the analyst’s tenure at Morgan Stanley is at most 36 months. Junior (Profession) equals one if the analyst’s industry experience across all brokerages is at most 3 years (identified from the full IBES price-target detail file). Columns (3)–(4) use continuous experience measures in years. Column (5) includes both MS and profession experience simultaneously. All columns are restricted to the analyst-matched subsample. All specifications include firm and year fixed effects. Standard errors are clustered by firm. *t*-statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	MacroBearGap				
	(1)	(2)	(3)	(4)	(5)
	Junior (MS)	Junior (Prof.)	Exp. MS	Exp. Prof.	Horse Race
Junior (MS)	−0.6924* (−1.66)				
Junior (Profession)		0.0759 (0.15)			
Experience MS (years)			0.0750 (0.94)		0.0914 (0.79)
Experience Prof. (years)				0.0223 (0.48)	−0.0142 (−0.21)
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	17,257	17,257	17,257	17,257	17,257
R^2	0.0005	0.0000	0.0003	0.0001	0.0003

Table A.7 Analyst Fixed Effects and Macro Bear Bias

Note: This table reports the incremental R^2 from adding analyst fixed effects to a specification with firm and year fixed effects. The dependent variable is MacroBearGap (bear minus base macro attribution, 0–100 scale). Firm effects are absorbed by within-firm demeaning. Year effects are captured by year dummies. Model (B) adds 232 analyst dummies. Both models are estimated on the analyst-matched subsample. ΔR^2 is the incremental R^2 from adding analyst fixed effects.

	(A)	(B)
	Baseline	+ Analyst FE
R^2	0.0028	0.0233
ΔR^2		0.0205
Firm FE	Yes	Yes
Year FE	Yes	Yes
Analyst FE		Yes
Observations	17,257	17,257
Firms	1,274	1,274
Analysts		232

Table A.8 MacroBearGap and the Base Case Relative to Consensus

Note: This table reports panel regressions of MacroBearGap (bear minus base macro attribution, 0–100 scale) on indicators and continuous measures of the Morgan Stanley base-case target's position relative to the IBES consensus. Columns (1)–(2) use the discrete indicators $\mathbf{1}\{P_{MS}^{base} > P^{cons}\}$ and $\mathbf{1}\{P_{MS}^{base} < P^{cons}\}$ separately. Columns (3)–(4) use the signed continuous deviation $(P_{MS}^{base} - P^{cons})/P^{cons}$ and its absolute value. The sample is restricted to observations with an available IBES consensus match within 30 days. All specifications include firm and year fixed effects. Standard errors are clustered by firm. t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	MacroBearGap			
	(1)	(2)	(3)	(4)
$\mathbf{1}\{P_{MS}^{base} > P^{cons}\}$	1.0491*** (4.47)			
$\mathbf{1}\{P_{MS}^{base} < P^{cons}\}$		−1.0938*** (−4.66)		
$(P_{MS}^{base} - P^{cons})/P^{cons}$			0.0865 (1.34)	
$ (P_{MS}^{base} - P^{cons})/P^{cons} $				0.0428 (0.61)
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Observations	26,858	26,858	26,380	26,380
Within R^2	0.0012	0.0013	0.0000	0.0000